



BLOCK 3

DATA ANALYSIS

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Block 3 on Data Analysis consists of six units.

Unit 7, “Hypothesis Testing,” discusses the concept, meaning, importance, and process of hypothesis testing. The stages of hypothesis testing, level of significance, two types of errors (type I and type II), and use of SPSS for hypothesis testing are also explained.

Unit 8, “Correlation and Simple Linear Regression,” explains the computation of the correlation coefficient by Karl Pearson and Spearman’s method. The estimation and interpretation of regression results, the applications of correlation and regression, and the use of SPSS are also highlighted.

Unit 9, “Multiple Regression Analysis,” explains the concept, applications, and computational aspects of multivariate regression model through an illustrative case example. The problems encountered in multiple regression analysis are also discussed.

Unit 10, “Discriminant Analysis and Logistic Regression Analysis,” discusses that both discriminant analysis and logistic regression are applied to predict the probability of a specific categorical dependent variable based on several explanatory or predictor variables. The key difference lies in the population distribution; when the population is normally distributed, discriminant analysis is used; however, no such assumptions about the population are required for logistic regression analysis. The calculation and interpretation of discriminant analysis and logistic regression methods are explained using SPSS and hypothetical case examples.

Unit 11, “Factor Analysis and Cluster Analysis,” discusses that factor analysis is a technique whose purpose is data reduction and summarization, whereas the objective of cluster analysis is to find a group that shares some common characteristics. Various inputs required, applications, and approaches for factor and cluster analysis are also explained.

Unit 12, “Conjoint Analysis and Multidimensional Scaling,” explains two of the techniques of multivariate analysis, viz., conjoint analysis and Multidimensional Scaling (MDS). Conjoint Analysis is essentially a data decomposition technique that attempts to plot the output data on the joint space of each attribute’s importance. The main steps in using conjoint analysis are explained. Multidimensional scaling technique, also known as the perceptual map or positioning map, that is particularly suitable for measuring human perceptions and preferences for products and brands. Steps in the application of conjoint analysis and multidimensional scaling are explained and use of SPSS are illustrated with the help of case example.

UNIT 7 HYPOTHESIS TESTING

Learning Outcomes

After going through this unit, you should be able to:

- explain the concept and meaning of hypothesis
- appreciate the importance of the hypothesis in marketing research
- describe the steps involved in conducting a test of hypothesis
- apply the concept learned to a practical marketing decision -making situations.

Structure

- 7.1 Introduction
- 7.2 Concept and Meaning
- 7.3 Importance of Hypothesis in Marketing Research
- 7.4 Concept of Hypothesis Testing
- 7.5 Process of Hypothesis Testing
- 7.6 Use of SPSS for Hypothetical Case Analysis
- 7.7 Summary
- 7.8 Key Words
- 7.9 Self-Assessment Questions
- 7.10 Project Question
- 7.11 Further Readings

7.1 INTRODUCTION

A decision maker may have a probability of being misled while making a decision based on data. This can be the result of faulty assumptions or observations, erroneous or inadequate data, or the existence of unidentified variables. This is a risk since crucial strategic decisions based on incomplete information may result in resource waste, missed opportunities, and disastrous results. The true benefit of hypothesis testing in business is that it enables experts to validate their hypotheses before acting on them. As a result, a business can essentially confirm that its analysis is accurate before allocating resources to the implementation of a more comprehensive strategy. We will discuss the concept, meaning, importance, and process of hypothesis testing in this unit.

7.2 CONCEPT AND MEANING

In general, open-ended research questions provide a wider perspective on the problem. In order to focus data gathering on a few key aspects of the research questions, research questions are transformed into hypotheses.

This means that the more general idea of a research question is stated in terms of particular variables as a potential response to the research question. In addition, a hypothesis can be a statement that links two or more variables generated from one or more research questions. This relationship must also be investigated to determine whether it is true or untrue. So, we can say that hypotheses are expressed in terms of quantifiable variables. Let us use an example to explain.

Research question: Identify the factors influencing the performance of a salesperson.

Hypothesis: Marital status of salesperson has significant effect on sales performance

The hypothesis has limited the scope of data collection, as opposed to the research question, which would have required information on a number of additional variables, including gender, age, work experience, and others, in order to determine the profile of a salesperson. The aforementioned hypothesis can be expressed as follows to show a relationship between two or more research questions:

Hypothesis: Sales training programs would influence unmarried sales people to perform better.

Therefore, the hypothesis helped to focus the effort of data collection in order to test the relation between the variables. A hypothesis is actually the assumption regarding the anticipated direction of the research findings. For example, it can be expected that the research will focus on determining whether there is a direct relationship between sales training and salespeople's performance. The more effectively sales training imparts selling skills, the better the salesperson performs in turning prospective consumers into actual purchasers of goods and services.

Kerlinger (1986) defines a hypothesis as "... a conjectural statement of the relationship between two or more variables." According to Grinnell (1993), 'A hypotheses is written in such a way that it can be proven or disproven by valid and reliable data—it is in order to obtain these data that we perform our study'.

It is clear from the aforementioned definitions that a hypothesis is any assumption a researcher makes on the likely course of outcomes that could be attained after the research process has been completed. The hypotheses are always in declarative statement form, in contrast to the research problem, which typically takes the form of a question. The statements constructed in this way are subject to empirical investigation.

The characteristics of a good hypothesis

There are a few characteristics that the researcher must meet before developing any hypotheses.

- The formulation of a hypothesis must be straightforward, declarative, and explicit. It might not be possible to empirically test a broad hypothesis. Therefore, it could be wise to simplify the hypothesis and simply test one relationship between two variables at a time.

- In order to establish the statistical validity of the relationship, a hypothesis must be measurable and quantifiable.
- A hypothesis needs to be linked to the body of existing knowledge rather than the researcher's intuition or subjective judgement.
- Testing the statistical significance of the hypothesised relationship is a necessary step in the validation of the hypothesis. This indicates that it can be described in terms that can be measured. If something cannot be measured, it cannot be tested, and hence no conclusions can be made. So, it may be useless to frame such hypotheses that cannot be tested because there are no methods for doing so. This does not imply, however, that you should avoid formulating a hypothesis for which there is no means of verification. While conducting your investigation, you might come up with some fresh ways to confirm it.

Activity 1

Give an example from your experience where you think the usage of testing hypotheses can be beneficial.

7.3 IMPORTANCE OF HYPOTHESIS IN MARKETING RESEARCH

Researchers are frequently questioned about the significance of having hypotheses in marketing research. Why cannot you just go out and gather data, analyse it, and then formulate hypotheses? If you limit yourselves to evaluating only a few possibilities before conducting the research, won't it limit you? There is a distinction between fascinating data and significant data, as market researchers like to say. While some researchers think that having a hypothesis is necessary in order to perform a study, having a hypothesis is not necessary in situations when the researcher lacks the support of relevant literature or market expertise to turn a research question into a hypothesis. Hypotheses should be carefully defined because they focus the data collection, otherwise it may result in erroneous data being collected. Nevertheless, a hypothesis is crucial for clarifying the research problem. In research, if you don't create hypotheses to test, you run the risk of collecting a lot of fascinating data without any definitive direction for what to do next. So, when you develop hypotheses, you ultimately have the hypotheses verified or rejected, so you are in a better position than when you first began the investigation.

Here are some additional reasons why having hypotheses in marketing research is crucial. A hypothesis specifically accomplishes the following tasks:

1. **The development of a hypothesis maximises the value of your study.** You could believe that the sole purpose of conducting market

research is to obtain data. There may be other objectives that could be translated into hypotheses, though, within that basic demand for market knowledge. Why did you initially wish to conduct research? Is it because you're unsure about your next step or which participants to approach? Is it because you're contemplating trying something new but are unsure if you should? It's critical to be clear about your needs when conducting market research. The following may be some needs for marketing research:

- The research problem may be uncertainty about how to diversify into new markets. The concern with this hypothesis may be how successfully the brand's core message will translate. The outcome may be that packaging design significantly affects the entry into a new market. For example, shampoo in a sachet at a very lower price was introduced in the rural market in India.
- The problem may be wanting to change but not knowing how to proceed. Hypotheses assume that the product line extension would be accepted. The results may be that brand image has significantly increased confidence in the product line extension.
- Uncertainty about the market to target. The target market is price-sensitive, it may be hypothesized. However, it may not be true.
- Taking into account your competitors' market share and positioning. The hypothesis may be to determine how well that brand position resonates with different target audiences. The results may be that the brand is well positioned in certain target market and has an opportunity to significantly increase market share in selected region or territory.

Marketing research can help you more if you can think more clearly about your marketing needs and what you hope to gain from the study. The early phases are crucial because they set the framework for the remaining research, and if a problem is poorly stated at the outset, it can be difficult to correct it afterwards. Thus, the construction of a hypothesis gives a study direction. It explains which precise facets of a study subject to look into. Also, decides what elements of the research question would be taken into account in a study.

2. **To prevent delusion at the conclusion of the study.** Your research may occasionally provide more, unexpected information, which is always exciting. The testing of particular hypotheses does not eliminate that. However, by formulating hypotheses, you can be certain that the new knowledge resulting from your work won't make your original concerns and uncertainties any more complicated. By formulating hypotheses in advance and making sure to put them up for study to get clarification on them, it is simpler to ensure that you will at least have answers to the main concerns (true, false, or unclear). Thus, the development of a hypothesis improves objectivity in a study by giving it a focus.

3. **Having a structure for formulating and assessing survey questions.**

Using hypotheses also aids in keeping surveys focused and concise, which is an additional advantage. The data quality won't suffer if you're clear on the hypotheses you need to test with the survey before you "go in and get out" and locate the answers you need. Too frequently, people disagree passionately about survey question choices and wording. Without the framework of understanding what hypotheses, you are testing; it is difficult to identify what constitutes "excellent" questions. The purpose of survey questions is to test the hypothesis. If you approach it in that manner, choosing whether to include a certain question and how to frame it becomes lot simpler. So, the direction of data collection and the study's focus are established by a hypothesis, which guides you on what data to gather and what not to.

4. **Having a plan for assessing the outcomes.** If you begin your research with hypotheses, you will already be aware of the topic of the research presentation, which will be testing and analysing the hypotheses. You can be prepared to describe what the study revealed regarding the hypotheses and to propose subsequent steps in light of those findings. You might be able to improve a theory by using a hypothesis. It allows you to determine precisely what is true or false.

Thus, marketing research can be conducted for a variety of purposes, including lowering the risk associated with a choice, assisting in setting issues aside when deciding between possibilities, and assisting in resolving issues that are preventing you from moving forward with your marketing decisions. But regardless of your objective, developing hypotheses will help you get more out of your research.

7.4 CONCEPT OF HYPOTHESIS TESTING

We will now discuss about some ideas that will be helpful when we try to put up a process for testing hypotheses. To further comprehend the fundamental ideas of hypothesis testing, let's look at an example. Let's imagine that the XYZ corporation is considering updating the product's packaging. The item was formerly kept in a glass jar. The company wants to replace them with plastic containers. However, it wants to hear from its customers first to see if they prefer the new packaging or if they're more inclined to stick with the present one.

For this example, the hypotheses can be put forth in the following ways.

- There is no difference in consumers preferences for glass jar and plastic containers
- There is difference in consumers preferences for glass jar and plastic containers
- Glass jars are more preferred than plastic containers as packaging material
- Plastic containers are more preferred than glass jars as packaging material

The way the first hypothesis has been presented in the aforementioned sets of hypotheses suggests that there is no difference between glass jars and plastic containers for packaging. A null hypothesis is a type of hypothesis that states that there is no difference between two situations, groups, outcomes, or occurrence. It is typically represented by the letters H_0 . Null hypotheses are those that are put forth with the intention of being rejected.

The null hypothesis in respect of preferences for glass jar and plastic container is represented as:

$$H_0: \mu_g = \mu_p$$

where H_0 represents the null hypothesis, μ_g is the mean score for glass jar preference, and μ_p is the mean score for plastic container preference

Rejection of null hypotheses leads to the acceptance of alternative hypotheses. The rejection of null hypothesis indicates that the relationship between variables, in case of our example the difference between mean scores for glass jar and plastic container preferences have statistical significance and the acceptance of the null hypotheses indicates that these differences are due to chance. The alternative hypotheses can cover a whole range of values rather than a single point. The alternative hypotheses are denoted by H_1 .

The alternate hypothesis for the above example would statistically be set as:

$$H_a: \mu_g \neq \mu_p$$

where H_a represents the alternate hypothesis and μ_g and μ_p are the mean score for the preference for glass jar and plastic containers, respectively.

The alternate hypothesis can also be denoted as:

$$H_a: \mu_g < \mu_p$$

or

$$H_a: \mu_g > \mu_p$$

Type I and Type II Errors

It is likely that the finding may not be accurate since the results of testing a hypothesis are based on data from a sample, and variability among samples may never be completely eliminated until the sample is as big as the population. There are two different types of errors that can occur, as shown in Table 7.1 below, and for ease of reference, these errors have been categorised as Type I and Type II errors.

Table 7.1: Types of Errors in Testing of Hypothesis

		Decision based on Sample	
		Accept (Reject H_1)	Reject (Accept H_1)
States of Population	H_0 is True (H_1 is False)	Correct	Wrong (Type I Error)
	H_0 is False (H_1 is True)	Wrong (Type II Error)	Correct

A type I error occurs when we incorrectly conclude that H_0 is false when in fact H_0 is true. Similar to this, when we incorrectly accept H_0 while H_0 is untrue, the error is referred to as a type II error.

Think about the packaging decision-making dilemma for the marketer. Marketing manager would commit a type I error if he rejects the null hypothesis (H_0) and chooses a plastic container while consumers' mean preferences for glass jars are actually no better. A type II error might involve not selecting a plastic container when consumer preference is actually higher.

Both these errors are bad and should be reduced to a minimum. However, they cannot be completely eliminated without looking at the entire population, in which case the testing process would be useless. These two errors, however, cancel each other out for a given sample size. This suggests that if the testing process is made to make type I errors less likely to occur, type II errors are also more likely to occur and vice versa. A good balance between these two errors is the best that can be done.

The Significance Level

Since type I error is expected to be more significant than type II error in all tests of hypotheses, the likelihood of type I error must be carefully controlled. This is accomplished by defining a significance level at which a test will be performed. Thus, the probability of type I error is constrained by the significance level, and test procedures are created to obtain the lowest probability of type II error subject to the significance level.

The sign α (read as alpha) is typically used to denote the probability of a type I error, while the symbol β (read as beta) denotes the chance of a type II error.

Activity 2

For an example of your experience that you identified as part of Activity 1, state null and alternative hypotheses, or take some other situations.

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Activity 3

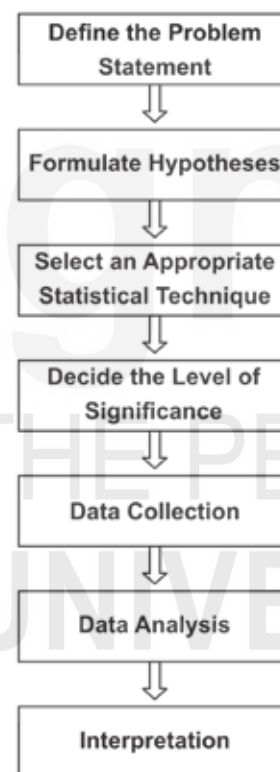
A new product was tested in Indore with a coupon for Rs 100 and in Bilaspur with a coupon for Rs 150. A sample of 100 people was contacted in each test city. A total of 40 percent of those contacted in Bilaspur had tried the new product, whereas only 30 percent of those contacted in Indore had tried it, a 10 percent difference. Prior to making the decision as to which coupon to use in the marketing program, a hypothesis was suggested.

- What should the null hypothesis be?
- What should the alternative hypothesis be?

7.5 PROCESS OF HYPOTHESIS TESTING

As we discussed in Unit 1, the scientific research process follows a step-by-step, logical, systematic, and rigorous method to find a solution to a problem. Similarly, the hypothesis testing process provides a useful, systematic approach for generating knowledge to solve basic and managerial problems. The eight-step process in the hypothesis testing method is exhibited in Figure 7.1.

Figure 7.1 Process of Hypothesis Testing



Identify a Problem

The first step in the process is to identify a broad problem area. The decision-maker might become interested in things like a decline in sales, customer switching, and similar things. For instance, a retailer of fruits and vegetables would inquire as to why sales of organic produce have fallen short of expectations.

Define the Problem Statement

The process of defining problem statement begins with the formulation of the research objectives. Research objectives are a general definition of the information needed for marketing research studies (e.g., understanding the effectiveness of advertisement). At the beginning of the research process, it might not be feasible to state the problem clearly because, at that point

often only the symptoms of the problem are visible. Gathering preliminary data on the variables that may be relevant to the problem (why are sales declining? What factors can be contributing to a falling market share?) will enable us to specify the problem statement and narrow the broad problem area. In order to define the problem situation, some exploratory research may be done prior to formulating a more detailed problem statement. By paying close attention to how the problem is defined, the researcher can establish appropriate study objectives, which in turn helps in formulating hypotheses and make it easier to collect useful and affordable data. Surveys of secondary data, experience surveys, or pilot studies are the methods that are most commonly used.

Develop Hypotheses

Variables are investigated in this step to determine their influence or contribution in describing the problem's cause and potential solutions. Null and alternative hypotheses are formulated. As we have discussed earlier in this unit that a null hypothesis is a statement of the status quo, one of no difference or no effect. The null hypothesis is tested. If the null hypothesis is not rejected, no changes will be made. The expectation of a difference or impact constitutes an alternate hypothesis. Changes will result if the alternative theory is accepted. As a result, the null hypothesis and the alternative hypothesis are the opposite. In marketing research, the null hypothesis is constructed in a way that if it is rejected, the desired conclusion will also be accepted.

Select an Appropriate Statistical Technique

we will not be able to test our hypotheses, until the variables in the theoretical framework are measured in some way. A suitable statistical method must be chosen in order to test the null hypothesis. The sample statistic's sampling distribution and the methodology used to calculate the test statistic should both be taken into account by the researcher. The test statistic quantifies the sample's degree of agreement with the null hypothesis. A well-known distribution, such as the normal, t, or chi-square distribution, is frequently followed by the test statistic.

Guidelines for selecting an appropriate test or statistical technique and their applications were discussed in detail in courses MMPC-005, and MMPC-0015. In Unit 11 of the course MMPC-005 (<https://egyankosh.ac.in/handle/123456789/79301>), you have been introduced to the general class of testing of hypotheses. These tests, popularly known as parametric tests, assume that parameters such as mean, standard deviation, etc. exist and are used to develop a test. For example, a t-test is based upon the comparison of means of two samples. The parametric tests are developed with an assumption that the form of the population distribution was known and that a test concerning a parameter of the distribution was to be made. Tests of hypotheses which deal with population parameters are called parametric tests. There are situations, particularly in psychological or in market research studies, where in the basic assumptions underlying the parametric tests are not valid or one does not have the knowledge of the distribution of the population parameter being tested. The tests which handle problems

of these types are known as nonparametric tests or distribution free tests. You have been introduced the nonparametric tests in Unit 9 of the course Research Methodology for Management Decisions (MMPC-015) (<https://egyankosh.ac.in/handle/123456789/90737>).

Decide the Level of Significance

There is a chance that we will come to the wrong conclusion whenever we infer something about a population based on a sample. As we have discussed, there are two types (Type 1 and Type 2) of errors that might occur. The level of significance affects how confidently a researcher rejects or accepts a null hypothesis. Prior to taking any samples, the level of significance indicated by the symbol α is decided. The probability of rejecting the null hypothesis when it is true is indicated by the level of significance α . Although the value of α varies from problem to problem, it is typically taken to be either 5% or 1%. A 5 % level of significance means that there are 5 chances out of 100 that a null hypothesis will get rejected when it should be accepted. This means that the researcher is 95 per cent confident that a right decision has been taken.

Data Collection

It is necessary to collect data with regard to each variable in the hypothesis after we have decided how to measure our variables. These data are then used as the basis for data analysis. We have discussed in detail the methods of data collection in Unit 5 of this course.

Data Analysis

To determine whether the developed hypotheses have been supported, the data collected are statistically analysed in the data analysis step. For instance, we would want to perform a correlational analysis to find the relationship between employee inactivity affects customer switching. We will discuss various tools for data analysis in Block 3 of the course. You can also refer to statistical analysis used to test hypotheses in Unit 11 of the course MMPC-005, (<https://egyankosh.ac.in/bitstream/123456789/79301/3/Unit-11.pdf>) and Unit 9 of the course MMPC-015 (<https://egyankosh.ac.in/handle/123456789/90737>).

Interpretation of Data

The results of the data analysis are interpreted in order to determine whether or not our hypotheses are validated. Depending on whether the result of the test statistic falls in the rejection or the acceptance zone, the hypothesis may be accepted or rejected. If the observed results have a probability of less than 5% and the hypothesis is being tested at a 5% level of significance, it would be rejected. In this situation, it is thought that there is a significant difference between the sample statistic and the population parameter that is hypothesised. Alternatively, if the hypothesis is accepted, the difference between the sample statistic and the hypothesised population parameter is not considered significant and can be attributed to chance.

For example, if data analysis revealed that employee responsiveness was negatively correlated with customer switching (let's assume the value of

static, in this case, the coefficient of correlation comes to be -0.43), then we can conclude that our employees need to be more responsively trained if customer retention is to be increased. By drawing these conclusions, we are able to provide the decision-makers with recommendations for how to solve the “customer switching” problem.

7.6 USE OF SPSS FOR HYPOTHETICAL CASE ANALYSIS

Suppose that the marketing manager of a firm has decided to launch a new ready-to eat snack. There are two minor variations of the product which have been developed. Both of these are basically similar but a bit different in their colour, flavour and crispness. Also, both of these are highly perishable and have a shelf life of about 48 hours.

The marketing manager decides to conduct a field trial of both the product variants to find out if one is liked better by the customers as compared to the other. He selects 20 shops which are similar in respect of their sizes, locations, clientele; etc. He introduces the first variant of the product (say X) in 10 of these shops and similarly, he introduces the second variant (say Y) in the other 10. Complete records are kept of the movement of these products for 15 days. The total sales in Kilogram (Kg) of X and Y in these shops in a period of 15 days is found to be as follows:

Table 7.2 Sales records in Kg of Product X and Y

Shop	1	2	3	4	5	6	7	8	9	10
Product X	13	16	11	8	12	14	12	12	9	8
Product Y	11	11	11	10	15	11	15	16	9	10

Both X and Y are priced equally. The marketing manager now wants to conclude whether there is any significant difference between X and Y. Using a significance level of 1%, what can he conclude?

Let us analyse the process of hypothesis testing for the given case.

Step 1 Identification of Broad Problem Area

How to test the sales (market) potential of two variants of new ready-to eat snack products launched in the market

Step 2 Defining Problem Statement

Which variant of product has higher sales potential? (Which variant of product will be more accepted by customers in the market?)

Step 3 Hypothesis Formulation

Null hypothesis (H_0): There is no difference in sales of X and Y variant of products

Alternative hypothesis (H_1): There is difference in sales of X and Y variant of products

Step 4 Select an Appropriate Statistical Test- The independent sample t test will be the most appropriate statistical technique in this situation because the data was gathered on a ratio (metric) scale and we need to compare sales of two different product variants.

Step 5 Decide the Level of Significance – Level of significance is decided as 1%

Step 6 Data Collection- For the purpose of collecting data, 20 shops that were comparable in terms of their sizes, locations, customers, etc., were chosen. In 10 of these shops, the first product variant (let's say X) was introduced, and in the same way, the second product variant (let's say Y) was introduced in the other 10. The total sales of both product variants (X and Y) in kilogrammes (Kg) made in these shops during the course of 15 days were compiled as shown in Table 7.2.

Step 7 Data Analysis

The collected data were entered into SPSS, and as shown in Figure 7.2, product X is coded as 1 and Y as 2.

Figure 7.2 Data entered in SPSS

	Product	Sales
1	1.00	13.00
2	1.00	16.00
3	1.00	11.00
4	1.00	8.00
5	1.00	12.00
6	1.00	14.00
7	1.00	12.00
8	1.00	12.00
9	1.00	9.00
10	1.00	8.00
11	2.00	11.00
12	2.00	11.00
13	2.00	11.00
14	2.00	10.00
15	2.00	15.00
16	2.00	11.00
17	2.00	15.00
18	2.00	16.00
19	2.00	9.00
20	2.00	10.00
21		

For analysis in SPSS, go to Analyze and select Independent Sample T Test as shown in Figure 7.3. After this, select grouping variables 1 for Product X and 2 for Product Y, and sales as a test variable (Figure 7.4).

Figure 7.3- An appropriate tool selected for analysis

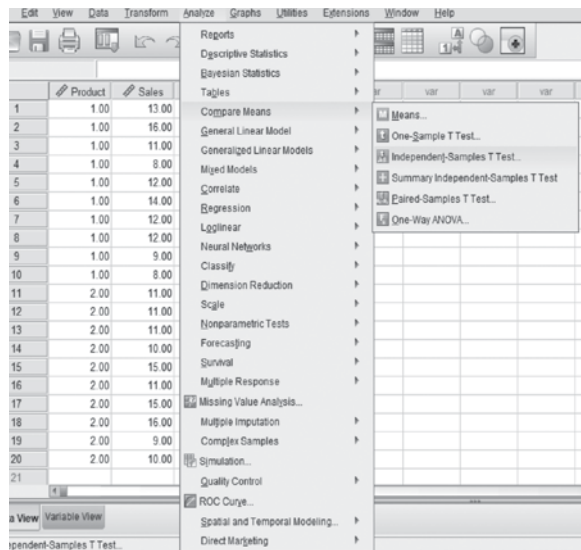
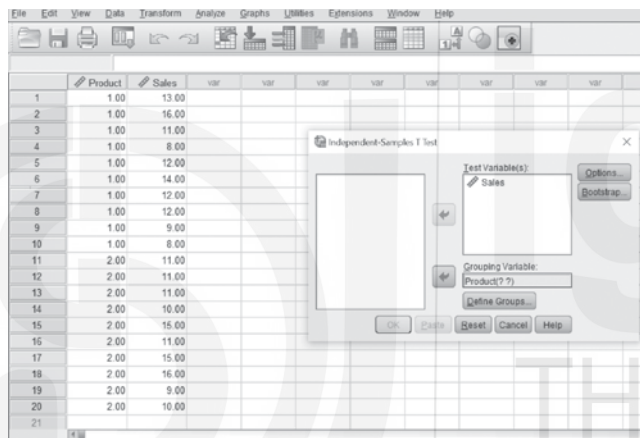


Figure 7.4: Groups were classified for two variants of product.



The mean and standard deviation of sales for products X and Y are given in Table 7.3.

Table 7.3 Mean and Standard Deviation for Two Variants of Product

	Product Variant	Sample Size	Mean	Std. Deviation	Std. Error Mean
Sales	X	10	11.500	2.59272	.81989
	Y	10	11.900	2.46982	.78102

The calculated t-value and level of significance (p value) are exhibited in Table 7.4.

Table 7.4 : Static and Probability (significant or p) Value

	t	df	Sig. (2-tailed)	95% Confidence Interval of the Difference
Sales	-.353	18	.728	-2.77898 1.97898
	-.353	17.958	.728	-2.77939 1.97939

Step 8 Interpretation- The results of the data analysis revealed that the sales of the two product variants that were launched into the market did not differ significantly. The calculated t value (-0.353) has a significant value of 0.728, which is higher than 0.01 (the 1% level of significance was earlier set as the decision criteria), as can be seen in Table 7.7. Depending on the results, the marketing manager may decide that both product variations can be offered.

7.7 SUMMARY

In this unit, we have discussed the concept, meaning, importance, and process of hypothesis testing. The true benefit of hypothesis testing in business is that it enables experts to validate their hypotheses before acting on them. We discussed the stages of the hypothesis-testing process. The null and alternative hypotheses are set up such that one of them, and only one of them, is always true. In the absence of strong evidence to the contrary, the decision-maker would be willing to accept the null hypothesis. Of the two errors that can occur in any hypothesis testing, type I error, i.e., incorrectly rejecting the null hypothesis, is considered to be more serious and thus is subject to explicit control. All tests are performed at a significance level, which defines the highest probability of a type I error. The unit was concluded by illustrating the use of SPSS for hypothesis testing.

7.8 KEY WORDS

Alternative Hypothesis: is contrary to null hypothesis, which establishes a statistically important relationship between two variables.

Hypothesis: is a statement that links two or more variables generated from one or more research questions.

Level of Significance: is the probability of wrong elimination of null hypothesis when in fact, it is true. It defines whether the null hypothesis is assumed to be accepted or rejected.

Null Hypothesis: is a type of hypothesis that states that there is no difference between two situations, groups, outcomes, or occurrence.

Type I Error: occurs when we incorrectly conclude that null hypothesis is false when in fact it is true.

Type II Error: occurs when we incorrectly accept null hypothesis while it is untrue.

7.9 SELF-ASSESSMENT QUESTIONS

1. What is meant by hypothesis? Explain the types and significance of hypothesis in marketing research
2. Discuss the stages of hypothesis testing process with an example.
3. Using an example, discuss the importance of level of significance in hypothesis testing.
4. Comment on “In hypothesis testing, it is implicitly assumed that type

I error is much more severe than type II error and so needs to be controlled.”

5. A food processor is concerned that the 16 gram can of sliced pineapple is being overfilled. The quality control department took a random sample of 50 cans and found that the arithmetic mean weight was 16.05 grams, with a sample standard deviation of 0.03 grams. At the 5 percent level of significance, can the hypothesis that the mean weight is equal to 16 grams be rejected?

7.10 PROJECT QUESTION

A questionnaire was designed to compare the level of students' familiarity with two types of products. For a random sample of 120 students, the mean familiarity level with CCTV cameras was found to be 3.355, and the sample standard deviation was 2.03. In an independent random sample of 100 students, the mean familiarity level for smart television was 9.5, and the sample standard deviation was 2.1. Assuming that the two population distributions are normal and have the same variance, test the null hypothesis that the population means are equal.

7.11 FURTHER READINGS

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<https://www.purebranding.com/the-importance-of-hypotheses-in-natural-products-market-research/>

UNIT 8 CORRELATION AND SIMPLE REGRESSION

Learning Outcomes

After going through this unit, you should be able to:

- compute the correlation coefficient between two variables
- test the significance of the correlation coefficient
- compute the rank correlation coefficient when rankings rather than actual values for variables are known
- explain the role of simple linear regression in establishing relationships between dependent and independent variables.
- appreciate some practical applications of correlation and simple regression analysis

Structure

- 8.1 Introduction
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8.1 INTRODUCTION

We frequently come into situations where data appears as pairs of figures related to two variables. A correlation problem considers the joint variation of two measurements and the degree of the association between two or more sets of variables, neither of which is restricted by the experimenter. Examples of correlation problems include the study of the relationship between IQ and the total percentage of marks a student scored in a

competitive examination, the relationship between blood pressure and metabolism, or the relationship between an individual's height and weight. In these examples, both variables are observed as they naturally occur, since neither variable is fixed at predetermined levels.

Regression, on the other hand, considers the frequency distributions of one variable (called the dependent variable) when another (the independent variable) is held fixed at each of several levels. A regression problem is used to explain the variations in one variable (called the dependent variable) by a set of independent variables. Examples of regression problems can be found in the study of the yields of crops grown with different amounts of fertiliser, the lifespan of specific animals subjected to various levels of radiation, the hardness of plastics that are heat-treated for different periods of time, and so on. In these problems, the variation in one measurement is studied for particular levels of the other variable selected by the experimenter.

Correlation and regression analysis are generally performed together. Understanding the concept and applications of correlation and regression statistical methods will help you better comprehend the nature, direction, and importance of the bivariate relationships between the variables used in marketing research studies, such as advertising expenditure and sales, sales volume at various pricing levels, and so on. This is a matter for discussion in this unit.

8.2 CHARACTERIZING RELATIONSHIPS BETWEEN VARIABLES

Let us explain the characteristics of relationship between variables with an example.

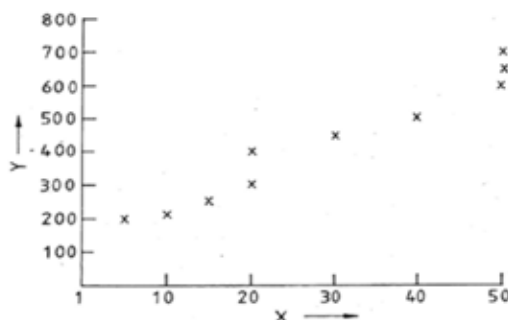
For example, we may have figures on advertisement expenditure (X) and sales (Y) for a firm for the last ten years, as shown in Table 8.1.

Table 8.1: Year-wise data on Advertisement Expenditure and Sales

Year	Advertisement Expenditure in Thousand Rs. (x)	Sales in thousand Rs. (y)
2013	5	200
2014	10	210
2015	15	250
2016	20	300
2017	20	400
2018	30	450
2019	40	500
2020	50	600
2021	50	650
2022	50	700

When this data is plotted on a graph, as in Figure 8.1, we obtain a scatter diagram.

Figure 8.1: Scatter Diagram



A scatter diagram gives two very useful types of information. First, we can observe patterns between variables that indicate whether the variables are related. Secondly, if the variables are related, we can get an idea of what kind of relationship (linear or non-linear) would describe the relationship. Correlation examines the first question of determining whether an association exists between the two variables and, if so, to what extent. Regression examines the second question of establishing an appropriate relationship between the variables.

8.3 THE CORRELATION COEFFICIENT

The correlation coefficient measures the degree of association between two variables X and Y. Karl Pearson's formula for correlation coefficient is given as:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y}$$

8.1

$$r = \frac{\sum XY}{n \sigma_x \sigma_y}$$

8.2

where

$$X = x - \bar{x}$$

$$Y = y - \bar{y}$$

r	→	Correlation Coefficient
σ_x	→	standard deviation of dataset X
σ_y	→	standard deviation of dataset Y
$\bar{x}$	→	mean of dataset X
$\bar{y}$	→	mean of dataset Y

Let us compute the correlation coefficient between two variables for a company's advertisement expenditure (x) and sales (y) over a 10-year period, as shown in Table 8.1. The calculations are completed in the manner depicted in Table 8.2.

Table 8.2: Calculation of Correlation Coefficient

Year	Advertisement Expenditure in Thousand Rs. (x)	Sales in thousand Rs. (y)	X = x - $\bar{x}$	Y = y - $\bar{y}$	X ²	Y ²	XY
2013	5	200	21	274	441	75076	5754
2014	10	210	21	224	441	50176	4704
2015	15	250	21	174	441	30276	3654
2016	20	300	11	74	121	5476	814
2017	20	400	1	24	1	576	24
2018	30	450	-9	-26	81	676	234
2019	40	500	-9	-126	81	15876	1134
2020	50	600	-14	-176	196	30976	2464
2021	50	650	-19	-216	361	46656	4104
2022	50	700	-24	-226	576	51076	5424
Total	290	4260	0	0	2740	306840	28310

$$\bar{x} = \frac{290}{10} = 29$$

$$\bar{y} = \frac{4260}{10} = 426$$

$$r = \frac{\sum xy}{\sqrt{\sum x^2} \sqrt{\sum y^2}} = \frac{28310}{\sqrt{2740} \times \sqrt{306840}} = 0.976$$

This value of $r (= 0.976)$ indicates a high degree of association between the variables X and Y. For this particular problem, it indicates that an increase in advertisement expenditure is likely to yield higher sales.

You should note that r is a dimensionless number whose numerical value lies between +1 and -1. Positive values of r indicate a positive (or direct) correlation between the two variables X and Y, i.e., as X increases, Y will also increase, or as X decreases, Y will also decrease.

Negative values of r indicate negative (or inverse) correlation, thereby meaning that an increase in one variable results in a decrease in the value of the other variable.

A zero correlation means that there is no association between the two variables. Figure 8.2 shows a number of scatter plots with corresponding values for the correlation coefficient r .

Figure 8.2: Different Types of Association Between Variables

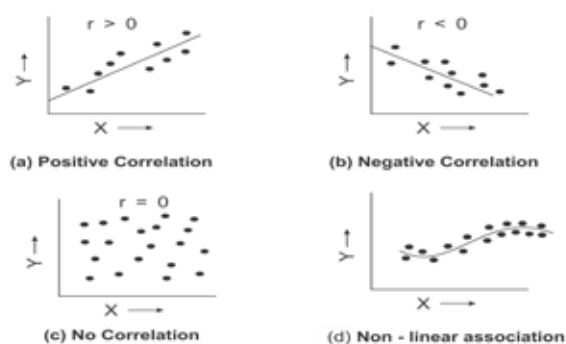


Figure 8.3 shows you the various strengths of the correlation coefficients.

Figure 8.3: Strength of Positive and Negative Correlation Coefficient

Perfect	+1	−1
Strong	+0.9	−0.9
	+0.8	−0.8
	+0.7	−0.7
Moderate	+0.6	−0.6
	+0.5	−0.5
	+0.4	−0.4
Weak	+0.3	−0.3
	+0.2	−0.2
	+0.1	−0.1
Zero	0	

Source: Dancey and Reidy, Statistics without Maths for Psychology, 7th ed.

Activity 1

Suggest five pairs of variables related to marketing that you expect to be positively correlated.

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8.4 TESTING FOR THE SIGNIFICANCE OF THE CORRELATION COEFFICIENT

Once the correlation coefficient has been calculated from sample data, one is normally interested in asking the question: Is there an association between the variables? Or with what confidence can we make a statement about the association between the variables?

Such questions are best answered statistically by testing the null hypothesis that population correlation coefficient equals zero vs. the alternative hypothesis that it does not, by using the *t*-statistic. The use of this procedure is now illustrated.

The null and alternative hypothesis to be tested is stated below:

$$H_0 : \rho = 0 \quad H_1 : \rho \neq 0$$

Test statistic is given by,

$$t_{n-2} = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \quad (15.3)$$

where, ρ = Population correlation coefficient between the variables X and Y

r = Sample correlation coefficient between the variables X and Y

$n - 2$ = The degrees of freedom

8.3

Given the values of r and n , the value of the test statistic t could be computed. The null hypothesis that X and Y do not correlate is rejected if calculated

$|t|$ is greater than tabulated $|t|$ with $n - 2$ degrees of freedom for a specified level of significance.

To illustrate further, consider the following hypothetical case: the correlation coefficient between respondents' preference for packaging and income was calculated, and it was discovered that the correlation coefficient between preference and income is equal to .4597. Now, to test the significance of the population correlation coefficient, we may use the t test described below.

H_0 (Null Hypothesis): There is no correlation between preference and income ($\rho = 0$)

H_1 (Alternative Hypothesis): There is correlation between preference and income ($\rho \neq 0$).

The significance of the correlation coefficient was tested by using the formula 8.3, and the computations are as given below.

$$t = \frac{0.46\sqrt{58}}{\sqrt{1-.2116}} = \frac{0.46 \times 7.616}{0.8879} = \frac{3.50336}{0.8879} = 3.9459$$

For a given level of significance (say, .05), the table value of t is 1.96. This value being less than the computed value of t (3.9459), we reject the null hypothesis that there is no correlation between the preference and actual income.

8.5 RANK CORRELATION

Data frequently comes in the form of rankings for various factors. Rankings are frequently used in fields like food testing, competitive activities (such as games, fashion shows, or beauty contests), and attitudinal surveys. In such cases, the main goal of determining a correlation coefficient is to assess the degree of agreement between the two sets of rankings. The coefficient that is determined from these ranks is known as Spearman's rank correlation coefficient, r_s .

This is determined by the formula below

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2-1)}$$

Here n is the number of pairs of observations and d_i is the difference in ranks for the i^{th} observation set.

Let us illustrate with a hypothetical case example. Suppose, advertisement cost and sales data of a company for 10 years are given in Table 8.3.

Table 8.3: Year-wise data on Advertisement Expenditure and Sales

Year	Advertisement Expenditure in Thousand Rs. (x)	Sales in thousand Rs. (y)
2013	38	48
2014	64	54
2015	61	59

2016	89	87
2017	81	63
2018	74	69
2019	24	61
2020	97	92
2021	35	52
2022	77	85

Let us compute the Spearman's correlation coefficient between two variables for a company's advertisement expenditure (x) and sales (y) over a 10-year period, as shown in Table 3. The calculations are completed in the manner depicted in Table 8.4.

Table 8.4: Calculation of Spearman's Correlation Coefficient

Year	Advertisement Expenditure in Thousand Rs. (x)	Sales in thousand Rs. (y)	R ₁	R ₂	Difference d=R ₁ -R ₂	d ²
2013	38	48	8	10	2	4
2014	64	54	6	8	2	4
2015	61	59	7	7	0	0
2016	89	87	2	2	0	0
2017	81	63	3	5	2	4
2018	74	69	5	4	-1	1
2019	24	61	10	6	-4	16
2020	97	92	1	1	0	0
2021	35	52	9	9	0	0
2022	77	85	4	3	-1	1
Total					$\sum d^2$	30

Using the formula (8.4) we obtain

$$r = 1 - \frac{6 \cdot 30}{10(10^2 - 1)} = 0.82$$

The Spearman's rank correlation coefficient between the advertisement cost and sales is 0.82. Thus, we can infer that there is a strong correlation between advertisement expenditure and sales.

8.6 PRACTICAL APPLICATIONS OF CORRELATION

The purpose of correlation analysis is to assess how closely two sets of quantitative data are related. For instance, what relationship exists between the sales of products A and B? Or how does the advertising expense relate to other promotional expenses? Practically every dataset with two or more variables can be used to apply correlation analysis. A different question is whether it makes sense to investigate a variable's association with another variable and how to interpret the results. For instance, there is evidence to suggest that spending more money on clothing of greater quality is correlated with income levels. However, one must be cautious of fictitious or

absurd correlations that might be discovered between completely unrelated variables by pure accident. It is the duty of the researcher to ensure proper application of correlation analysis. Therefore, the main goal of correlation is to identify a relationship between any two random variables.

Correlation and regression are generally performed together. In many applications, correlation is usually followed by regression analysis. Correlation analysis is used as a starting point for selecting useful independent variables for regression analysis. For instance, a construction company could identify factors like population, construction employment, and building permits issued last year that it feels would affect its sales for the current year. These and other factors that may be identified could be checked for mutual correlation by computing the correlation coefficient of each pair of variables from the given historical data (this kind of analysis is easily done by using an appropriate computer routine). Only variables having a high correlation with the yearly sales could be singled out for inclusion in a regression model.

Correlation is also used in factor analysis, where attempts are made to resolve a large set of measured variables in terms of relatively few new categories, known as factors. The results could be useful in the following three ways: 1) to reveal the underlying or latent factors that determine the relationship between the observed data, 2) to make evident relationships between data that had been obscured before such analysis, and 3) to provide a classification scheme when data scored on various rating scales have to be grouped together. Another major application of correlation is in forecasting with the help of time series models.

8.7 SIMPLE LINEAR REGRESSION

One of the limitations of correlation analysis is that it only measures the degree of association between two variables and does not deal with cause-and-effect relationships between variables. Regression analysis solves this problem by assuming a causal relationship between variables (regression analysis only assumes and does not establish a cause-and-effect relationship). Assuming Y is a function of X , we are treating X as a causal variable and Y as a dependent variable. This means it is the variable X that influences the variable Y , not the other way around. We have taken a very simplistic case of a situation where the dependent variable Y is explained only by one independent (causal variable) X . In a real-life situation, there may be a host of variables influencing the dependent variable Y . The discussion of such a case comes under the multiple regression model and will be taken up in Unit No. 9 of this course. Here our concern is with a simple linear regression model where there is only one dependent and one independent variable. Assuming a linear relationship between the dependent and the independent variable, the relationship between Y and X may be written as

$$Y = b_0 + b_1X + u$$

Where b_0 = intercept term

b_1 = slope term

u = random error

The estimation of b_0 and b_1 is carried out by using ordinary least squares method of estimation (for details on this method, please consult Unit no. 15 of MMPC -005 Course.) The ordinary least squares method (OLS) 'aims at minimizing the error sum of squares to obtain the estimated value of b_0 and b_1 . The estimated values are obtained using the following formulae:

$$\hat{Y}_i = b_0 + b_1 X_i$$

where $\hat{Y}_i$ is called the predicted value of Y_i .

The values of the least-squares estimates b_0 and b_1 are calculated using the formulas

$$b_1 = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}$$

and

$$b_0 = \bar{y} - b_1 \bar{x}$$

where

$$\bar{y} = \frac{\sum y_i}{n} \quad \bar{x} = \frac{\sum x_i}{n}$$

Let us illustrate with a hypothetical case example of advertisement expenditure effect on sales. The calculations are completed in the manner depicted in Table 8.5.

Table 8.5: Calculation of Regression Coefficient

Year	Advertisement Expenditure in Thousand Rs. (X)	Sales in thousand Rs. (Y)	X *Y	X ²
2013	38	148	5624	1444
2014	64	200	12800	4096
2015	61	120	7320	3721
2016	89	150	13350	7921
2017	81	160	12960	6561
2018	74	125	9250	5476
2019	24	61	1464	576
2020	97	210	20370	9409
2021	35	51	1785	1225
2022	77	85	6545	5929
SUM	640	1310	91468	46358

$$\bar{X} = \frac{640}{10} = 64$$

$$\bar{Y} = \frac{1310}{10} = 131$$

$$\begin{aligned} b_1 &= [10(91468) - (640)(1310)] / [10(46358) - (640)^2] \\ &= [914680 - 838400] / [463580 - 409600] \end{aligned}$$

$$76280 / 53980 = 1.41$$

$$\begin{aligned} b_0 &= 131 - 1.41 (64) \\ &= 131 - 90.24 = 39.76 = 40 \text{ (APPROX.)} \\ &= 40 + 1.41 X \end{aligned}$$

The above estimated regression equation indicates that as the advertisement expenditure increases by one rupee the sales revenue goes up by 1.41 points. Further the positive sign of the slope coefficient indicates that the relationship is positive.

Suppose advertising expenses is Rs 81000, then estimated sales will be

$$\begin{aligned} &= 40 + 1.41 (81000) \\ &= 40 + 114210 = 114250 \end{aligned}$$

Thus, if advertising expenditures of Rs 81000 are planned, our estimated sales will be of Rs 114250.

The goodness of fit of the regression equation is estimated by r^2 , the coefficient of determination. The value of r^2 is always between 0 and 1, with higher values indicating better goodness of fit. The value of r^2 also represents the explanatory power of the regression model. If the value of r^2 is 0.84, it means that the independent variable explains 84% of the variations in the dependent variable. We have very briefly touched upon the simple regression model in this section. For details on regression analysis, please refer to Unit No. 15 Regression of the Course MMPC-005 Quantitative Analysis for Managerial Application <https://egyankosh.ac.in/handle/123456789/79307>, and for correlation, Unit 14 (<https://egyankosh.ac.in/handle/123456789/79306>)

8.8 PRACTICAL APPLICATIONS OF REGRESSION

A number of research objectives could be served by regression analysis.

- It might be employed to investigate the existence of a overall relationship between the dependent variable and a number of independent variables.
- It can be used to gauge how important each independent variable is in explaining the dependent variable.
- Regression analysis is also used to anticipate the values of the dependent variable; that is, one can forecast the values of the dependent variable by knowing the values of the independent variables. For instance, family income and family size might be used as independent variables in a regression to predict the amount of money households spend on food products.
- Regression analysis can be used to determine how the location, parking facility, prices, mode of payment, credit policy, product quality, and service response time of a retail store affect the amount that a customer spends there.

- Another example, one may involve estimating the sales volume of a microwave oven utilising a number of independent factors, such as the size of the sales force, the size of the advertising budget, and consumer perceptions of the company's goods. Similar to this, a small business owner who wants to export his goods may look into export opportunities based on the strengths of his workforce, sales revenue from the home market, and years of experience operating on the domestic market.

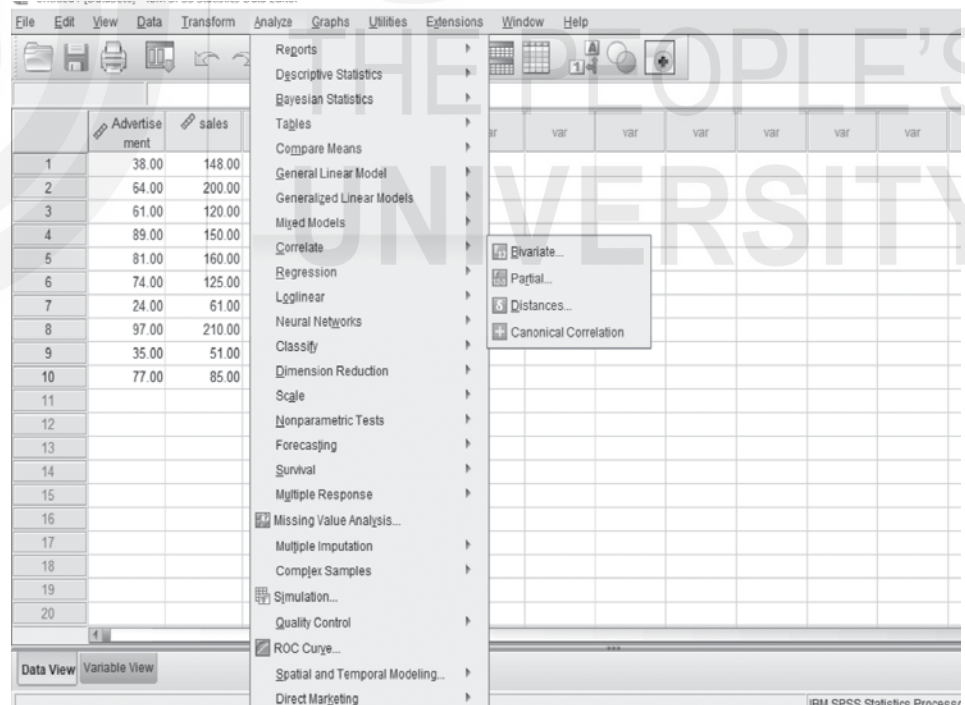
8.9 USE OF SPSS FOR CORRELATION AND THE SIMPLE LINEAR REGRESSION ANALYSIS

Let us explain the use of SPSS for correlation and simple regression analysis by taking our hypothetical case example of advertisement expenditure and sales revenue discussed in Section 8.6 for simple linear regression. Advertisement expenditure and sales revenue data presented in Table 8.5 (columns 2 and 3) was entered into the SPSS file along with the variable labels.

8.9.1 Use of SPSS For Correlation Analysis

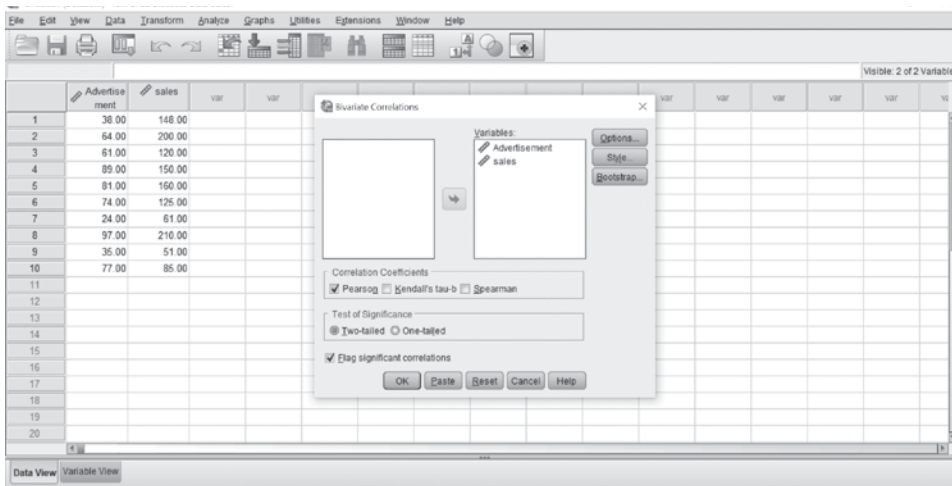
To get the output for a correlation problem, go to Analyze and select Correlate, followed by Bivariate as shown in Figure 8.4.

Figure 8.4: An appropriate tool selected for analysis



After this, select the variables advertisement expenditure and sales by clicking on the right arrow to transfer them from the variable list on the left. Then select Pearson under the heading Correlation coefficients, and select 2-tailed under the heading Tests of Significance (Figure 8.5).

Figure 8.5: Variables transferred for the analysis



By clicking OK, we will get the matrix of the pair-wise Pearson correlations between advertisement and sales, along with the two-tailed significance as shown in Table 8.6. The results indicate that the correlation between advertisement and sales is 0.643. The p value for the correlation coefficient is 0.045, which is less than 0.05, the assumed level of significance. This implies that the correlation coefficient between the advertisement and sales is positive, moderate and statistically significant.

Table 8.6: Correlation Matrix

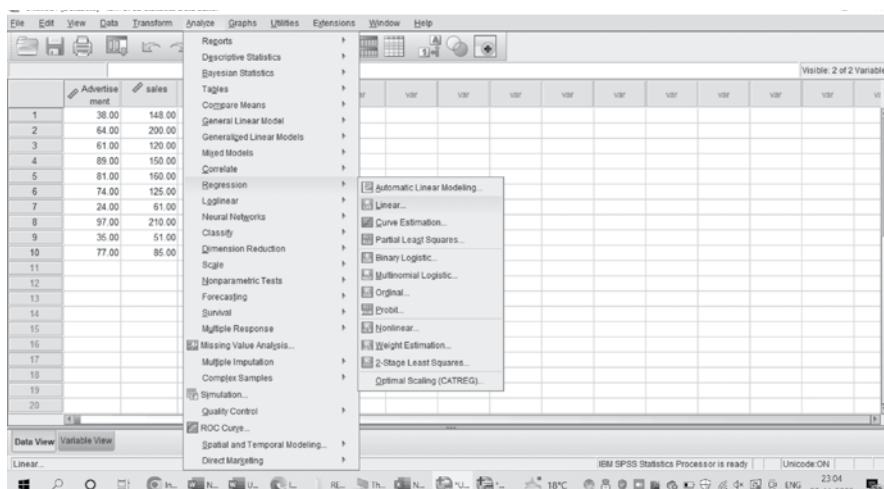
		Advertisement	sales
Advertisement	Pearson Correlation	1	.643*
	Sig. (2-tailed)		.045
	N	10	10
sales	Pearson Correlation	.643*	1
	Sig. (2-tailed)	.045	
	N	10	10

*. Correlation is significant at the 0.05 level (2-tailed).

8.9.2 Use of SPSS For the Simple Linear Regression Analysis

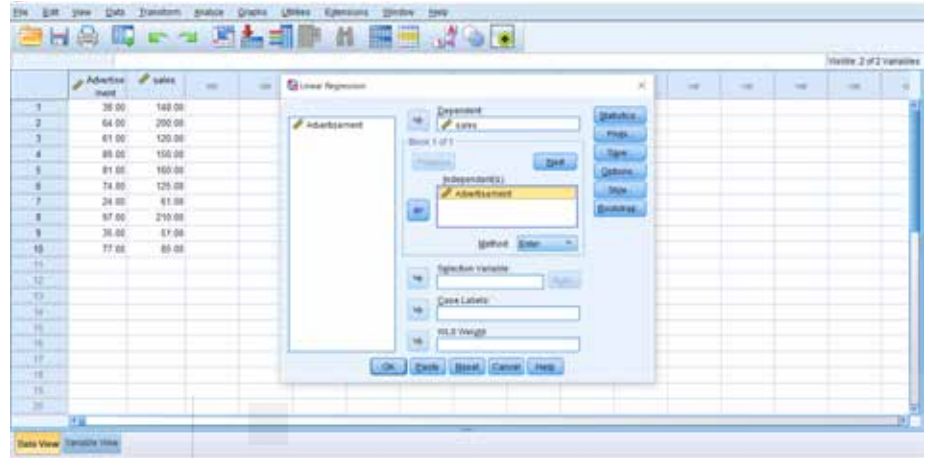
To get the output for a correlation problem, go to Analyze and select Regression, followed by Linear as shown in Figure 8.6.

Figure 8.6: An appropriate tool selected for analysis



After this, select the variable advertisement expenditure by clicking on the right arrow to transfer it from the variable list on the left to right side box under Independent variables or Independents. Similarly, select the variable sales by clicking on the right arrow to transfer it from the variable list on the left to right side box under Dependent variables or Dependents (Figure 8.7).

Figure 8.7: Variables transferred for the analysis



By clicking OK, we will get the output for regress as shown in Tables 8.7 to 8.9.

Table 8.7: Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.643 ^a	.414	.340	43.71320
a. Predictors: (Constant), Advertisement				

Table 8.8: ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	10779.249	1	10779.249	5.641	.045 ^b
	Residual	15286.751	8	1910.844		
	Total	26066.000	9			
a. Dependent Variable: sales						
b. Predictors: (Constant), Advertisement						

Table 8.9: Coefficients ^a						
Model		Unstandardized Coef- ficients		Standard- ized Coef- ficients	t	Sig.
		Std. Error	Beta			
1	(Constant)	40.561	40.510		1.001	.346
	Advertise- ment	1.413	.595	.643	2.375	.045
a. Dependent Variable: sales						

By using the results presented in Table 8.9, we can write the estimated regression equation as:

$$\text{Sales} = 40.561 + 1.413 \text{ Advertisement}$$

We note that the intercept and the slope terms are 40.561 and 1.413, respectively, which

is exactly the same as when the problem was worked out manually (as 40, and 1.41, respectively).

The value of the t statistic corresponding to the coefficient of advertisement is 2.375, which is significant at the value of 0.045 (Table 8.9).

The value of $r^2 = 0.414$ as presented in Table 8.7. The F statistic used to test the significance of r^2 as given in Table 8.8 equals 5.641, which is significant as indicated by the p value (sig.) as given in the last column.

8.10 SUMMARY

In this unit, the concepts of correlation and simple linear regression between two variables have been discussed. A scatter plot of the variables may suggest that the two variables are related, but the value of the Pearson correlation coefficient r quantifies this association. The correlation coefficient r may have values between -1 and 1. The sign indicates whether the association is direct (+ve) or inverse (-ve). A numerical value of r equal to unity indicates perfect association, while a value of zero indicates no association. Tests for the significance of the correlation coefficient have been described. Spearman's rank correlation for data with ranks is outlined. However, the correlation coefficient does not deal with cause-and-effect relationships, and to overcome this, regression analysis is used. We have discussed the estimation and interpretation of regression results. The use of SPSS and applications of correlation and regression have also been highly highlighted.

8.11 KEY WORDS

Correlation: Degree of association between two variables.

Correlation Coefficient: A number lying between -1 (Perfect negative correlation) and + 1 (perfect positive correlation) to quantify the association between two variables.

Linear regression: Fitting of any chosen mathematical model, linear in unknown parameters, to a given data.

Regression: Relating of a dependent (or response) variable to a number independent variables, based on a given set of data.

Scatter Diagram: An ungrouped plot of two variables, on the X and Y axes.

8.12 SELF ASSESSMENT QUESTIONS

1. What do you understand by the term “correlation”? Explain how the study of the correlation coefficient helps in understanding the relationship between variables under consideration.

2. Use an example to differentiate between correlation and regression. How do the concepts work together? Explain
3. The producer of a herbal soap brand was curious to know the correlation between soap sales and the shelf space given to that brand by various retailers. The following data was gathered from 10 stores:

Retail Store	Shelf space (Square feet)	Sales (Thousand rupees)
1	6	30
2	4	20
3	7	35
4	6	33
5	5	22
6	4	21
7	3	17
8	6	25
9	5	24
10	6	31

- (a) Assess the correlation between the sales and the shelf space and test it at a 5 percent level of significance.
- (b) Is it possible to forecast sales using shelf space? Explain
4. Ten salesmen are ranked for efficiency and the length of service as below:

Salesmen	Efficiency	Length of service
1	1	2
2	2	1
3	3	5
4	5	3
5	7	9
6	8	7
7	9	6
8	10	4
9	11	11
10	12	10

Find the value of Spearman's rank correlation coefficient.

5. Data about a firm's advertising expenditure and the corresponding sales figure over a period of five months are given in the following table.

Month	Advertising Expenditure (Thousand Rupees)	Sales (Thousand Rupees)
1	8	12
2	9	13
3	8	14
4	9	15
5	10	16

Estimate the linear regression of sales on the advertising expenditure and interpret the results.

8.13 PROJECT QUESTION

Suppose you are looking to buy an apartment in your city. You contacted 25 real estate agents in your city and gathered details regarding the neighbourhood, price of a unit, build-up area, carpet area, number of rooms, distance to the closest market, hospitals, and school for your child. Based on the information collected, develop a suitable regression model and test it by changing the value of the variable that most influences your decision to buy a residential apartment.

8.14 FURTHER READINGS

Beri, G. C. “Marketing Research - Text and Cases” Tata McGraw-Hill Publishing Co. Ltd.

Kinney, Thomas C. and James R. Taylor, “Marketing Research - An Applied Approach” McGraw-Hill International Editions (3rd Edition).

Luck, David J. and Ronald S. Rubin, “Marketing Research “ Prentice-Hall of India Pvt. Ltd.

Malhotra, N.K., Marketing Research: An Applied Orientation, seventh edition, Pearson Education.

UNIT 9 MULTIPLE REGRESSION ANALYSIS

Learning Outcomes

After going through this unit, you should be able to:

- appreciate the application of the multiple regression analysis in the context of marketing.
- explain how multiple regression analysis is computed.
- interpret the findings of multiple regression analysis.
- assess the considerations for multiple regression analysis.
- apply the concept learned in real-world marketing decisions.

Structure

- 9.1 Introduction
- 9.2 Multiple Regression Analysis
- 9.3 Computational Aspect of Multiple Regression
- 9.4 Use of SPSS for Multiple Regression Analysis
- 9.5 Considerations for Multiple Regression Analysis
- 9.6 Summary
- 9.7 Key Words
- 9.8 Self-Assessment Questions
- 9.9 Project Question
- 9.10 Further Readings

9.1 INTRODUCTION

The main objective of regression analysis is to explain the variation in one variable (called the dependent variable) based on the variation in one or more other variables (called the independent variables). The number of independent variables in a regression analysis could be one or more. In the case of one independent variable, we classify it as a simple regression, whereas in the case of more than one independent variable, it is called a multiple regression analysis. We have discussed simple linear regression with one independent variable in unit 8. In this unit, we will discuss multiple regression analysis.

9.2 MULTIPLE REGRESSION ANALYSIS

In general, a multivariate technique of analysis is required rather than a univariate or bivariate technique when there are two or more variables involved in, or simultaneously affecting, another variable. For instance, using a multivariate method known as multiple regression, we can assess

the simultaneous impact of pricing, sales promotion, and advertising on sales.

Regression analysis finds out the degree or relationship between a dependent variable and a set of independent variables by fitting a statistical equation through the method of least square. Whenever we are interested in the combined influence of several independent variables upon a dependent variable our study is that of multiple regression. For example, demand may be influenced not only by price but also by growth in industrial production, the extent of import prices of other goods, consumer's income, taste, and preferences etc. Marketing researchers could use regression for explaining per cent variation in dependent variables caused by a number of independent variables and also problems involving prediction or forecasting.

Regression analysis is probably the most widely applied technique among the analytical models of association used in marketing research. Regression analysis attempts to study the relationship between a dependent variable and a set of independent variables (one or more). For example, in demand analysis, demand is inversely related to the price of normal commodities. We may write $D = A - BP$, where D is the demand, which is the dependent variable, and P is the unit price of the commodity, an independent variable. This is an example of a simple linear regression equation. The multiple linear regression model is the prototype of the single criterion/multiple predictor association models, where we would like to study the combined influence of several independent variables upon one dependent variable. If P is the consumer price index and Q is the industrial production index in the preceding example, we may be able to study demand as a function of two independent variables, P and Q, and write

$D = A - BP + CQ$ as a multiple linear regression model.

9.2.1 Objectives

The objectives of the marketing researchers in using regression analysis are:

- 1) To study a general underlying pattern connecting the dependent variable and independent variables by establishing a functional relationship between the two. In this equation, the degree of relationship is derived, which is a matter of interest to the researcher in his study.
- 2) To use the well-established regression equation for problems involving prediction and forecasting.
- 3) To study how much of the variation in the dependent variable is explained by the set of independent variables. This would enable him to remove certain unwanted variables from the system. For example, if 95% of the variation in demand in a study could be explained by price and consumer rating index, the researcher may drop other factors like industrial production, the extent of imports, substitution effect, etc., which may contribute to only 5% of the variation in demand, provided all the causal variables are linearly independent.

9.2.2 Assumptions of the model

- 1) The relationship between Y and X_1 is linear.
- 2) Y is a random variable which follows a normal distribution from which sample values are drawn independently.
- 3) X_1 is fixed and is non-stochastic (non-random).
- 4) The means of all these normal distributions of Y as conditioned by X_1 lie on a straight line with slope β_1 .
- 5) ϵ is the error term $\cap$ IND (0, σ^2) and independent of X_1 .

9.3 COMPUTATIONAL ASPECT OF MULTIPLE REGRESSION

In the multiple regression model, there are at least two independent variables. The general form of the multiple regression model can be expressed as

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon \quad 9.1$$

To understand the concept clearly, let us study a case of two independent variables. Estimated regression line based on sampling is written as

$$Y = b_0 + b_1 X_1 + b_2 X_2 + U \quad 9.2$$

In the model 9.2, there are three parameters b_0 , b_1 , and b_2 that are to be estimated. b_0 , b_1 , and b_2 are estimates of β_0 and β_1 , and β_2 obtained through the method of least square by minimising the error sum of squares.

We state the normal equations without going into any derivations. The normal equations are:

$$\sum Y = n\hat{b}_0 + b_1 \sum X_1 + \hat{b}_2 \sum X_2 \quad 9.3$$

$$\sum X_1 Y = \hat{b}_0 \sum X_1 + \hat{b}_1 \sum X_1^2 + \hat{b}_2 \sum X_1 X_2 \quad 9.4$$

$$\sum X_2 Y = \hat{b}_0 \sum X_2 + \hat{b}_1 \sum X_1 X_2 + \hat{b}_2 \sum X_2^2 \quad 9.5$$

By solving these three simultaneous equations we will get the values of $\hat{b}_0, \hat{b}_1$ and $\hat{b}_2$

$$\hat{b}_0 = \bar{Y} - \hat{b}_1 \bar{X}_1 - \hat{b}_2 \bar{X}_2 \quad 9.6$$

$$\hat{b}_1 = \frac{(\sum X_1 Y) (\sum X_2^2) - (\sum X_2 Y) (\sum X_1 X_2)}{(\sum X_1^2) (\sum X_2^2) - (\sum X_1 X_2)^2} \quad 9.7$$

$$\hat{b}_2 = \frac{(\sum X_2 Y) (\sum X_1^2) - (\sum X_1 Y) (\sum X_1 X_2)}{(\sum X_1^2) (\sum X_2^2) - (\sum X_1 X_2)^2} \quad 9.8$$

$$x_1 = X_1 - \bar{X}_1$$

$$x_2 = X_2 - \bar{X}_2$$

$$y = Y - \bar{Y}$$

After providing an overview of the regression model's underlying assumptions, let's take a numerical example to help make the concepts clear. For comprehending the principles, it should go without saying that data analysis and result interpretation are essential.

Suppose that a marketing researcher is interested in customers' preferences (Y) for a nutritious diet of ready-to-eat cereal based on the amount of protein (X_1) and vitamin D (X_2) per standard serving. On a preference rating scale ranging from 1, "dislike extremely," to 9, "like extremely well," the researcher collected the data from 10 ready-to-eat cereal customers. The data is presented in Table 9.1.

Table 9.1: Customers preference for ready to eat cereals

Respondent	Preference (Y)	Protein (X1)	Vitamin D (X2)
1	3	4	2
2	7	9	7
3	2	3	1
4	1	1	2
5	6	3	3
6	2	4	4
7	8	7	9
8	3	3	2
9	9	8	7
10	2	1	3
	43	43	40

Table 9.2: Calculation customers preference for ready to eat cereals

Respondent	Preference (Y)	Protein (X_1)	Vitamin D (X_2)	x_1	x_2	x_1^2	x_2^2	x_1x_2	y	x_1y	x_2y
1	3	4	2	-0.3	-2	0.09	4	0.6	-1.3	0.39	2.6
2	7	9	7	4.7	3	22.09	9	14.1	2.7	12.69	8.1
3	2	3	1	-1.3	-3	1.69	9	3.9	-2.3	2.99	6.9
4	1	1	2	-3.3	-2	10.89	4	6.6	-3.3	10.89	6.6
5	6	3	3	-1.3	-1	1.69	1	1.3	1.7	-2.21	-1.7
6	2	4	4	-0.3	0	0.09	0	0	-2.3	0.69	0
7	8	7	9	2.7	5	7.29	25	13.5	3.7	9.99	18.5
8	3	3	2	-1.3	-2	1.69	4	2.6	-1.3	1.69	2.6
9	9	8	7	3.7	3	13.69	9	11.1	4.7	17.39	14.1
10	2	1	3	-3.3	-1	10.89	1	3.3	-2.3	7.59	2.3
SUM	43	43	40			70.1	66	57		62.1	60

$$\bar{X}_1 = \frac{43}{10} = 4.3$$

$$\bar{Y} = \frac{430}{10} = 4.3$$

$$\bar{X}_2 = \frac{40}{10} = 4.0$$

Substituting the values in equation 9.7, we get

$$\begin{aligned}\hat{b}_1 &= (62.1*66 - 60*57) / (70.1*66 - 57*57) \\ &= (4098.6 - 3420) / (4626.6 - 3249) \\ &= 678.6 / 1377.6 \\ &= 0.493\end{aligned}$$

Substituting the values in equation 9.8, we get

$$\begin{aligned}\hat{b}_2 &= (60*70.1 - 62.1*57) / (70.1*66 - 57*57) \\ &= (4206 - 3539.7) / (4626.6 - 3249) \\ &= 666.3 / 1377.6 \\ &= 0.484\end{aligned}$$

Substituting the values in equation 9.6, we get

$$\begin{aligned}\hat{b}_0 &= 4.3 - (.493*4.3) - (.484*4) \\ &= 4.3 - 2.1199 - 1.936 \\ &= 4.3 - 4.0599 = .247\end{aligned}$$

By substitution the values of $\hat{b}_0$, $\hat{b}_1$ and $\hat{b}_2$ in equation 9.2 we get the regression model as

$$\hat{Y} = 0.244 + 0.493 X_1 + 0.484 X_2$$

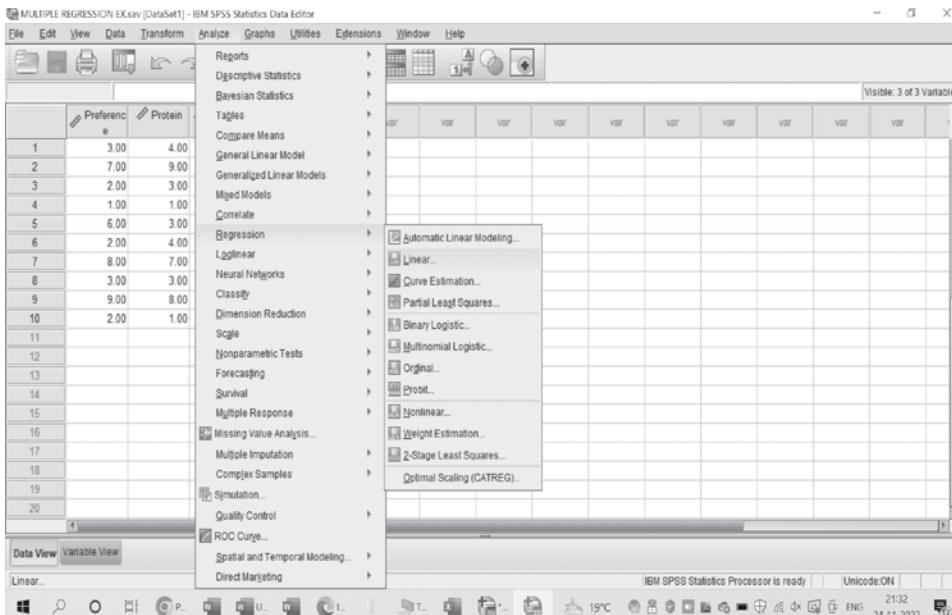
Here $\hat{b}_1$ and $\hat{b}_2$ are called partial regression coefficients, $b_1 = 0.493$ denotes the change in per unit change in X_1 when X_2 is held constant. Similarly, $b_2 = 0.484$ denotes the change in per unit change in X_2 when X_1 is held constant.

You must have already seen how complicated computations become as the number of variables increases, which become extremely difficult when the number of variables is more than 3. Regression models based on computers must be used. In reality, it should be noted that all multivariate methods need a computer's assistance when the variables and observations are large. As before, we can calculate the other coefficients, like the R^2 co-efficient of multiple determination and the R multiple correlation co-efficient, and also the ANOVA for hypothesis testing.

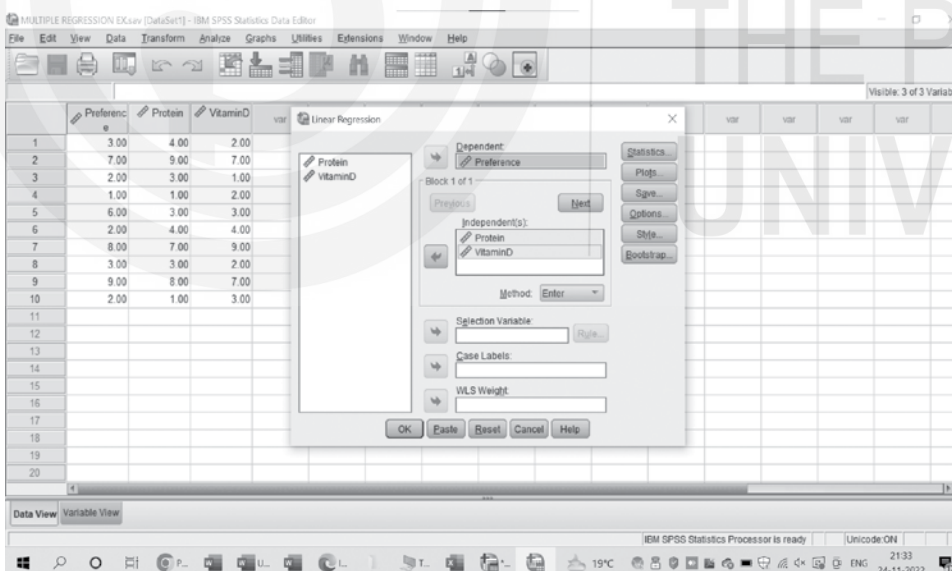
9.4 USE OF SPSS FOR MULTIPLE REGRESSION ANALYSIS

We will illustrate in this section how to estimate the regression model using the SPSS software for the data in Table 9.1.

To get the output for a multiple regression problem, go to Analyse and select Regression, followed by Linear as shown in Figure 9.1.

Figure 9.1: An appropriate tool selected for analysis

After this, select the variable preference by clicking on the right arrow to transfer it from the variable list on the left to right side box under independent variables or Independents. Similarly, select the variable Protein, and VitaminD by clicking on the right arrow to transfer these from the variable list on the left to right side box under Dependent variables or Dependents (Figure 9.2).

Figure 9.2: Variables transferred for the analysis

ENTER as the method, as we want all independent variables to be included in the model. By clicking OK, we will get the output for regression as shown in Tables 9.3 to 9.5.

For the validity of the regression equation, we look first at the co-efficient of multiple determination R^2 and multiple correlation co-efficient R . In our example $R^2 = 0.783$ and $R = 0.885$ which is a satisfactory one indicating that the preference rating Y is linearly related to protein intake X_1 and vitamin D

intake X_2 . It tells that 78.3 per cent of the variation in Y is explained jointly by the variations in X_1 and X_2 jointly (Table 9.3).

Table 9.3: Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.885 ^a	.783	.721	1.53482
a. Predictors: (Constant), VitaminD, Protein				

Hypothesis testing for linearity through ANOVA

The value of R^2 is significant as indicated by the p value (0.005) of F statistic as given in ANOVA Table 9.4.

$H_0 : \beta_1 \& \beta_2 = 0 \Rightarrow$ There is no linear relation between Y, X_1 and X_2 .

$H_1 : \beta_1 \& \beta_2 \neq 0 \Rightarrow$ There is linear relationship.

Look at the ANOVA

The calculated $F = 12.65$

The table $F(2, 7) = 4.74$ at 5% level. Reject H_0 and accept H_1 .

Table 9.4: ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	59.610	2	29.805	12.652	.005 ^b
	Residual	16.490	7	2.356		
	Total	76.100	9			
a. Dependent Variable: Preference						
b. Predictors: (Constant), VitaminD, Protein						

Table 9.5: Coefficients ^a						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		Std. Error	Beta			
1	(Constant)	.247	.942		.262	.801
	Protein	.493	.336	.473	1.466	.186
	VitaminD	.484	.346	.450	1.397	.205
a. Dependent Variable: Preference						

The estimated regression equation as obtained in Table 9.5 may be written as:

$$Y = 0.247 + 0.49X_1 + 0.48X_2$$

The above estimated regression equation indicates that the Protein intake is positively related with Preference for ready to eat cereals as is evident from the positive value of its coefficient (0.49). Similarly, the Vitamin D intake is positively related to the Preference for ready to eat cereals as the coefficient

for the Vitamin D variable is positive (0.48).

The results indicate that if the Protein intake goes up by one unit, the Preference rating for ready to eat cereals will go up by 0.49 units while keeping the Vitamin D intake constant. If the Vitamin D intake goes up by one unit, Preference rating for ready to eat cereals will go up by 0.48 units while keeping Protein intake constant.

We further note that the intercept and the slope terms are 0.247, 0.49 and 0.48, respectively, which is exactly the same as when the problem was worked out manually.

9.5 CONSIDERATIONS FOR MULTIPLE REGRESSION ANALYSIS

- 1) The equation should be validated statistically.
- 2) For forecasting the dependent variable, the independent variables should be forecast first. For example, if demand is a function of the price index and production index established by a multiple regression model, then to forecast demand for the next period, it is imperative first to forecast the price index and production index and then substitute them in the model to get the forecast for demand. This is one of the limitations of regression forecasting.
- 3) When the variables become too many the analysis is complex and very often the market researcher does not know which variables to retain. This problem could be overcome by doing stepwise regression on computer. For example, if demand is a function of 20 variables, we first fit the demand equation with 3 important variables which we think affect demand. Suppose $R^2 = 0.85$, that is, 85% of the variation in demand is explained by these variables, we add another two more variables of importance to make five independent variables. Now if $R^2 = 0.95$ we can as well stop adding further variables as the contribution may not appreciably improve the situation. We can thus visualise demand as a function of just 5 variables. The various permutations of changing and adding variables are possible only with the help of a computer. The important point to remember is that the cut-off points for the number of variables to be added should be based on the increase every time you get on R^2 . The moment the increase is marginal, stop adding variables.
- 4) If the independent variables among themselves are highly correlated, then we are facing the problem of 'multicollinearity'. Normally we say that the partial regression coefficient with respect to X_1 implies change in Y per unit change in X_1 provided $X_2, X_3, \dots$ are held constant. This poses a serious problem if there is multicollinearity. One way to overcome multicollinearity is to drop certain variables from the model if the corresponding standard error of the regression coefficient is unduly large. Another method is to see whether the original set could be transformed into another linear composite so that the new variables are uncorrelated.

9.6 SUMMARY

In this unit, we have discussed regression analysis, which is probably the most widely used technique among the analytical models of association. We first introduced the concept and assumptions of regression and then moved on to the multiple linear regression model. Multivariate regression models have been illustrated using the example of customer preference rating as a function of protein intake and vitamin D intake in the case of a ready-to-eat cereal. The need for resorting to computer solutions for a large number of variables and observations has been brought out with the use of SPSS software on the data set of the example already discussed and performed manually. Finally, the problems encountered in any regression analysis have also been explained.

9.7 KEY WORDS

Dependent Variable: The variable of interest or focus which is influenced by one or more independent variable(s).

Estimate: A value obtained from data for a certain parameter of the assumed model or a forecast value obtained from the model.

Independent Variable: A variable that can be set either to a desirable value or takes values that can be observed but not controlled. parameters of the model are estimated by minimising the sum of squares of error (discrepancy between fitted and actual value).

Linear Regression: Fitting of any chosen mathematical model, linear in unknown parameters, to a given data.

Model: A general mathematical relationship relating a dependent (or response) variable Y to independent variables $X_1, X_2, \dots, X_k$ by a force $Y=f(X_1, X_2, \dots, X_k)$

Multiple Regression: is a statistical technique that can be used to analyze the relationship between a single dependent variable and several independent variables.

Parameters: The constant terms of the chosen model that have to be estimated before the model is completely specified.

Regression: Relating of a dependent (or response) variable to a number independent variables, based on a given set of data.

9.8 SELF-ASSESSMENT QUESTIONS

1. What does building a regression model aim to achieve? Describe with an example from a marketing application.
2. A marketing manager for a consumer durable product has gathered information on advertising spending, promotional expenditure (other than advertising), and sales for the Indian states of Madhya Pradesh and Chhattisgarh from the company's records. The data are presented below.

Week	Advertising Expenditure (in Thousand Rupees)	Promotional Expenditure (Non-advertising) (in Thousand rupees)	Sales (in Thousand Units)
------	--	--	---------------------------

1	20	27	125
2	30	15	128
3	22	22	145
4	25	11	120
5	20	21	131
6	20	23	135
7	23	17	120
8	15	22	132
9	15	15	123
10	20	25	126

You were asked by the marketing manager to develop a multivariate regression model to predict sales based on the amount spent on advertising and promotion. Apply regression analysis to the provided data and explain the findings to the marketing manager.

3. A manufacturer of industrial supplies developed the following model for predicting the number of sales per month

$$Y = 41 + .3X_1 + .05X_2 - 7X_3 + 10X_4$$

where Y = Sales per month

X_1 = Number of manufacturing firms

X_2 = Number of wholesale and retail firms

X_3 = Number of competing firms

X_4 = Number of full-time company sales people.

- Explain the correct interpretation of all estimated parameters in the equation.
- If $R^2 = 0.49$, what does this figure mean to you?
- Explain how you will go about testing the validity of this multiple linear regression equation.

9.9 PROJECT QUESTION

Describe the independence assumption used in multiple regression. How can you check the independence of your data? In other words, how would you recognise the presence of multicollinearity and what steps would you take to eliminate it? In the same line, describe the other assumptions for multiple regression analysis.

9.10 FURTHER READINGS

F.E. Brown, 1980, Marketing Research, Addison-Wesley Publishing Company.

Paul E. Green and Donald S. Tull, 1986, Research for Marketing Decisions, Prentice-Hall of India Pvt. Ltd.

Boyd, Westfall and Stasch, 1988, Marketing Research Text and Cases, D. Irwin Inc. Homewood, Illinois.

UNIT 10 DISCRIMINANT ANALYSIS AND LOGISTIC REGRESSION ANALYSIS

Learning Outcomes

After going through this unit, you should be able to:

- appreciate the application areas of discriminant analysis and logistic regression analysis in the context of marketing.
- explain how discriminant analysis and logistic regression analysis are computed.
- interpret the findings of discriminant analysis and logistic regression analysis.
- differentiate between logistic regression and discriminant analysis.
- demonstrate the use of SPSS for discriminant analysis and logistic regression analysis.
- apply the concept learned to real-world marketing decisions.

Structure

- 10.1 Introduction
- 10.2 Discriminant Analysis
- 10.3 A Numerical Example of Discriminant Analysis
- 10.4 Use of SPSS for Discriminant Analysis
- 10.5 Applications of Discriminant Analysis
- 10.6 Logistic Regression
- 10.7 Use of SPSS for Logistic Regression Analysis
- 10.8 Applications of Logistic Regression
- 10.9 Difference Between Logistic Regression and Discriminate Analysis
- 10.10 Summary
- 10.11 Key Words
- 10.12 Self-Assessment Questions
- 10.13 Project Question
- 10.14 Further Readings

10.1 INTRODUCTION

We discussed multiple regression analysis in unit 9. In this unit, we will discuss two methods of multivariate analysis: discriminant analysis and logistic regression.

Discriminant analysis is useful in situations where a total sample could be classified into mutually exclusive and exhaustive groups on the basis of a set of predictor variables. Unlike the regression analysis, these predictor variables need not be independent. For example, one may wish to predict whether sales potential in a particular marketing territory will be 'good' or 'bad' based on the territory's personal disposal income, population density and number of retail outlets. You may like to classify a consumer as a user or non-user of one of the five brands of a product based on his age, income and length of time spent in his present job. Here, the interest is what variables discriminate well between groups.

Logistic regression helps in understanding the relationship among variables and predicting the outcome when the decision variable is binary in nature. Potential applications include determining whether a new product will succeed or fail and whether discounts will result in the purchase of the product or not, among other things.

10.2 DISCRIMINANT ANALYSIS

Market researchers often need to understand how groups of customers or items differ in terms of a set of explanatory or independent variables. For this purpose, market researchers often use discriminant analysis to assign an object (a customer or an item) to one group among the number of known groups. Discriminant analysis is a useful tool for situations where the total sample is to be divided into two or more mutually exclusive and collectively exhaustive groups on the basis of a set of predictor variables. For example, a problem involving classifying salespeople as "high performers" or "low performers," classifying customers as "loyal" or "not loyal," and so on.

Objectives of two group discriminant analysis:

- 1) Finding linear composites of the predictor variable that enable the analyst to separate the groups by maximising among groups relative to within groups variation.
- 2) Establishing procedures for assigning new individuals, whose profiles but not group identity are known, to one of the two groups.
- 3) Testing whether significant differences exist between the mean predictor variable profiles of the two groups.
- 4) Determining which variables account most for intergroup differences in mean profiles.

10.3 A NUMERICAL EXAMPLE OF DISCRIMINANT ANALYSIS

Let us return to the example involving ready-to-eat cereal that was presented in the regression analysis. However, in this problem, the ten consumer raters are simply asked to classify the cereal into one of two categories: likes or dislikes. The data is given in Table 10.1. Here, again

X_1 : The amount of protein (in grams) per standard serving.

X_2 : The percentage of minimum daily requirements of vitamin D per standard serving.

Also shown in Table 10.1 are the various sums of squares and cross products, the means on X_1 and X_2 of each group, and the total sample mean.

Table 10.1: Consumer evaluations of ten cereals varying in nutritional content

Person	Evaluation	Protein X_1	Vitamin D X_2	X_1^2	X_2^2	X_1X_2
1	Dislike	2	4	4	16	8
2	Dislike	3	2	9	4	6
3	Dislike	4	5	16	25	20
4	Dislike	5	4	25	16	20
5	Dislike	6	7	36	49	42
Mean		4	4.4	Sum 90	110	96
6	Like	7	6	49	36	42
7	Like	8	4	64	16	32
8	Like	9	7	81	49	63
9	Like	10	6	100	36	60
10	Like	11	9	121	81	99
Mean		9	6.4	Sum 415	218	296
Grand Mean		6.5	5.4			
Standard deviation		3.028	2.011			

The grand mean is $\bar{X}_1 = 6.5$ $\bar{X}_2 = 5.4$

We first note from the table that the two groups are much more widely separated on X_1 (Protein) than they are on X_2 (Vitamin D). If we were forced to choose just one of the variables, it would appear that X_1 is a better bet than X_2 . However, there is information provided by the group separation on X_2 , so we wonder if some linear composite of both X_1 and X_2 could do better than X_1 alone. Accordingly, we have the following linear function:

$Z = K_1X_1 + K_2X_2$ where K_1 and K_2 are the weights that we seek.

But how shall we define variability? In discriminant analysis, we are concerned with the ratio of two sums of squares after the set of scores on the linear composite has been computed. One sum of squared deviations represents the variability of the two group means on the composite around their grand mean. The second sum of squared deviations represents the pooled variability of the individual cases around their respective group means also on the linear composite. One can then find the ratio of the first sum of squares to the second. It is this ratio that is to be maximised through the appropriate selection of K_1 and K_2 . Solving for K_1 and K_2 involves a procedure similar to the one encountered in the multiple regression. However, in the present case, we shall want to find a set of sums of squares and cross products that relate to the variation within groups. For ease of calculation let us define $x_1 = x_1 - \bar{X}_1$, and $x_2 = X_2 - \bar{X}_2$, (i.e., each observation measured from its mean).

Solving for K_1 and K_2 .

Mean corrected sums of squares and cross products

	Dislikers	Likers	Total
$\sum x_1^2 = \sum (X_1 - \bar{X}_1)^2 = \sum X_1^2 - N\bar{X}_1^2 =$	10	10	20
$\sum x_2^2 = \sum (X_2 - \bar{X}_2)^2 = \sum X_2^2 - N\bar{X}_2^2 =$	13.2	13.2	26.4
$\sum x_1 x_2 = \sum (X_1 - \bar{X}_1)(X_2 - \bar{X}_2) = \sum X_1 X_2 - N\bar{X}_1 \bar{X}_2 =$	8	8	16

The normal equations are

$$K_1 \sum x_1^2 + K_2 \sum x_1 x_2 = \bar{X}_1 (\text{Likers}) - \bar{X}_1 (\text{dislikers})$$

$$K_1 \sum x_1 x_2 + K_2 \sum x_2^2 = \bar{X}_2 (\text{Likers}) - \bar{X}_2 (\text{Dislikers})$$

$$20 K_1 + 16 K_2 = 9 - 4 = 5$$

$$16 K_1 + 26.4 K_2 = 6.4 - 4.4 = 2$$

Solving these two simultaneous equations, we have $K_1 = 0.368$, $K_2 = -0.147$

$$\text{Discriminant function } Z = 0.368 X_1 - 0.147 X_2$$

We can also find discriminants scores for the means of the two groups and the grand mean.

$$Z (\text{dislikers}) = 0.368 (4) - 0.147 (4.4) = 0.824$$

$$Z (\text{likers}) = 0.368 (9) - 0.147 (6.4) = 2.368$$

$$Z (\text{grand means}) = 0.368 (6.5) - 0.147 (5.4) = 1.596$$

We note that the discriminant function “favours” X_1 by giving about 2.5 times the (absolute value) weight ($K_1 = 0.368$ versus $K_2 = -0.147$) to X_1 as is given to X_2 .

The discriminant scores of each person are shown below. Each score is computed by the application of the discriminant function to the persons original X_1 and X_2 Values.

Dislikers			Likers	
Person	Discriminant Score	Person	Discriminant Score	
1	0.148	6	1.691	
2	0.809	7	2.353	
3	0.735	8	2.279	
4	1.250	9	2.794	
5	1.176	10	2.721	
Mean	0.824	Mean	2.368	
Grand Mean 1.596				

$$\text{Between group variability: } 5 (0.824 - 1.596)^2 + 5 (2.368 - 1.596)^2 = 5.96$$

Within group variability

$$\text{Dislikers } (0.148 - 0.824)^2 + (0.809 - 0.824)^2 + \dots + (1.176 - 0.824)^2 = 0.772$$

$$\text{Likers } (1.691 - 2.368)^2 + (2.353 - 2.368)^2 + \dots + (2.721 - 2.368)^2 = \frac{0.772}{1.544}$$

$$\text{Discriminant criterion } C = \frac{5.96}{1.544} = 3.86$$

Since the normal equations for solving K_1 and K_2 are obtained by maximising the ratio between the group and within group variance the discriminant criterion as calculated above = 3.86 will be the maximum possible ratio. If we suppress X_2 in the discriminant function and calculate another C, it will be less than 3.86.

It is rather interesting that the optimal function $Z = 0.368 X_1 - 0.147 X_2$ is a different function in which X_2 (Vitamin D) receives a negative weight bringing thereby the importance of X_1 to the highest order. This means protein is much more important than Vitamin D.

Classifying the persons

It is all well and good to find the discriminant function, but the question is how to assign the persons to the relevant groups.

- Assign all cases with discriminant scores that are on the left of the midpoint (1.596) to the disliker group.
- Assign all cases with discriminant scores that are on the right of the midpoint (1.596) to the Liker group.

Dislikers	Likers	Total
0.824	2.368	1.596
	(Grand Mean)	
0.824	1.596	2.368
(Dislikers)	(Midpoint)	(Likers)

That is all true dislikers will be correctly classified as such and all true likers will be correctly classified. This can be shown by a 2 X 2 table

True State	Assigned by Rule		Total
	Disliker	Liker	
Disliker	5	0	5
Liker	0	5	5
Total	5	5	10

Activity 1

Speak with the marketing manager of your own company or any other organisation you are familiar with, and find out how they predict the performance of the sales executives who report to them. What factors do they consider as predictors of future performance to promote sales executives?

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10.4 USE OF SPSS FOR DISCRIMINANT ANALYSIS

You would have realised by now the complexity of calculations in the discriminant analysis involving two predictors which itself needs computer-based solutions when the number of observations increases considerably. Let us explain the calculation and interpretation of discriminant analysis by using SPSS and a hypothetical case example.

Let us assume the sales promotion manager of an online grocery store called Bigbasket wants to reward its loyal customers during festival season. The manager wishes to classify the customer as loyal or disloyal for this purpose. The basis of this grouping of customers is based on the frequency of purchases in a year, the average amount spent on each purchase in a year, and the number of years the person has been shopping at Bigbasket.

Now the market researcher runs discriminant analysis to set up a system

to separate loyal customers from disloyal customers. And the objective of building this discriminant function is not just to group them but also to identify which independent variable among frequency of purchase, average amount spent, or number of years of purchase is relatively better at discriminating between these two groups. And finally, how to classify a new customer into one of those two groups based on loyalty. Loyal customers are coded as 1, while disloyal customers are coded as 2.

Steps in SPSS

Step 1: Problem Formulation: Identifying the objectives

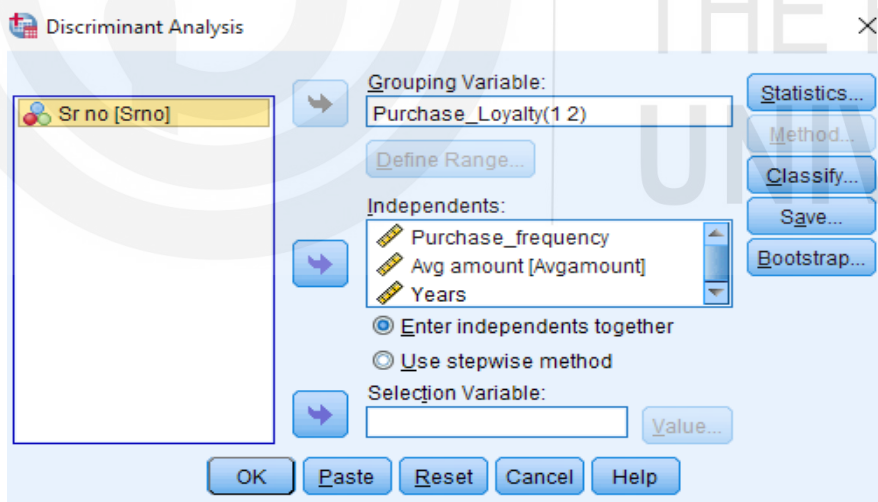
As mentioned above, the objective of the study is to classify the customer as loyal and disloyal customers using discriminant analysis using SPSS.

Step 2: Research Design for Discriminant Analysis

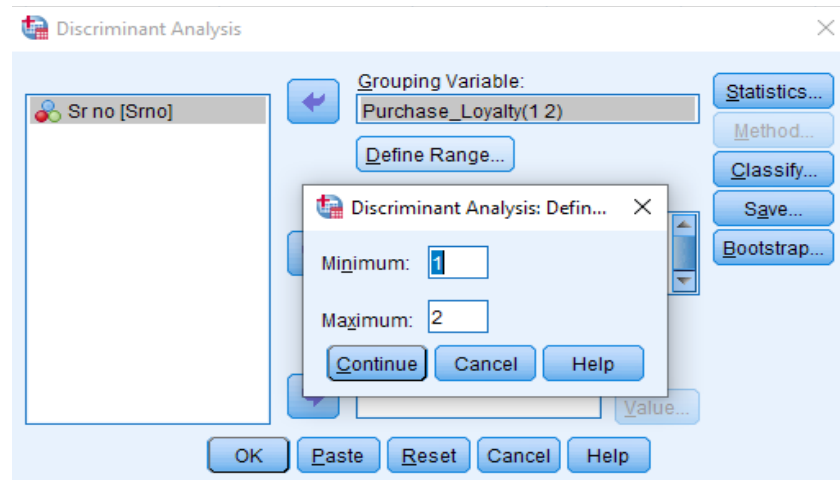
This stage involves the selection of dependent and independent variables. In order to classify the customers, the loyalty of customers named “Purchase_Loyalty” is considered the dependent variable. The independent variables are the customers’ purchase frequency in a year, the average amount spent on purchases in a year, and the number of years the customer has been purchasing groceries from Bigbasket.

Step 3: SPSS input data

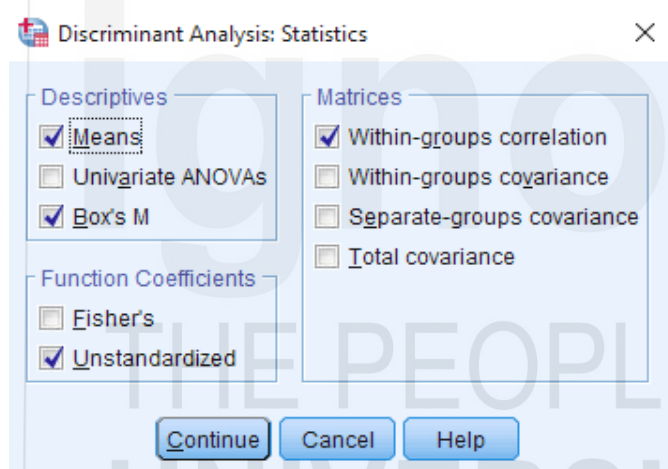
Select Analyze > classify > discriminant analysis. Move the dependent variable (Purchase_Loyalty) in the *variable box under Grouping variable*. Move the explanatory variables (Purchase_frequency, Avg amount and Years) into the *independents*. Select ‘enter independents together’ option.



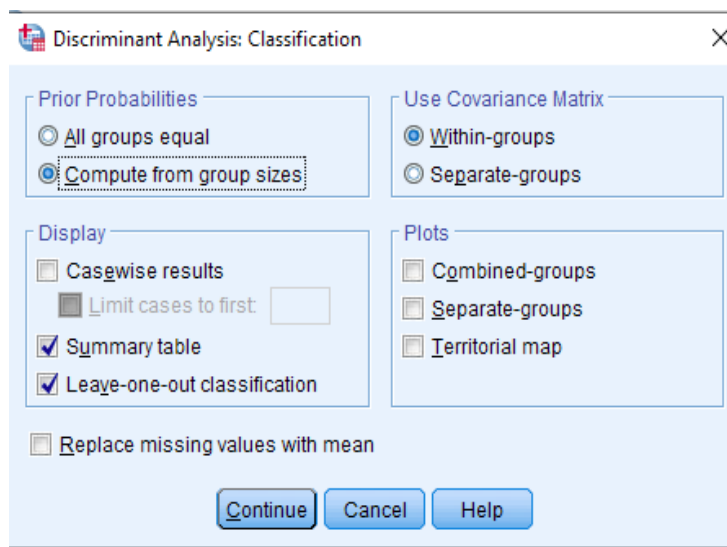
- Click *define range* > Enter 0 as the minimum range and 1 as the maximum range for two groups. Click on *continue*.



- Click on *statistics* > Select for *means*, *Univariate ANOVAs*, and *Box's M* in the *descriptive box* > Select *unstandardized* from *Functions coefficients* > Select *within-groups correlation* from *matrices* > Click on continue



- Click on *classify* option from the pop-up and select 'Compute from group sizes' and for display, select 'Summary table' and 'Leave one out classification'; Click on continue > Click on OK



Step 4: Interpreting and reporting the output of Discriminant Analysis:**1. Strength of the model**

From the output table “Classification Results,” we can observe that the discriminant function obtained can classify 76.7% of the 30 customers correctly. Of the 30 customers, 7 customers are misclassified.

Classification Results ^a					
		Predicted Group Membership			
	Purchase_Loyalty	Disloyal	Loyal		
Original	Count	Disloyal	11	4	15
		Loyal	3	12	15
%		Disloyal	73.3	26.7	100.0
		Loyal	20.0	80.0	100.0

a. 76.7% of original grouped cases correctly classified.

2. Statistical significance of the model:

From the table “Wilk’s lambda,” Wilks’ λ was found to be 0.560. Since the value is almost closer to 0, it indicates better discriminating power for the model. The significance is 0.002, which indicates that the chi-square test is significant at a confidence level of at least 99.8%.

Wilks' Lambda				
Test of Function(s)	Wilks' Lambda	Chi-square	df	Sig.
1	.560	15.389	3	.002

- The discriminant functions extracted accounted for nearly 50% (i.e., 0.664) of the variance in the result of the Eigenvalue table, confirming the hypothesis.

Eigenvalues				
Function	Eigenvalue	% of Variance	Cumulative %	Canonical Correlation
1	.787 ^a	100.0	100.0	.664

a. First 1 canonical discriminant functions were used in the analysis.

Box’s M test: The Box’s M test (also known as the Box’s Test for Equivalence of Covariance Matrices) is a parametric test that is used to compare variation in multivariate data. More specifically, the null hypothesis for this test is that the observed covariance matrices for the dependent variable are equal or homogeneous. The generated test box statistic is called Box’M statistic.

Test Results		
Box's M		4.342
F	Approx.	.639
	df1	6
	df2	5680.302
	Sig.	.699
Tests null hypothesis of equal population covariance matrices.		

In the analysis from the Box's M test we found that the null hypothesis stating that there is no difference among the groups that exist is true, since, $p\text{-value}=0.699 > 0.05$.

3. Predictor:

Standardized Canonical Discriminant Function Coefficients	
	Function 1
Purchase_frequency	.209
Avg amount	.755
Years	.366

In order to decide which among the three variables is a better predictor of “loyal” and “disloyal” customers, we need to look at the table “Standard coefficient for canonical variable,” which shows “average amount of purchase” in a year to be the best predictor variable with a coefficient of 0.755. The next dominant variable is the number of years the customer has been purchasing with a co-efficient of 0.366, followed by the “purchase frequency” in a year with a co-efficient of 0.209.

4. Classification of new customer

Functions at Group Centroids	
	Function 1
Purchase_Loyalty	1
Disloyal	-.857
Loyal	.857
Unstandardized canonical discriminant functions evaluated at group means	

From the output table “Functions at group centroid,” the mean canonical variable gives us the new means for transformed group centroids. The new

mean for group 1 (disloyal) is -0.857, and the new mean for group 2 (loyal) is 0.857. This means that the midpoint of the two is 0. If the discriminant score of the customer is less than 0, then he or she is a disloyal customer; if the discriminant score of the customer is more than 0, then we classify him or her as a loyal customer.

To compute the discriminant score of any customer, the unstandardized discriminant function has to be used.

Canonical Discriminant Function Coefficients

	Function	
	1	
Purchase_frequency		.028
Avg amount		.000006
Years		.354
(Constant)		-4.023

Unstandardized coefficients

$$D = (-4.023) + 0.028(\text{Purchase_frequency}) + 0.000006(\text{Avg purchase}) + 0.354(\text{years})$$

where D is the discriminant score of any customer whose purchase frequency and average amount spent in a year are known, along with the number of years for which the customer has been purchasing groceries from Bigbasket.

As mentioned above from the results, any positive discriminant score leads to the classification of the customer as “loyal,” and negative discriminant score customers are classified as “disloyal.”

Step 5: Validation of the Results:

The results are validated by assessing the hit ratios. The hit ratios are obtained by summing the number of objects in the diagonal of the classification matrix divided by the total number of objects. The hit ratio of the result is , which means 76.6% are correctly classified. Further, cross validation is carried out with a hit ratio of . therefore 73.3% of cross-vallidated groups are correctly classified. Thus, the results are validated.

Classification Results ^{a,c}					
		Predicted Group Membership			Total
		Purchase_Loyalty	Disloyal	Loyal	
Original	Count	Disloyal	11	4	15
		Loyal	3	12	15
	%	Disloyal	73.3	26.7	100.0
		Loyal	20.0	80.0	100.0
Cross-validated ^b	Count	Disloyal	11	4	15
		Loyal	4	11	15
	%	Disloyal	73.3	26.7	100.0
		Loyal	26.7	73.3	100.0

a. 76.7% of original grouped cases correctly classified.
b. Cross validation is done only for those cases in the analysis. In cross validation, each case is classified by the functions derived from all cases other than that case.
c. 73.3% of cross-validated grouped cases correctly classified.

10.5 APPLICATIONS OF DISCRIMINANT ANALYSIS

The major application area of this technique is when the objects (people, products, etc.) are classified into two or more groups based on the knowledge of some of their characteristics. These groups are the nonmetric dependent variable. Thus, the goal is to predict and explain the reasons for each object's group membership using a set of independent variables.

Discriminant analysis can be used heavily in marketing; potential applications include predicting the success or failure of a new product based on some explanatory variables. Furthermore, it can be used to categorise groups of people visiting a mall as purchasers or non-purchasers, to categorise sales executives as high performers or low performers, a banker can categorise them as high-risk lenders or low-risk lenders, and so on, based on a series of independent variables.

10.6 LOGISTIC REGRESSION

Logistics regression combines the independent variables to estimate the probability that a particular event will occur. It is a multiple regression but with a dependent variable (outcome variable) that is a categorical (nominal or nonmetric) and independent variables (predictor variables) can be metric or nonmetric. In Logistics regression analysis, independent variables do not need to be interval, normally distributed, linearly related, or of equal variance within each group.

The following equation represents logistic regression:

Where,

$$Y = \frac{e^{(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}}{1 + e^{(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}}$$

$x_1, \dots, x_n$ = input value

y = predicted output

β_0 = intercept term

β_1 = coefficient for input (x)

$x_1, x_2, x_3, \dots, x_n$ with n explanatory variables satisfying the two conditions

- The logistic probability function must always be positive (since $p \geq 0$).
- The logistic probability function must be less than 1 (since $p \leq 1$)

Parameter Estimation: The parameters in the equation of a 2-class classification in logistic regression is a $\hat{\beta}$ vector as in the log-odds equation.

$$l(\beta) = \sum_{i=1}^n y_i \beta_{xi} - \log(1 - e^{\beta_{xi}})$$

The equation is similar to linear regression, where the input values are combined linearly to predict an output value using weights or coefficient

values. However, unlike linear regression, the output values modelled here is a binary value (0 or 1) where class 0 means POOR and class 1 means GOOD.

Key properties of Logistic Regression include:

- Logistics Regression's dependent variable obeys 'Bernoulli Distribution'
- Estimation/ prediction is based on 'Maximum Likelihood'
- Logistics Regression does not evaluate the coefficient of determination (like the linear regression).

10.6.1 Types of Logistic Regression

1. **Binary (Binomial) Logistic Regression:** A binary output is a variable where there are only two possible outcomes. These outcomes must be opposite of each other and mutually exclusive. This predicts the probability that an observation falls into one of two categories of dichotomous dependent variables based on one or more independent variables.
2. **Multinomial Logistic Regression:** It is sometimes considered an extension of binomial logistic regression having the dependent variable with more than two categories. It also uses maximum likelihood estimation to evaluate the probability of occurrence of dependent variable. The independent variables can be either dichotomous (i.e., binary) or continuous (i.e., interval or ratio in scale).
3. **Ordinal Logistics Regression:** The dependent variable in ordinal logistic regression is not only categorical but they also follow an order. For example, the performance of sale manager is poor, fair, good, excellent or a customer is highly dissatisfied, dissatisfied, satisfied or highly satisfied. Just like the above two logistic regression, here also the objective is to determination of which among the independent variable has a statistically significant effect on the dependent variable.
 - For categorical independent variables which consist of three groups can lead to the easy interpretation of the odds in one particular group had a higher or lower value on the dependent variable compared to the second group.
 - For continuous independent variables, it is easy to interpret how a single unit increases or decreases in that variable, and was associated with the odds of the dependent variable having a higher or lower value.

10.6.2 Assumptions for implementing Binary Logistic Regression

1. **The dependent variable is binary or dichotomous:** The first assumption of binary logistic regression is that response variables can only take on two possible outcomes. For example customer, will purchase the product or not, is the customer loyal or not, product will be successful or not etc.

2. **No multicollinearity between the predictor or explanatory variables:** This assumption implies that the predictor variables should be independent of each other. Multicollinearity is related to two or more highly correlated independent variables. Such variables don't provide unique information in the regression model and lead to wrongful interpretation.
3. **The logistic regression does not require a linear relationship between the independent variables and the dependent variable:** It can address nonlinear effects even when exponential and polynomial terms are not explicitly added as additional independent variables because of the logistic relationship.

10.7 USE OF SPSS FOR LOGISTICS REGRESSION ANALYSIS

Let us explain the calculation and interpretation of discriminant analysis by using SPSS and a hypothetical case example.

Let us assume that a market researcher wants to understand the characteristics of an Indian customer that determine which one of the two brands of coffee—Bru and Nescafe—they prefer. Data on 60 customers is collected with age, gender, and income as three independent variables, as well as their brand preference, i.e., Bru or Nescafe. It can be noted that for the hypothetical case, the data is limited to 60; however, the overall sample size to achieve a good result with maximum likelihood estimation should be larger, say 400 or more.

Step 1: Objectives of logistics regression

As mentioned above, the objective of the study is to classify the customer based on their preferences for two brands of coffee, namely Bru and Nescafe, by using logistic regression with SPSS.

Step 2: Research design for logistic regression

This step involves the selection of dependent and independent variables. In order to classify the customers based on their coffee brand preferences, “Choice of Brand” is considered the dependent variable. The independent variables are the age, gender, and income of the customers in thousands of rupees per month.

Step 3: SPSS input data

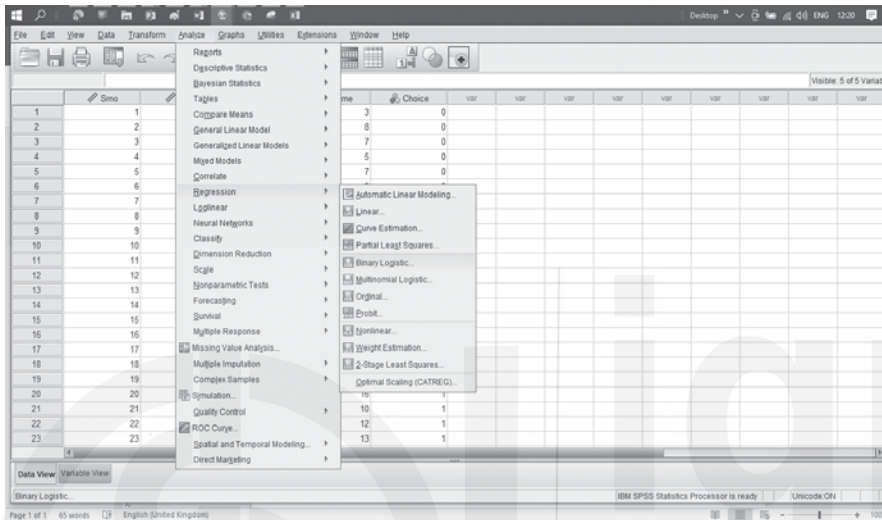
The survey data of 60 customers' preferences are imported into an SPSS data file, with the variable “Choice” coded as “0” for customers who prefer Bru and “1” for those who prefer Nescafe. In the independent variables, age and income are continuous data, and gender is categorical data coded as “0” for female and “1” for male.

**Dependent Variable
Encoding**

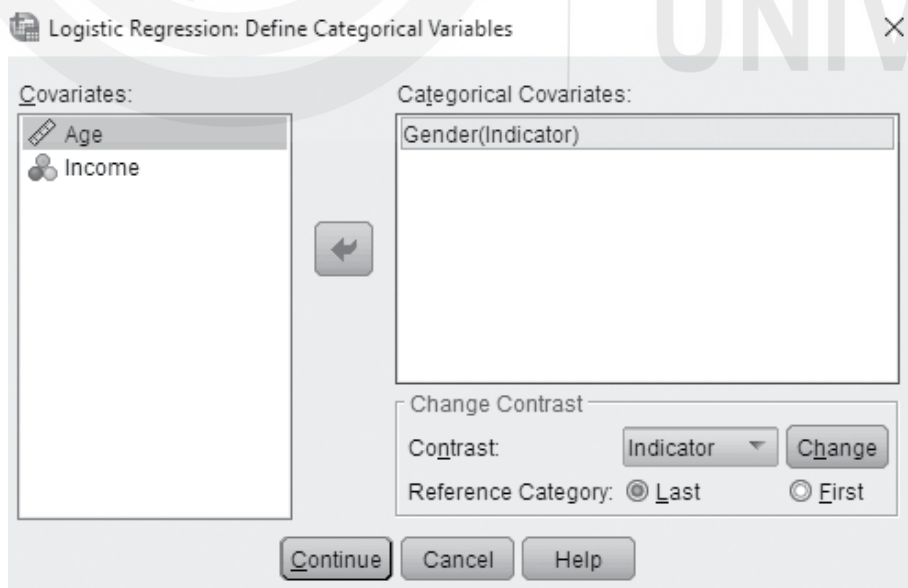
Original Value	Internal Value
Bru	0
Nescafe	1

After coding the data, the following steps are carried out.

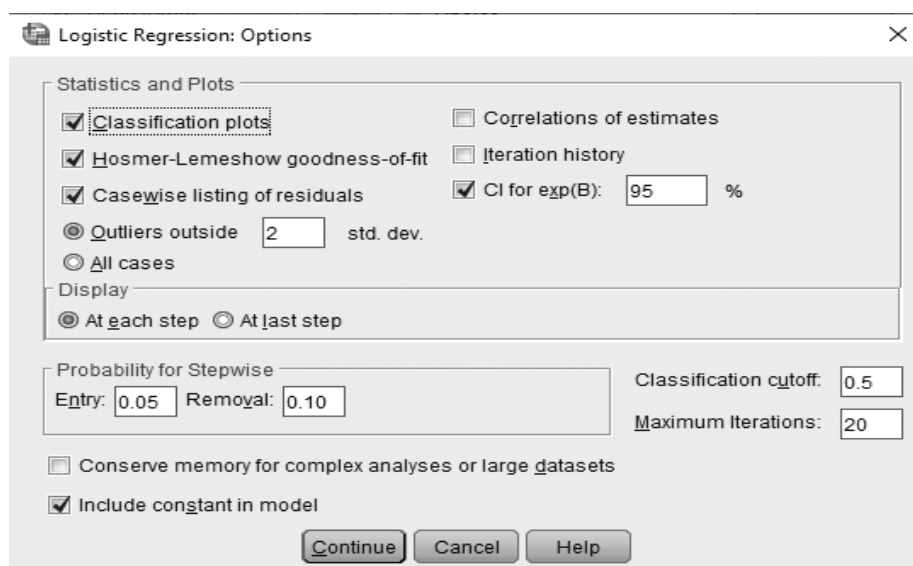
1. Click on Analyze > Regression > Binary Logistic Regression



2. Transfer the dependent variable, into the fields assigned and the independent variables into the covariate box.
3. For categorical variables > click on *logistic regression: define categorical variables* dialogue box > Transfer the categorical independent variables from covariate box to > categorical covariates



4. Click on options > *logistics regression: Options* dialogue box > *Statistics and Plots* > click on classification plots > click on Hosmer-Lemeshow goodness-of-fit > click on the case wise listing of residuals > click on CL for exp (B) > click Continue> click Ok



Stage 4: Estimation of the logistic regression model and assessing overall fit

SPSS Statistics generates many output tables when carrying out binary logistics regression. The tables required to understand the results from the binary logistics regression are:

1. Strength of the model:

From the output table “Classification Table,” we can observe that the logistic function is able to classify 85% of the 60 customers correctly into two groups: those who prefer Bru and those who prefer Nescafe coffee. Of 60 customers, 9 have been misclassified. Therefore, the model does a good job of classifying the customers correctly.

Classification Table^a

		Predicted Choice		Percentage Correct
		Bru	Nescafe	
Step 1	Observed Choice			
	Bru	26	5	83.9
	Nescafe	4	25	86.2
Overall Percentage				85.0

a. The cut value is .500

2. Variance explained:

In order to understand how much variation in the dependent variable can be explained by the model (the equivalent of R² in multiple regression), the “Model Summary” table needs to be interpreted.

Model Summary			
Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	39.375 ^a	.518	.690
a. Estimation terminated at iteration number 6 because parameter estimates changed by less than .001.			

The “Model Summary” table contains Cox & Snell R square and Nagelkerke R square values; both of these methods are used for calculating the explained variation. These values are sometimes referred to as “pseudo-R square” values. In the analysis, the explained variation in the dependent variable ranges from 51.8% to 69%. Researchers suggest that it is preferable to report the Nagelkerke R square values. Since 69% of the variance is explained, the developed model is good.

3. Measure of overall fit:

The Omnibus test is a likelihood ratio chi-square test performed to determine the overall fit of the model relative to the null model. The Omnibus tests of model coefficients are used to check that the new model (with explanatory variables included) is an improvement over the baseline model (without any explanatory variables, consisting of the only constant term). It uses the chi-square test to see if there is a significant difference between the log-likelihoods (specifically the -2LLs) of the baseline model and the new model. If the p-value is less than 0.05, then we have a new model that is significantly better than the baseline model and should be further interpreted. The table below shows the “Omnibus Test of Model Coefficients” based on the Chi-Square test. The results imply that the overall model is statistically significant (the focus is on row three—”Model”: 3 degrees of freedom, Chi-square = 43.736, $p < .001$), and the model is fitting the data significantly better than the base model.

Omnibus Tests of Model Coefficients				
		Chi-square	df	Sig.
Step 1	Step	43.736	3	.000
	Block	43.736	3	.000
	Model	43.736	3	.000

Hosmer and Lemeshow is the second statistical test to measure the overall fit. It is a commonly used test for assessing the goodness of fit of a binary logistic regression model and allows for any number of explanatory variables, which may be continuous or categorical. In Hosmer-Lemeshow tests, the null hypothesis states that predictions made by the model fit perfectly with observed data. A chi-square statistic is computed comparing the observed frequencies with those expected under the linear model. A non-significant chi-square ($p > 0.05$) indicates that the data fit the model well (Hosmer and

Lemeshow, 2000; Hosmer et al., 2013). The high p –value ($0.457 > 0.05$) on the Hosmer and Lemeshow test suggests that goodness of fit is satisfactory and the observed data adequately fits the model.

Hosmer and Lemeshow Test

Step	Chi-square	df	Sig.
1	7.762	8	.457

Step 5: Interpreting and reporting the output of Binary Logistics Regression Analysis

Variables in the equation:

Variables in the Equation

	B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I. for EXP(B)	
							Lower	Upper
Step 1 ^a								
Age	.056	.042	1.767	1	.184	1.058	.974	1.149
Gender(1)	-3.731	1.261	8.748	1	.003	.024	.002	.284
Income	.417	.142	8.639	1	.003	1.517	1.149	2.004
Constant	-5.895	1.779	10.977	1	.001	.003		

a. Variable(s) entered on step 1: Age, Gender, Income.

- From the “Variables in the Equation” table, it is noted that the variable “Age” is not significant ($p\text{-value} > 0.05$), therefore the variable “Age” is not a dominating and significant factor in the model that impacts the customers’ choice preferences for considered coffee brands. Hence, it is not included in the developed model.
- The variable “Gender” (Male) has a negative influence of (-3.731) in the model, indicating that they are likely to prefer Bru, the first brand considered, which is coded as “0” in the data file.
- The next independent variable, “income,” has a positive co-efficient (0.417), indicating a positive influence closer to one, indicating that higher-income customers are more likely to prefer Nescafe, the second brand considered, which is coded “1” in the data file.

- Therefore, the prediction equation for the customers preferring the 2nd brand ‘Nescafe’ that is coded as ‘1’ is

Log (Odds of choice 1, i.e. Nescafe)

$$= - 5.895 - 3.731 (\text{Gender}) + 0.417 (\text{income})$$

- For example:

The preference to the coffee brand of a male customer with a monthly income of 40,000 can be determined with the developed model i.e.,

Log (odds of choic 1, i.e. Nescafe)

$$= - 5.895 - 3.731 (1) + .417 (40)$$

$$= 7.054$$

$$P = \frac{e^{7.054}}{1 + e^{7.054}}$$

$$= 0.99$$

Since the probability is greater than 0.5, the customer is classified as preferring the second brand, i.e., Nescafe. If the probability was less than 0.5, then the customer is classified as preferring the first brand, i.e., Bru coffee.

Stage 6: Validation of the results

External validity is supported with the hit ratio for the selected holdout sample, also referred to as the validation sample, which is separate from the analysis sample used to estimate the model. For the holdout sample, the hit ratio is = 85%, thus the result is valid.

Classification Table ^a					
		Predicted		Percentage Correct	
		Choice Bru	Choice Nescafe		
Step 1	Choice Bru	9	2	81.8	
	Choice Nescafe	1	8	88.9	
	Overall Percentage			85.0	

a. The cut value is .500

10.8 APPLICATIONS OF LOGISTIC REGRESSION

Logistics regression is one of the multivariate analysis techniques that provides a data driven decision making support system to marketing researchers. It is easier to implement and interpret, as no assumptions regarding multivariate normality or equal variance is required. Thus, in social science research like marketing when the data may not be normal the applicability of logistic regression is quite high. Typical application areas are cases where one wishes to predict the likelihood of an entity belonging to one group or another such as in response to credit worthiness, to issue credit cards the marketers/bankers can classify the prospects into high risk and low risk. Further, binary logistic regression can be used to understand which factors influence the purchase of smart phone and predict who can be buyers and nonbuyers of smartphones, what factors will predict a sports person to be successful or not etc.

Activity 2

Speak with the marketing manager of your own company or any other organisation you are familiar with and discuss how they predict whether a person whose demographics are known would be a buyer or a non-buyer of a given brand or product.

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10.9 DIFFERENCE BETWEEN LOGISTIC REGRESSION AND DISCRIMINATE ANALYSIS

- Both Logistic regression and discriminant are multivariate statistical techniques that are applied to predict the probability of a specific categorical dependent variable based on several explanatory or predictor variables. Key difference lies in the population distribution, when the population is normally distributed, discriminant analysis is used however, no such assumptions about the population is required for logistic regression analysis. As mentioned earlier logistic regression is a form of multiple regression when the dependent variable is categorical, dichotomous or discrete while all the explanatory variables can be of any kind.
- Both the techniques require categorical variable, the logistic regression is preferred when the dependent variable is dichotomous while discriminant analysis is preferred when dependent variable is nominal (for more than two groups).
- Logistic regression is often used by researcher when they are interested in finding the most important predictors or in finding the role different variables play in prediction. So, while logistic regression emphasis on the predictors, discriminant analysis emphasis on the group prediction itself.

10.10 SUMMARY

In this unit, we have discussed two multivariate statistical techniques: discriminant analysis and logistic regression. Both discriminant analysis and logistic regression are applied to predict the probability of a specific categorical dependent variable based on several explanatory or predictor variables. The key difference lies in the population distribution; when the population is normally distributed, discriminant analysis is used; however, no such assumptions about the population are required for logistic regression analysis. Assumptions involved in applications of these statistical methods have also been discussed. Further, the calculation and interpretation of discriminant analysis and logistic regression methods were explained using SPSS and hypothetical case examples.

10.11 KEY WORDS

Centroid: Mean value for the discriminant Z scores of all objects within a particular category or group.

Classification matrix: Means of assessing the predictive ability of the model.

Discriminant function: It is a linear combination of independent variables developed by discriminant analysis that will best discriminate between the categories of the dependent variable.

Discriminant scores: The unstandardized coefficients are multiplied by

the values of the variables. These products are summed and added to the constant term to obtain the discriminant scores.

Eigenvalues: For each discriminant function, the eigenvalue is the ratio of between groups to within-group sum of squares.

Hit ratio: Percentage of objects correctly classified by the discriminant function or logistic regression model.

Likelihood value: Measure used in logistic regression to represent the lack of predictive fit.

Logistic coefficient: Coefficient in the logistic regression model that acts as the weighting factor for the independent variables in relation to their discriminatory power.

Logistic regression or Logit analysis: A special form of regression in which the dependent variable is a nonmetric, dichotomous variable. Although some differences exist, the general manner of interpretation is quite similar to linear regression.

Odds: The ratio of the probability of an event occurring to the probability of the event not happening, which is used as a measure of the dependent variable in logistic regression.

Standardized discriminant function coefficients: They are the discriminant function coefficients that are used as the multipliers when the variables have been standardized to a mean of 0 and a variance of 1.

Wald statistic: Test used for the significance of the logistic coefficient. Its interpretation is like the *F* or *t* values used for the significance testing of regression coefficients.

Wilk's or U statistics: Wilk's for each dependent variable is the ratio of the within-group sum of squares to the total sum of squares

10.12 SELF-ASSESSMENT QUESTIONS

1. What is discriminant analysis? Discuss the applications of discriminant analysis in marketing.
2. What is logistic regression? Discuss the applications of logistics regression in marketing.
3. "Discriminant analysis is used to predict the group membership for a new data point." Comment on this statement.
4. In what circumstances, as a market researcher, would you prefer discriminate analysis over logistic regression? Explain.
5. How does logistic regression handle the relationship between dependent and independent variables? Discuss.
6. Write short notes on the assumptions of logistic regression analysis.
7. The following discriminant function was developed to classify salespersons into the categories of successful and unsuccessful salespersons:

$$Z = 0.53 X_1 + 2.1 X_2 + 1.5 X_3$$

Where, X_1 = No. of sales call made by salesperson

X_2 = No. of customers developed by salesperson

X_3 = No. of units sold by salesperson

The following decision rule was developed.

If $Z \geq 10$, classify the salesperson as successful.

If $Z < 10$, classify the salesperson as unsuccessful.

Salespersons A and B were considered for promotion on the basis of being classified as successful or unsuccessful.

Only the successful salesperson would be promoted. The relevant data on A and B is given below. Whom will you promote?

	A	B
X_1	10	11
X_2	2	1.5
X_3	1	0.5

10.13 PROJECT QUESTION

During festival season, a retailer wants to reward its loyal customers based on purchases made in the last five years. Assuming three independent variables: frequency of purchase, average purchase amount, and number of years a person has been the customer (from 1 to 5). Collect data and find the loyal and disloyal customers to help the retailer reward them accordingly. How many minimum data sets should you have? Which method will you use to identify such groups? Justify your answer.

10.14 FURTHER READINGS

1. 'Discriminant Analysis in SPSS' Prof. Jogendra Kumar Nayak, NPTEL COURSE (You-tube tutorial)
2. 'Logistics Regression Analysis' Prof. Jogendra Kumar Nayak, NPTEL COURSE (You-tube tutorial) Education Limited.
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UNIT 11 FACTOR ANALYSIS AND CLUSTER ANALYSIS

Learning Outcomes

After going through this unit, you should be able to:

- appreciate the application areas of factor analysis and cluster analysis in the context of marketing.
- explain how factor analysis and cluster analysis are computed.
- interpret the findings of factor analysis and cluster analysis.
- differentiate between logistic regression and discriminant analysis.
- demonstrate the use of SPSS for factor analysis and cluster analysis.
- apply the concept learned to real-world marketing decisions.

Structure

- 11.1 Introduction
- 11.2 Factor Analysis
- 11.3 A Numerical Example of Factor Analysis
- 11.4 Use of SPSS for Factor Analysis
- 11.5 Applications of Factor Analysis
- 11.6 Limitations of Factor Analysis
- 11.7 Cluster Analysis
- 11.8 Inputs for Cluster Analysis
- 11.9 Different Approaches to Cluster Analysis
- 11.10 Factors to be Considered While Using Cluster Analysis Technique
- 11.11 Use of SPSS for Cluster Analysis
- 11.12 Applications of Cluster Analysis
- 11.13 Summary
- 11.14 Key Words
- 11.15 Self-Assessment Questions
- 11.16 Project Question
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11.1 INTRODUCTION

In the marketing domain, when collecting the data through survey methods, the number of items is generally very large. Many of the items represent the same concepts or constructs. Analyzing each of the items and their

causal relationships is extremely time-consuming and may not give the desired meaningful insights to aid in the decision-making process. Thus, before doing any statistical modelling, there is a need to reduce the data into meaningful variables. For this, marketers use factor analysis. Factor analysis provides an approach that reduces a set of variables into one or more underlying variables. The technique groups together those variables that seem to belong together and simultaneously supplies the weighing scheme. For example, one may be interested in the identification of factors that determine the company's image.

Market segmentation is another very important decision-making task for a marketing manager. One of the crucial tasks of a marketing manager is to select a target market for a given product or service and formulate the marketing mix strategy specific to that target market. Cluster analysis is a very powerful technique for segmenting the market. It is an exploratory technique used to group individuals or objects into clusters so that objects in the same cluster are more similar to one another than they are to objects in other clusters. The objective is to maximize the homogeneity of objects within the clusters while also maximizing the heterogeneity between the clusters. In most of the analytical projects in marketing, cluster analysis is often carried out before predictive or other statistical or analytical modelling.

In this unit, we will discuss the concept, methodology, applications, benefits, and limitations of factor analysis and cluster analysis.

11.2 FACTOR ANALYSIS

Factor analysis is the generic name given to a class of techniques whose purpose is data reduction and summarization. Very often, market researchers are overwhelmed by the plethora of data. Factor analysis comes to their rescue by reducing the number of variables. Factor analysis does not entail partitioning the data matrix into criterion and predictor subsets; rather, interest is centred on relationships involving the whole set of variables. In factor analysis:

- 1) The analyst is interested in examining the “strength” of the overall association among variables in the sense that he would like to account for this association in terms of a smaller set of linear composites of the original variables that preserve most of the information in the full data set. Often, his interest will emphasise the description of the data rather than statistical inference.
- 2) No attempt is made to divide the variables into criterion and prediction sets.
- 3) The models are primarily based on linear relationships.

Factor analysis is a “search” technique. Typically, the researcher-decision maker lacks a clear priori structure for the number of factors to be identified. Cut-off points with respect to stopping rules for the analysis are often determined ad hoc as the output becomes available. Even where the procedures and rules are stipulated in advance, the results are more descriptive than inferential.

Factor analysis provides the tools for marketers to analyse the structure of the interrelationships (correlations) among a large number of variables (e.g., consumer satisfaction, perceived service quality) by defining sets of variables that are highly interrelated, known as factors. These groups of variables (factors), which are by definition highly inter-correlated, are assumed to represent dimensions within the data. The dual objectives of factor analysis are as follows:

- simplifies the data by reducing a large number of variables to a small set of variables with minimal loss of information.
- analyses the interdependence of interrelationships among a total set of variables.

Assumptions of Factor Analysis

- A basic assumption of factor analysis is that some underlying structure does exist in the set of selected variables.
- The sample must be homogeneous with respect to the underlying factor structure.
- If the sample is heterogeneous with regard to at least one subset of the variables, then the first factors will represent those variables that are more homogeneous across the entire sample
- The variables must be sufficiently intercorrelated to produce representative factors.

11.3 A NUMERICAL EXAMPLE OF FACTOR ANALYSIS

The term “factor analysis” embraces a variety of techniques. Our discussion focuses on one procedure: principal component analysis and the factors derived from the analysis are expressed as linear equations. These linear equations are of the form

$$F_i = a_{i1} x_1 + a_{i2} x_2 + a_{i3} x_3 + \dots + a_{im} x_m$$

The i factors are derived, and each variable appears in each equation. The coefficients indicate the importance of each variable with respect to a particular factor. Co-efficient of zero indicating the variable is of no significance for the factor. In principal component analysis, the factors are derived sequentially, using criteria of the maximum reduction in variance and non-correlation among factors.

Let us go to a specific example to explain factor analysis and its output.

Suppose a manufacturer of fabrication units is interested in identifying the determinants of a successful salesperson. The manufacturer has on file information on 14 salespersons, as shown in Table 11.1. He is wondering whether he could reduce these seven variables to two or three factors for a meaningful appreciation of the problem.

Table 11.1: Data Matrix for Factor Analysis of seven variables (14 salespeople)

Sales person	Height (X ₁)	Weight (X ₂)	Educations (X ₃)	Age (X ₄)	No. of Children (X ₅)	Size of household (X ₆)	IQ (X ₇)
1	67	155	12	27	0	2	102
2	69	175	11	35	3	6	92
3	71	170	14	32	1	3	111
4	70	160	16	25	0	1	115
5	72	180	12	36	2	4	108
6	69	170	11	41	3	5	90
7	74	195	13	30	1	2	114
8	68	160	16	32	1	3	118
9	70	175	12	45	4	6	121
10	71	180	13	24	0	2	92
11	66	145	10	39	2	4	100
12	75	210	16	26	0	1	109
13	70	160	12	31	0	3	102
14	71	175	13	43	3	5	112

Can we now reduce the seven variables into three factors? Intuition might suggest the presence of three primary factors: a maturity factor revealed in the age of household members or children, physical size as shown by height and weight, and intelligence or training as revealed by education and IQ.

The SAS programme has analysed the salespeople's data. This program accepts data in the original units, automatically transforming them into standard scores. The three factors derived from the salesperson data by a principal component analysis (SAS program) are presented in Table 11.2.

Table 11.2 Three-factor results with seven variables

Variable	Sales people characteristics			Communality
	I	II	III	
Height	0.59038	0.72170	-0.30331	0.96140
Weight	0.45256	-0.75932	-0.44273	0.97738
Education	0.80252	0.18513	0.42631	0.86006
Age	-0.86689	0.41116	0.18733	0.95564
No. of children	-0.84930	0.49247	0.05883	0.96730
Size of household	-0.92582	0.30007	-0.01953	0.94756
IQ	0.28761	0.46696	0.80524	0.94918
Sum of squares	3.61007	1.85136	1.15709	
Variance summarised	0.51572	0.26448	0.16530	0.94550

Factor Loading: The coefficients in the factor equations are called “factor loadings” They appear above in each factor column, corresponding to each variable. The equations are:

$$F_1 = 0.59038x_1 + 0.45256x_2 + 0.80252x_3 - 0.86689x_4 - 0.84930x_5 - 0.92582x_6 + 0.28761x_7$$

$$F_2 = 0.72170x_1 + 0.75932x_2 + 0.18513x_3 + 0.41116x_4 + 0.49247x_5 + 0.30007x_6 + 0.46696x_7$$

$$F_3 = -0.30331x_1 - 0.44273x_2 + 0.42631x_3 + 0.18733x_4 + 0.05883x_5 - 0.01953x_6 + 0.80524x_7$$

Variance summarised: Factor analysis employs the criterion of maximum reduction of variance—the variance found in the initial set of variables. Each factor contributes to the reduction. In our example, factor I accounts for 51.6 percent of the total variance. Factor II for 26.4 percent and Factor

III for 16.5 percent Together, the three factors “explain” almost 95 percent of the variance.

Communality: In the ideal solution, the factors derived will explain 100 percent of the variance in each of the original variables; “communality” measures the percentage of the variance in the original variables that is captured by the combination of factors in the solution. Thus, a communality is computed for each of the original variables. Each variable’s communality might be thought of as showing the extent to which it is revealed by the system of factors. In our example, the communality is over 85 percent for every variable. Thus, the three factors seem to capture the underlying dimensions involved in these variables.

There is yet another analysis called varimax rotation that we will discuss in the context of a factor analysis example using SPSS. However, it should be mentioned here that there are two important subjective issues that should be properly resolved before employing the factor analysis model. They are:

- 1) How many factors should be employed in attempting to reduce the data? What criteria should be used in establishing that number?
- 2) The labelling of the factors is purely intuitive and subjective.

11.4 USE OF SPSS FOR FACTOR ANALYSIS

Let us assume that the marketing manager at the automobile service centre in Bangalore wants to measure the quality of after-sales service provided at the center. For this purpose, he created a structured questionnaire to identify the factors that influence service quality and its impact on customer satisfaction. A sample of data from 250 car users was collected. The questionnaire consists of 21 questions; responses to each question were measured on a 5-point Likert scale from strongly agree (coded as 5) to strongly disagree (coded as 1). The coded data was then subjected to factor analysis; the following are the steps followed in SPSS:

Step 1: Objectives of factor analysis

As mentioned above, the objective of the study was to measure the impact of service quality on customer satisfaction. However, the prime objective of the factor analysis is to summarize and reduce the data. Coded data obtained from the survey is subjected to exploratory factor analysis in order to understand and examine the inter-relationships of 21 variables, using SPSS.

Step 2: Selecting a factor method

The below steps need to be followed for selecting a factor method for extracting the factors.

1. In the SPSS select: Analyse → Dimension Reduction → Factor
2. Select the variables from the left-hand side dialog box and transfer them to the box labelled ‘Variables’, in this case all the 21 coded items were transferred.
3. Click on the ‘Descriptives’ button which brings up a dialog box. In the Statistics section, make sure that ‘Initial Solution’ is ticked. In the section marked Correlation Matrix, select the

options Coefficients and KMO and Bartlett's test of sphericity. Click on Continue.

- Click the 'Extraction' to specify the methods and number of factors to be extracted. In this PCA is the method adopted to perform Factor Analysis.

Extraction → Methods → PCA(Principal Component Analysis → → Continue

- Select options and check the coefficient display format and restrict the 'absolute value' to be 0.5 for obtaining factors with loadings greater than 0.5.

Click 'OK' to run the SPSS and to obtain the output file.

Interpretation of the output results from SPSS

Based on various stages described in the previous section, the results from the output file generated are to be checked and analysed. It can be understood from the following points:

- 1. Check the KMO and significance of Bartlett's test:** to decide whether the data available can be used for Factor Analysis.

Table 11.1.2: KMO and Bartlett's test output

KMO and Bartlett's Test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.893
Bartlett's Test of Sphericity	Approx. Chi-Square	3460.667
	df	210
	Sig.	.000

Select the 'KMO and Bartlett's Test' from the left side output file to interpret the sample adequacy and to understand the correlation among the variables. The above Table 11.1.2 shows the KMO and Bartlett's Test results for this case.

Check the KMO:

KMO value for this case is 0.893. As a rule of thumb KMO value between 0.8 and 1 indicates that the sampling is adequate. Therefore, the sample data is adequate to run the factor analysis.

Check the Bartlett's test

A significant result of Bartlett's test of sphericity (Sig. < 0.05) for this case as shown in Table 11.1.2 indicates the variables do relate to one another enough to run a meaningful EFA.

Since the above KMO and Bartlett's test are significant, the number of factors to be extracted are determined by the following step.

- 2. Determination of number of factors to be extracted:**

Check scree plot

Select the number of factors with the help of scree plot. For the chosen

case the below Figure 11.1.2 Scree plot suggests that four factor solution contributes the larger part of the data since at the component number five, the curve first begins to straighten out and with Eigen value going below 1.

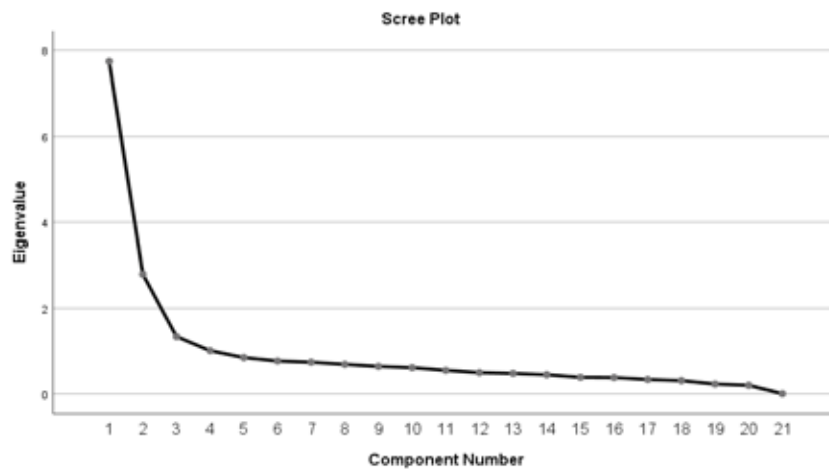


Figure 11.1.2: Scree plot

The Eigen values, variance and cumulative variance table needs to be checked from the 'Total Variance Explained Table'. For instance, for the chosen case the total variance of the four factors extracted in this case is 61 % as shown in above Table 11.1.3.

Table 11.1.3: Total Variance for the extracted factors

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	7.741	36.860	36.860	7.741	36.860	36.860	3.956	18.839	18.839
2	2.798	13.324	50.184	2.798	13.324	50.184	3.944	18.782	37.621
3	1.340	6.380	56.564	1.340	6.380	56.564	2.893	13.775	51.396
4	1.010	4.809	61.373	1.010	4.809	61.373	2.095	9.977	61.373
5	.849	4.042	65.415						
6	.768	3.658	69.073						
7	.739	3.521	72.593						
8	.689	3.280	75.874						
9	.642	3.058	78.932						
10	.612	2.916	81.848						
11	.550	2.617	84.465						
12	.496	2.361	86.826						
13	.477	2.273	89.099						
14	.446	2.126	91.225						
15	.386	1.840	93.065						
16	.380	1.810	94.874						
17	.335	1.597	96.471						
18	.309	1.470	97.941						
19	.229	1.091	99.032						
20	.200	.950	99.982						

21	.004	.018	100.000					
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Extraction Method: Principal Component Analysis.

Step 3: Selecting the Factor Rotation

To avoid variables clubbed in one factor, factor rotation is employed.

After interpretation of the output file and finalising the number of factors to be extracted, go back to the SPSS window to select the factor rotation through the steps below.

Analyse → Dimension Reduction → Factor

Click on 'Rotation' in the lower part of the main dialogue box and select the type of rotation under the 'Method'. There are five factor rotation methods provided by SPSS as mentioned in the above session. However, Varimax is the most common method used for factor rotation.

Factor rotation → Method → Varimax → Continue

Click 'OK' to run the SPSS and to obtain the output file.

Step 4: Interpretation the Rotated factor matrix:

Check the Rotated component matrix and communalities

- Select the Rotated component matrix and communalities tables shown in the left side of the output file
- Based on the factor loadings, the factors are extracted as per the Rotated Component matrix. If there is no problem of cross loadings and if the factor loadings are significant (i.e., if the loadings are > 0.5) the factors and factor loadings extracted are finalised.
- Along with the factor loadings, the communality of each variables are also checked to understand how much of the extracted factors are explaining the variables in terms of percentage.
- Once the factors extracted are significant, they are named and defined based on the traits of the variables fallen in each factor.
- If the obtained factor loadings are insignificant (i.e., if the loadings are < 0.5) and if the extracted factors are unable to be interpreted, step 2 is repeated with deletion of certain variables.

Step 5: Factor Reconfirmation:

This is a decision-making step required before proceeding to the final validity analysis. If there are any variables deleted in the factor extraction process or a need to change the number of factors or the type of rotation due to poor distribution of factor loadings, step 4 is repeated and the same

process is executed with a change in the factor numbers; if not, then step 6 is carried out.

Step 6: Reliability analysis:

- The reliability of the inter-item correlation is measured by Cronbach's alpha using the following step: feeding the variables categorized in each factor.
- Analyse → Scale → Reliability Analysis
- Compute for all the variables under one factor

Press "continue" to check and obtain the Cronbach's alpha. Cronbach's alpha has a generally accepted lower limit of .70, though it may drop to .60 in exploratory research.

The summated scales of the variables under each factor are further performed to develop the model.

Step 7: Reporting and interpreting the results

Based on the factor loadings, four factors were extracted as per the rotated component matrix. The identified factors are labelled as responsiveness, dependability, and tangibility, with the fourth factor combining assurance and empathy dimensions.

Factor 1: Responsiveness

Items that fall under Factor 1 are: the customers feel that the automobile service centre informed them exactly when the services were going to be executed, the service rendered was fast and efficient, additional services and their cost, and the centre responds to requests promptly, hence labelled as 'Responsiveness' dimension of service quality.

Factor 2: Reliability

Items that came under Factor 2: the customers feel that the automobile service centres' work is done as per the promised time, vehicles are diagnosed effectively, original parts were replaced for repair during service, customer profile records are maintained correctly, and personal information about customers is safe. Therefore, this factor was termed as the "Reliability" dimension of service quality.

Factor 3: Tangibility

Factor 3 consists of items like: the customers feel that the service centres use the latest tools and equipment, the facilities are visually attractive, and the employees are well dressed. This factor was named as the 'Tangibility' dimension of service quality.

Factor 4: Assurance and Empathy dimensions

Items under Factor 4 includes: Customers believe the employees are trustworthy; they feel safe during transactions; the employees are humble and polite; the employees are able to perform their job effectively; they receive individual attention; they express sympathy and reassurance for problems; and they suggest the best solution. This factor combined two

dimensions of service quality and was hence named as ‘Assurance and Empathy’.

The extracted factors can be further subject to regression analysis to measure the impact of these service quality factors on customer satisfaction.

11.5 APPLICATIONS OF FACTOR ANALYSIS

In marketing, factor analysis could be used for any of the following purpose:

Explore the data patterns:

Factor analysis can be done to explore the data and reveal patterns among the interrelationships of the items. It can bring out the hidden or latent dimensions relevant to the study.

Data Reduction:

Factor analysis is extremely useful to reduce a large number of items or variables to a smaller number of factors for statistical modelling purposes, which is otherwise difficult to model. As a result, factor analysis can be combined with other multivariate analysis techniques, such as cluster analysis, conjoint analysis, SEM (structure equation modeling), and so on, to aid in the development of the latent variables model.

Confirms Hypothesis of Factor structure:

Factor analysis can be used to test whether a set of items designed to measure certain variables in fact reveal the hypothesized factor structure (i.e., whether the underlying latent factor truly causes the variance in the observed variables and how certain we can be about it).

Theory Testing

Factor analysis can be used to test an a priori hypothesis about the relationship among a set of observed variables; for this, generally confirmatory factor analysis is used.

11.6 LIMITATION OF FACTOR ANALYSIS

1. When data changes due to changes in the sample, the data collection process, or the various types of measurement errors, the results of the factor analysis may also change. The results of factor analysis are therefore less than perfectly dependable.
2. During the analysis, sometimes more relevant factors may be left out due to cross-factor loadings or poor factor loadings on those factors.
3. The naming of the factors can be difficult as multiple attributes can be highly correlated.
4. If the observed variables are completely unrelated, the factor analysis is unable to produce meaningful pattern.

Activity 1

Speak with the marketing manager of your own company or any other

organisation you are familiar with and discuss the application areas for factor analysis.

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11.7 CLUSTER ANALYSIS

Cluster analysis is a technique that is used to segment a market. The objective is to find a group of customers in the market that are homogeneous, i.e., they share some characteristics so that they can be classified into one group. The cluster or group so identified should be large enough so that the company can develop it profitably, as the ultimate objective of a company is to serve the customer and earn profits. The group of customers that the company hopes to serve should be large enough for the company to make it a financially viable proposition. This is also true for the customer, who will not pay more than a certain price for a particular product (price, of course, is a function of product positioning, cost of production, and so on).

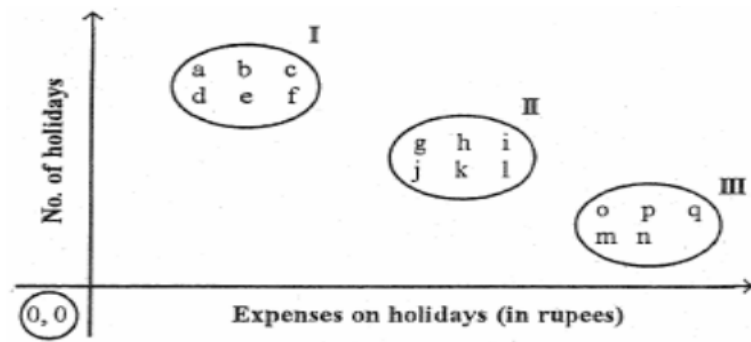
As an example, let us consider the watch industry. There could be many ways in which the watch industry could be segmented, which are as follows:

- a) Gender (Male/Female)
- b) Technology (Digital/Analog)
- c) Design Features
- d) Occasion of Use (Formal/Casual/Party)
- e) Price (Low/Medium/High/Jewellery)

Some of the above segmentation factors are demographic (price, gender), whereas others are psychographic (occasion of use).

This, therefore, presents a problem to the market researcher or company as to how to identify combinations of factors that can be used to segment the market. It is not always possible to segment a market on the basis of one single factor. Thus, a combination of factors must be used to segment the market. And this is where the Cluster Analysis technique specifically deals with how objects (people, places, and products) should be assigned to groups so that there should be similarity within the groups and as much difference between the groups as possible.

Let us look at an example.



In the above example, cluster analysis has identified three distinct clusters (groups) as follows:

Group I: “People who spend less money but have more vacation days” (adventure seekers, students)

Group II: People who spend an average amount of money and have an average number of vacation days (first- class train travel, 3-star hotels, etc.)

Group III: People who have very few vacation days but spend heavily during these few days (five-star luxury hotels, air travel, foreign trips, etc.).

These groupings would help a travel and tourism company in the following ways:

- a) The travel and tourism company can study the various clusters of customers that have emerged and decide which customer group it would like to serve, depending upon the company’s own resources and capabilities, and the volume of business in each cluster group that will generate sufficient business for the company’s own survival.
- b) Once a cluster has been selected by the company, it can tailor various tour programmes for its cluster of customers.
- c) The company can keep a profile of its customers and identify any new emerging clusters.
- d) The company can decide to serve either only one group of customers (niche marketing) or all groups of customers at the same time, but with a range of travel and tour programmes.

11.8 INPUTS FOR CLUSTER ANALYSIS

Since cluster analysis is used for the purpose of segmenting the market, one has to go through the following procedure on a very broad level:

- a) company identifies its own product capabilities, financial strength, and distribution network (who am I?)
- b) The company identifies the basis on which the customers are to be clustered, e.g., one-factor segmentation (age, income, gender, geographic location), two-factor basis of segmentation of the market place (age and income, quality and price, nutrition value and price), or multi-factor basis of segmentation of the market place (who am I going to serve?).

The company or market research agency has to decide which of the attributes

are to be employed because they would generate natural groupings in the market. The choice of variables that are to be used to cluster the objects is one of the most critical decisions that analysis makes. The best way is to select those variables that make conceptual sense rather than using any or all variables. Ideally, the variables should be selected within the framework of an explicitly stated theory that is used to support the classification.

c) Respondents who participate in the research are given a questionnaire to complete. The questionnaire is typically an interval-scaled (5-point, 7-point) questionnaire.

The cluster analysis program that is run on the computer identifies the different groups (clusters) or respondent groups such that the members of any one group (cluster) are similar to each other but different from the respondents in the other groups. The computer program for cluster analysis searches through the answers that the respondents give to the various questions in the questionnaire and groups together those respondents who have given identical or almost identical answers to a certain combination of the questions. These respondents with similar responses are grouped together. The procedure is then continued till more “clusters” are formed.

11.9 DIFFERENT APPROACHES TO CLUSTER ANALYSIS

There are various approaches adopted for cluster analysis.

One extreme approach is to cluster respondents in terms of all available information about the respondent; although this is very thorough, it may lead to difficulty in interpretation and marked differences between groups, i.e., the clusters or groups of respondents that are produced may be a hotch-potch rather than a clearly marked description of the target population.

Another extreme approach involves grouping respondents only based on data that directly represents the specific consumer need. As an example, consider the product’s quality and price. This has the advantage of clarity and accuracy, but may be quite limited in describing the target population.

The way one approaches cluster analysis depends upon the objective of the study. Suppose the objective of the company is the physical development of an existing product, then the consumers’ needs regarding the product are the most suitable variables for forming clusters.

Suppose an advertisement for a product is to be developed, then the attitude of the consumer towards the product gains more importance, as there are groups of people who have a positive, negative, or neutral attitude towards the product or company. And, depending on the product or communication need, the developed advertisement would target specific clusters of consumers.

Suppose a company wishes to develop a new product concept or make product changes in the existing product offering; then, the attitude of the consumer towards the product, company, or product category gains more importance over other attributes when developing clusters.

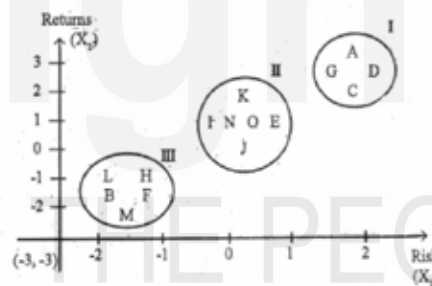
Therefore, depending upon the objective of the study, various characteristics can be used to describe the consumer, including his attitudes, needs, and demographic characteristics.

A general principle followed in cluster analysis is to omit behavioural and brand image data. One should concentrate on characteristics describing the consumer and the circumstances, e.g., specific needs, demographic variables, general attitude, and select any variables that would interact with each other to influence consumer behaviour. Once attributes are identified on the basis of which one is trying to form natural groupings of consumers, one has to scale the attributes so that one has a basis to measure the similarity or differences between or within groups.

Let us look at an example.

Suppose one wants to understand the consumers who operate in the finance market. With regard to the financial market, there are a few salient aspects like risk, returns, and liquidity. To form consumer clusters, any two attributes could be used at the same time, or more than two attributes could be used.

There are 15 consumers, whose responses have been analyzed with respect to risk and return in the financial market, as shown in the following figure:



Cluster I is a group of consumers who believe in high-risk, high-return instruments, e.g., equity.

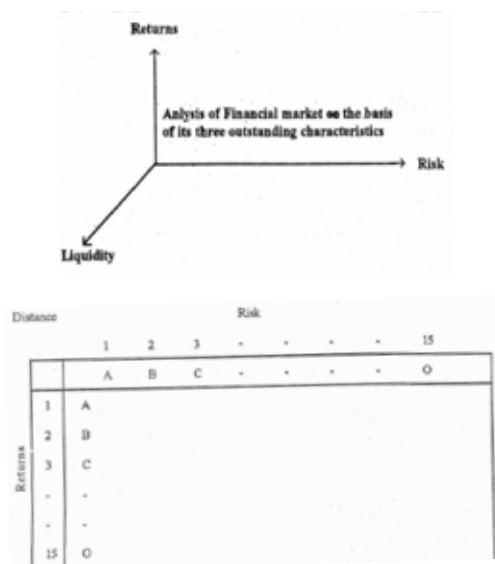
Cluster II is a group of consumers who believe in average risk and average return (mutual funds, company fixed-deposits).

Cluster III is a group of consumers who believe in low-risk, low-return investments (savings banks, fixed deposits).

An obvious way to measure similarity is to calculate the Euclidean distance between points. In the above figure, the distance between points A and C would be calculated as follows:

$$(A, C) = \sqrt{(X_{C_1} - X_{A_1})^2 + (X_{C_2} - X_{A_2})^2}$$

And so on, one could calculate distance between various points.



The distance between any two points is an inverse measure of similarity. Thus, the larger the distance between two points, the greater the difference or dissimilarity between the two points. The closer two points are to each other, the more similar they are. In addition, the number of distance calculations increases as the number of respondents to be classified into clusters increases. Hence, for large sample sizes of respondents, computers would have to be used. This is what we will see when we use SPSS for cluster analysis.

After calculating distances, one has to decide how to form clusters. There are various methods of doing it, as follows:

i) **Nodal Method**

In this method, one selects a point or points that will serve as nodes for forming clusters or focal points. The rest of the points are then allocated to each cluster on the basis of their distance from the focal point. The above method is demonstrated as follows:

- Choose as nodes those objects (points) that have the greatest distance between them.
- Consider these two objects as focal nodes and allocate all the remaining objects to one cluster or the other based on their distance from the polar nodes.
- Split the resulting two clusters in the same way. Continue this process until the collection of points is split into its original members.

In the above example, the points “D” and “M” have the largest distance between them, and they will be considered nodes for the two clusters. Each of the remaining objects would then be allocated to each cluster on the basis of its shortest distance to either D or M. Thus, the original cluster would be:

Group1: DACEGK

Group 2: MBFHIJLNO

In Group 1, the least similar (greatest distance) are D and K. They would thus be considered new nodes, and the items in Group 1 would be allocated to each of the new clusters on the basis of their distance from these new nodes. The new groups would be:

Group 1A: DACG

Group 1B: KE

Similarly, Group 2 would be divided (using B and J as new nodes to yield)

Group 2A: BFHLM

Group 2B: JINO

Thus, at every stage, one has to find out if the sub-groups that have emerged can be further combined on the basis of some measure of similarity between the objects.

ii) Factor Analysis Method

Another way of developing clusters is by using the method called “Q-factor analysis.” By this method, one can determine which objects logically belong together. This is also called “inverse factor analysis.”

iii) Linkage Method

There are three methods in this: the single linkage method, the complete linkage method, and the average linkage method.

Using the previous example of analyzing financial markets, the method starts by finding the points with the shortest Euclidean distance. Therefore, initially, the following groups emerge:

C-D - Group 1

A-G - Group 2

N-O - Group 3

F-M - Group 4

The remaining objects are then assigned to the above clusters based on the cut-off distance. The final clusters that emerge are:

Group 1 - AGDC

Group 2 - NOKELT

Group 3 - MFLBH

The complete linkage option starts out in the same manner by clustering the two closest points. However, the distance between two clusters is the longest distance from a point in the first cluster to a point in the second cluster.

In the average linkage option, the distance between the two clusters is the average distance from points in the first cluster to points in the second cluster.

Activity 2

Study the products and the consumers of your company (or the company you are familiar with) and try to develop clusters of customers depending on the different requirements of the customers.

.....

.....

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.....

11.10 FACTORS TO BE CONSIDERED WHILE USING CLUSTER ANALYSIS TECHNIQUE

The following aspects should be kept in mind while using the cluster analysis method:

- a). A number of clusters may emerge after the analysis. However, there is a limit to the number of clusters that a company can consider due to- limitation of market potential within a cluster, and the distinction between clusters is not clearly defined.
- b). Cluster analysis provides a way of segmenting the market, but these segments are not watertight compartments. Products that are developed for a particular segment may attract people from other segments too.
- c). The characteristics of a cluster may change over time as the consumers' economic status, education, lifestyle, etc. change. Then the company has to take another look at the market.
- d). The clusters that have been identified are used to develop further marketing strategies in areas like product developments, advertising research, distribution strategies, pricing strategies, etc.
- e). The most important assumption in cluster analysis is that the basic measure of similarity on which clustering is based is a valid measure of the similarity between objects. A second major assumption is that clustering objects is theoretically justified and supported. As, with other multivariate techniques, there should be theory and logic underlying the cluster analysis.
- f). The major limitation of cluster analysis is the difficulty in evaluating the quality of the clusters. It is very difficult to know exactly which clusters are very similar and which objects are dissimilar, and it is also difficult to select clustering criteria.

Limitations of Cluster Analysis:

- Different clustering methods usually give different results due to the differences in the criteria for merging clusters. Therefore, it becomes a challenging task for the marketing data analyst to choose the best method to meet the objectives.
- The analysis is unstable when the cases are dropped because clusters

are formed by measuring the inter-object (variable) similarity of one cluster to another. Dropping one case can drastically affect the course in which the analysis progresses.

11.11 USE OF SPSS FOR CLUSTER ANALYSIS

Let us assume that a market research institute in Germany wants to measure the consumers' attitudes and stated consumption habits. In order to segment the customer based on their consumption habits, they created a structured questionnaire with 68 statements on a five-point Likert scale of 1 (completely disagree) to 5 (completely agree). The data from the German household panel data, with a sample size of 10,001 households, was recorded annually for the years 2005–2008 (Müller and Hamm, 2013). The coded data was then subjected to cluster analysis in SPSS. The following are the steps followed in SPSS:

These 68 statements were subjected to exploratory factor analysis for the year 2005; later, the developed model was also subjected to confirmatory factor analysis for the years 2006–2008 with the same specifications as 2005.

40 statements mentioned below were extracted with factor loadings >0.5 , and 12 factors were extracted. The extracted 12 factors were later named based on their traits and characteristics, which are used as the input variables for cluster analysis.

1. In my household, I pay attention to a light and balanced diet
2. I pay strict attention to eating little fat
3. I pay strict attention to low salt
4. I pay strict attention to the sugar content of foods
5. I watch what I eat and drink, because I have to consider my health
6. The topic of food cholesterol plays a role when shopping
7. I avoid eating anything unhealthy
8. I would like to see a wider range of organic products in stores
9. I am willing to spend more money for organic products
10. When grocery shopping I prefer organic food
11. I would like more information about organic products
12. I often have new products earlier than my friends
13. I already buy many items that other housewives do not know yet
14. I like to try out new products
15. I'm always on the lookout for new products that meet my needs more
16. If I have a choice, I definitely buy food from Germany
17. For me food products from Germany are qualitatively the best

18. I pay attention to purchasing food products from my region
19. It's fun to try foreign specialties
20. I'm enthusiastic about specialties from other countries
21. I like to cook fancy foods and dishes
22. The simpler the cooking the better I like it
23. I prefer to cook dishes that don't take too long
24. Food products from well-known brands are better than products with unknown names
25. Branded products are better than products with unknown names
26. I have no real confidence in food without a brand name
27. I often go eat something on the way
28. I often eat fast food (fast food, takeaways, pizza delivery service)
29. I eat mostly snacks
30. New products are often more expensive than the old, but not better
31. I greatly distrust advertising statements
32. To buy brand new products often is a failure
33. I think that most products that continually enter the market are unnecessary
34. I regularly use vitamin and mineral supplements to keep myself physically fit
35. I like to buy food/drinks that contain added minerals and vitamins (ACE, calcium, etc.)
36. Multivitamin juices are an important addition to the diet
37. I love to eat home cooking
38. I prefer to stick to established recipes when cooking
39. I would not want to miss personal shopping service
40. I love the atmosphere of small shops and specialty stores

Input data for Cluster Analysis :

Factor 1: Health consciousness

Factor 2: Organic product

Factor 3: New products

Factor 4: Regional origin

Factor 5: Foreign specialties

Factor 6: Fast and simple cooking

Factor 7: Brands

Factor 8: Snacks

Factor 9: Scepticism

Factor 10: Food supplements

Factor 11: Established and traditional dishes

Factor 12: Corner shops

Step 1: Objectives of Cluster analysis

The first stage in applying cluster analysis involves determining the objectives to be achieved. Once the objectives have been agreed upon, the Market Research team must select the clustering variables to be used in the clustering process. In this case, the primary objective is to segment the customers based on their attitudes and consumption habits based on the 12 factors extracted from EFA and CFA.

Step 2: Selecting Cluster Method

SPSS offers three methods for the cluster analysis:

- Two – Step Cluster
- Hierarchical Cluster
- K-means

The most common method used in segmenting the market is the Combination method i.e., using 'Hierarchical Cluster' to determine the number of cluster solution with the help of dendrogram or 'agglomeration schedule', a tabular column obtained in the analysis and extract the clusters using the method 'K-means'

The steps to be followed to perform Cluster Analysis in SPSS is documented below:

Determine the number of clusters by Hierarchical method

Go to analyse:

- Analyse → Classify → Hierarchical Cluster
- Insert all the variables to analyse from left to right side Variables window

The dialog box allows to specify the distance measure and the clustering method.

For interval data (Continuous data): most common method is Square Euclidian Distance.

For ordinal data (non-continuous data): common method is Chi-square

In this case since the data is Continuous, thus, Square Euclidian Distance is adopted.

Next, we have to choose the Cluster Method. Most common method is the Ward's method.

- Method → Cluster Method: select Ward's method

→ Measure: Interval: select Squared Euclidean Distance

In this case, the initial cluster solution was determined using the most common method called The Wards, where the similarity between two clusters is measured as the sum of squares within the clusters summed over all variables along with “Squared Euclidean distances” and all other default settings followed by the K-means analysis.

- Select ‘OK’ after the above settings

Finally, at the bottom of the dialog box is the option to standardize our data. Standardizing data is a good idea when they are non-metric variables. There are a number of ways in which data can be standardized but the most easily understood is to convert to a Z-score.

Step 3: Interpretation of Hierarchical Method result

The cluster solutions can be obtained graphically with Dendrogram, but as the sample size increases, the procedure becomes complex to interpret the clustered solutions. Therefore, agglomeration schedule is checked to easily determine the clustered solutions.

- Check the agglomeration schedule to determine the number of clusters.
- Go to coefficients column, check where the difference between the current and the preceding respondent’s coefficient varies drastically from last respondent to determine the number of clusters.
- Number of clusters = Number of respondents – coefficient where the value starts changing drastically (from last)

In this case, the clusters and the distance among each member clusters were examined, out of which 6- cluster solutions turned out to be the best.

Step 4: Determine the cluster solutions using K-means

Once the number of clusters to be formed are determined, the non-hierarchical clustering is performed to determine the cluster solutions. In SPSS, the non-hierarchical clustering routine is commonly referred to as K-means.

Go to analyse:

- Analyse → Classify → K-means Cluster
- Insert all the variables to analyse from left to right side Variables window
- Number of cluster → Input the number of clusters determined in the step 1 Hierarchical method to be analysed (here in this case it is 6 clusters)

Go to options and select the ANOVA Table, the ANOVA results are used to analyse if the the number of clusters determined is statistically significant or not.

- Select 'continue'
- Select 'OK' after the above settings

Step 5: Reporting and interpreting the results

Interpretation and profiling of clusters:

The rating of the statements per cluster for the 6-cluster solution are described in Figure 11.2.3, followed by profiling of the derived clusters' properties and characteristics.

Table 6

Average rating of the statements per factor in the clusters 2005 and changes in the confirmatory solution of 2008.

	C1		C2		C3		C4		C5		C6	
	Friends of foreign cuisine		Sceptical conservatives		Alternative gourmets		Health-conscious traditionalists		Fast food fans		Health-conscious modernists	
	2005	2008	2005	2008	2005	2008	2005	2008	2005	2008	2005	2008
Share of identical assignments ^a	51.6%		62.2%		54.4%		60.2%		54.8%		53.8%	
N factor	1467	1397	1877	1917	1609	1622	1707	1676	1446	1425	1681	1750
F1 health consciousness	2.5	-0.1	2.6	±0.0	3.1	-0.1	3.6	±0.0	2.5	±0.0	3.4	-0.1
F2 organic products	1.9	-0.2	1.9	-0.2	3.4	-0.1	3.1	-0.2	2.4	-0.1	3.2	-0.2
F3 new products	2.9	-0.1	2.0	-0.1	2.5	-0.1	2.4	-0.2	3.2	-0.1	3.4	-0.1
F4 regional origin	2.3	+0.2	2.9	+0.2	3.6	+0.2	3.9	+0.1	2.7	+0.2	3.3	+0.2
F5 foreign specialties	4.0	-0.1	2.4	-0.1	3.8	-0.1	2.6	-0.1	3.3	-0.1	3.8	-0.1
F6 fast and simple cooking	2.7	±0.0	3.7	-0.1	2.6	±0.0	3.3	-0.1	3.9	-0.1	2.9	±0.0
F7 brands	1.6	+0.1	1.7	+0.1	2.1	+0.1	2.8	+0.1	2.0	+0.2	2.5	+0.1
F8 snacks	1.6	-0.1	1.6	-0.1	1.4	±0.0	1.4	-0.1	2.9	±0.0	1.8	±0.0
F9 scepticism	3.4	±0.0	3.8	±0.0	3.8	±0.0	3.7	±0.0	3.3	±0.0	3.0	+0.1
F10 food supplements	2.2	-0.2	2.1	-0.2	1.9	-0.1	2.7	-0.2	2.6	-0.1	3.1	-0.2
F11 established and traditional dishes	2.6	+0.2	3.8	+0.1	2.8	±0.0	4.0	±0.0	3.3	±0.0	3.0	+0.1
F12 corner shops	3.0	±0.0	2.9	±0.0	3.8	±0.0	3.8	±0.0	3.1	±0.0	3.4	+0.1

^a Share of individuals, who were assigned to the same cluster in 2005 and 2008 as percentage of the cases of the respective cluster in 2005. Rating scale of the statements from 1 (totally disagree) to 5 (totally agree) $n = 9787$ (after elimination of 214 outliers).

Figure 11.2.3: Cluster Analysis Source: Müller and Hamm(2013)

Labelling and Interpretation of the Clusters:

Cluster 1 – Friends of foreign cuisine

The customers segmented in Cluster 1 are particularly enthusiastic about foreign specialties. Accordingly, they attach no importance to the regional food (authentic German food). For them, cooking does not have to be fast or simple, and they do not care about particular brands; they like to try new and unusual recipes.

Cluster 2 – Sceptical conservatives

In Cluster 2, it is observed that the segmented customers are quite sceptical about anything new but do not prefer small shops with personal service. They do not value specialties, new products, organic foods, or dietary supplements. For them, cooking does not necessarily have to be healthy, but fast and simple. They like to stick to tried-and-tested recipes.

Cluster 3 – Alternative gourmets

Cluster 3 highlights those customers who prefer organic products. The regional or German origin of the food is important to them, but they also like foreign specialties. Fast, simple food is not their favourite and nutritional supplements are equally unnecessary for them. They prefer shopping in smaller shops with personal service.

Cluster 4 – Health-conscious traditionalists

The traditional health-conscious customers are segmented in the fourth cluster and attach particular importance to regional origin and a healthy diet. For this, they rely on branded products. They appreciate home cooking and like to shop in smaller shops with personal service. They do not care for new products, foreign dishes, snacks, or fast food.

Cluster 5 – Fast food fans

The fast-food lovers segmented in cluster 5 like to eat snacks or fast food, and they appreciate quick and easy meals. They are receptive to new products. They are unconcerned about healthy nutrition or regional food origin.

Cluster 6 – Health-conscious modernists

The modern, health-conscious customer focuses on a healthy diet and likes to make use of new products as well as food supplements in Cluster 6. They are open-minded about novelties, organic products, and foreign specialties.

The results of the cluster analysis indicate that over time, consumers tend to move toward other clusters.

Testing internal stability:

The stability of the above clustered solution is tested by choosing random variables in the dataset. This is done once in descending order and once in ascending order. With each sorting, a K-Means analysis with K clusters is calculated. The agreement of the solutions can be measured in crosstabs using the Rand, the adjusted Rand Index, and Kappa. Kappa values > 0.74 show very good or excellent accordance; values from 0.60 to 0.74 show satisfactory or good accordance as per the rule of thumb.

The case highlights that the data analyst needs to design segment-specific marketing strategies that include opportunities for flexible adaptation so that changes in consumer preference, such as an increasing interest in regional food production or a reduction in the importance attached to food supplements, can be properly integrated.

11.12 APPLICATIONS OF CLUSTER ANALYSIS

Cluster analysis could be used for any of the following purposes in marketing:

Segmenting the market:

Market segmentation is one of the most important applications of cluster analysis. Marketers need to formulate marketing strategies and various marketing programs based on the unique requirements of their target segments. Cluster analysis helps them identify similar segments; for example, consumers may be clustered based on the benefits sought from the purchase of a product or any other common characteristic. Based on the sales and customer data, a multi-brand retailer in the fashion and lifestyle product category wants to segregate its customers into three income groups: high-income, medium-income, and low-income customers. Such segregation will help the retailers develop better marketing strategies. In such cases, cluster analysis is applied.

Understanding buyer behaviour:

Cluster analysis identifies homogenous groups of buyers and aids marketing managers in better understanding buyer behaviour. This is very important while framing marketing mix strategies to capture the target market more efficiently.

Identifying new product opportunities:

Clustering brands and products can help to categorise the competitive sets within the market.

Selecting test markets:

It would be easier to compare cities if they were organised into homogeneous clusters. The findings of one test market or city can be applied to another belonging to the same cluster; thus, it will aid the researcher in reducing the number of test markets required.

Data Aggregation:

By developing clusters of data, the large-size data is reduced into meaningful aggregates, thereby easing the analysis.

11.13 SUMMARY

In this unit, we have discussed two of the techniques of multivariate analysis, viz., factor analysis and cluster analysis. Factor analysis is a technique whose purpose is data reduction and summarization. It analyses the structure of the interrelationships among a large number of variables and reduces them into a small set of variables known as factors. Cluster analysis is mainly used for market segmentation. Some of its other applications include test market selection and the grouping of products within a product line. Various inputs required for cluster analysis have been discussed in the unit. Further applications to factor and cluster analysis have been explained. Finally, some of the important considerations that should be kept in mind while using factor and cluster analysis have been highlighted.

11.14 KEY WORDS

Agglomerative clustering: Two clusters that are most similar are combined to build a new aggregate cluster. The process is repeated until all objects are finally combined into a single cluster

Average linkage method: The inter object similarity of clusters is based on the average distance from all objects in one cluster to all objects in another.

Cluster centroid: Average value of the objects contained in the cluster on all the variables in the cluster variate.

Cluster seed: Initial value or starting point for a cluster.

Communality: Total amount of variance an original variable shares with all other variables included in the factor analysis. Mathematically, it is the sum of the squared factor loadings for a variable across the factors.

Composite linkage method: The inter object similarity of clusters is based on the maximum distance between objects in two clusters.

Cross loadings: When a variable is found to have more than one significant loading, it is termed as cross-loadings.

Dendrogram: Graphical representation (tree graph) of the results of a hierarchical procedure in which each object is arrayed on one axis, and the other axis portrays the steps in the hierarchical procedure.

Diverse clustering: begins with all objects in a single cluster, which is then divided at each step into two additional clusters that contain the most dissimilar objects

Eigen value: It represents the amount of variance accounted for a factor. Calculated as the column sum of squared loadings for a factor.

Factor analysis: It is an interdependent technique, which examines and explores the inter-relationship of large variables simultaneously, often termed as data reduction technique.

Hierarchical procedure: Stepwise clustering procedures involving a combination (or division) of the objects into clusters.

K-means: group of non-hierarchical clustering algorithms that work by partitioning observations into a user-specified number of clusters and then iteratively reassigning observations until some numeric goal related to cluster distinctiveness is met.

Non-hierarchical cluster analysis: Procedures that produce only a single cluster solution for a set of cluster seeds and a given number of clusters.

Principal Component Analysis (PCA): In this method, factors are selected one at a time such that each factor best fits the data.

Scree plot: A plot where the point at which the curve first begins to straighten out is considered to indicate the maximum number of factors to extract.

Single linkage method: The inter object similarity of clusters is based on the distance between the closest objects in two clusters.

Varimax: orthogonal factor rotation method that is focused on simplifying the column in a factor matrix.

Ward's method: The inter object similarity of clusters is based on the sum of squares between the two clusters summed over all variables.

11.15 SELF-ASSESSMENT QUESTIONS

1. What is factor analysis? What are its possible applications?
2. Comment: "Factor analysis provides the tools for marketers to analyse the structure of the interrelationships among a large number of variables."
3. What is cluster analysis? What are its possible applications?
4. What are the different approaches to cluster analysis? Discuss.
5. What factors are to be considered while using the cluster analysis

technique? Explain.

6. An airline marketing manager wants to segment his customer base. How can cluster analysis be used for this purpose?
7. In what circumstances, as a market researcher, would you prefer cluster analysis over factor analysis? Explain.

11.16 PROJECT QUESTION

You are a marketing analyst of a cosmetic company, and your General Manager wants you to cluster its present and potential accounts. How will you go ahead with this?

11.17 FURTHER READINGS



UNIT 12 CONJOINT ANALYSIS AND MULTIDIMENSIONAL SCALING

Learning Outcomes

After going through this unit, you should be able to:

- explain how conjoint analysis and Multidimensional scaling are computed.
- interpret the findings of conjoint analysis and Multidimensional scaling.
- appreciate the application areas of conjoint analysis and Multidimensional scaling in the context of marketing.
- differentiate between conjoint analysis and Multidimensional scaling.
- demonstrate the use of SPSS for conjoint analysis and Multidimensional scaling.
- apply the concept learned to real-world marketing decisions.

Structure

- 12.1 Introduction
- 12.2 What is Conjoint Analysis
- 12.3 Steps in The Application of Conjoint Analysis
- 12.4 Use of SPSS for Conjoint Analysis
- 12.5 Multidimensional Scaling (MDS)
- 12.6 Attribute Space - Objective Space and Perceived Space
- 12.7 Applications of Multidimensional Scaling
- 12.8 Use of SPSS For Multidimensional Scaling
- 12.9 Difference Between Multidimensional Scaling and Conjoint Analysis
- 12.10 Summary
- 12.11 Key Words
- 12.12 Self-Assessment Questions
- 12.13 Project Question
- 12.14 Further Readings

12.1 INTRODUCTION

In marketing, the need to understand consumer behaviour is paramount. Although consumer behaviour has many facets that have to be understood, the need to discover the attributes that the consumer considers important is quite common. Following an understanding of these attributes, marketers

attempt to comprehend consumers' perceptions of the products or brands in relation to these critical attributes. Many research and analytical techniques are available and often employed for these purposes. In this unit, we will discuss two multivariate techniques: conjoint analysis and multidimensional scaling (MDS).

When the marketer is interested in getting answers to several "What if?" kind of questions. What if I reduce one attribute? What if the competitor changes its brand position? What if the salience of one attribute changed through some consumer education efforts? etc. Most of these questions are answered by the conjoint analysis technique, which also guides the marketer in optimising his decisions. It helps him understand the trade-offs that consumers make between different attributes. In fact, it is with these trade-offs that the marketer finds his greatest opportunities to optimise his marketing efforts.

MDS is found to be superior to more traditional methods of attribute-wise data collection because of its lesser demands on data scaling. It can work with metrics as well as ranking data. This feature is also very consumer-friendly, as consumers, in most cases, compare products in terms of "better" or "worse" instead of assigning absolute values to products or their attributes. With the help of MDS techniques, marketers can understand the significant attributes of products or services and obtain the plots of competing products or brands in the perceptual space of consumers. These details go a long way toward helping marketers make product positioning decisions.

12.2 WHAT IS CONJOINT ANALYSIS

Conjoint analysis is essentially a data decomposition technique that attempts to plot the output data in the joint space of each attribute's importance. It seeks data from the consumers in the form of their overall response to the totality of products (or their descriptions), while the output is in terms of the scores that the consumer has implicitly assigned to each of the attributes and their levels.

The important thing to note is that the consumer is not asked to assign scores to different attributes separately. In fact, the consumer is presented with the stimulus in the form of the totality of the product, as in the case of MDS. However, there is one difference between MDS and conjoint analysis. In conjoint analysis, the stimuli are created by the researcher himself. while in the case of MDS, already existing products or brands are used.

Conjoint analysis derives the importance weights (referred to as "part-worth utilities") assigned by each consumer to respective levels of attributes in a way that is directly comparable. This feature of the technique allows for the determination of the trade-offs that consumers make in their minds. The relative importance of the attributes can also be derived from the output of "part-worth utilities." Thus, starting with very simple input data (just the ranking of some predesigned product alternatives), conjoint analysis provides the part-worth utilities for each of the product attribute levels for every consumer individually.

12.3 STEPS IN THE APPLICATION OF CONJOINT ANALYSIS

The main steps involved in the application of conjoint analysis are the following:

1. Identification of the key attributes of a given product from the perspective of the consumers
2. Assigning a set of discrete levels or a range of continuous values to each of the attributes
3. Designing the stimuli for the experiment using a fractional factorial design of experiment.
4. Physically designing the stimuli
5. Ranking or Rating Data Collection
6. Conjoint analysis and determination of part-worth utilities
7. Applying the output of conjoint analysis to different marketing decisions

We shall now describe each of these steps in some detail.

1. Identification of Salient Attributes

Unlike MDS or factor analysis, conjoint analysis requires the salient attributes to be given. These attributes can be selected based on the marketer's experience, in-depth interviewing, or focused group discussions. In some cases, factor analysis, or MDS, may also be used for labelling the salient factors. Only those attributes around which differences do exist among brands or through which future differentiations can be achieved should be selected here. For example, if it is not possible to differentiate products based on a specific attribute, the managerial value of that attribute falls dramatically.

2. Assignment of levels to selected attributes

The real products can be described as specific combinations of the attributes, where each attribute can take on different values. Sometimes the attribute can take only discrete values. Examples of such attributes may be the brand name, the colour of the product, or the nature of the technology used. Out of the entire range of values that may be theoretically possible, the marketer may choose only a few for active consideration. For example, a soap can technically be produced in all colours. But some colours, like dark black, may be ruled out. Some attributes, such as price or product life, may have continuous values. Conjoint analysis can treat them as well. Some range of values may be specified for them, or the range may be kept open-ended.

3. Fractional Factorial Design of Experiment

In conjoint analysis, the profiles of different products are presented to the consumers for their responses. These profiles are generated by varying the levels of their attributes. For example, suppose we are conducting a conjoint analysis-based study of washing machines. Let us assume that the most important attributes considered by its customers are brand, price, washing capacity, colour, and loading

type. Let us further assume that the levels of attributes shown in Table 12.1 are considered relevant and interesting by the marketer for the study.

Table 12.1: Attributes of Washing Machines

	Attributes	Levels
1	Brand	A B C D
2	Price	Rs. 15,000/- Rs. 20,000/- Rs. 25,000/-
3	Washing Capacity	High Medium Low
4	Colour	Steel grey White Light Blue Pink
5	Loading Type	Front-loading Top-loading

Since the 5 attributes can have 4, 3, 3, 4, and 2 levels, the total number of possible product concepts that can be generated by configuring these attributes is $4 \times 3 \times 3 \times 4 \times 2 = 288$. To determine the part-value utilities of each of these attributes' levels, we will need to solicit responses from 288 different product concepts. This number is certainly too large for any consumer. Therefore, we resort to the method of fractional factorial design of experiments to make it manageable.

A fractional factorial design of experiment is a statistical technique that determines the smallest number of product designs required for the study while still providing all of the information that we sought. These designs are also mutually independent (orthogonal) to avoid any redundancy in the data and allow the representation of each of the attributes and their respective levels in an unbiased manner. In the example of the washing machine considered here, this technique has given us only 16 designs out of the 288 possible washing machines. However, it should be noted that such a reduction in the number of product designs is possible only after making certain assumptions. For example, we had assumed that none of the attributes interacted among themselves, or, in other words, that the attributes are considered to be independent of each other. Only under this assumption did we arrive at the figure of 16 product concepts. On the other end, if we had allowed

all the attributes to interact with each other, the required number of product concepts would have remained at 288. With different types of assumptions, the number of concepts required would be in between these two extremes. The 16 product concepts found through this method are not unique. Many other sets of 16 cards would have also been equally good. However, all of these sets would have to be independent and represent all the attributes and their respective levels in an unbiased manner. We are illustrating one such set of 16 cards representing the product concepts of washing machines using the fractional factorial design of an experiment in Table 12.2.

Table 12.2: Description of Product Concept Cards

Card	Brand	Price	Capacity	Colour	Loading Type
1	A	15,000	High	Steel Gray	Front
2	B	25,000	Medium	Steel Gray	Top
3	C	20,000	Medium	Steel Gray	Top
4	D	20,000	Low	Steel Gray	Front
5	A	20,000	Medium	White	Top
6	B	20,000	Low	White	Front
7	C	25,000	High	White	Front
8	D	15,000	Medium	White	Top
9	A	25,000	Low	Light Blue	Top
10	B	15,000	Medium	Light Blue	Front
11	C	20,000	Medium	Light Blue	Front
12	D	20,000	High	Light Blue	Top
13	A	20,000	Medium	Pink	Front
14	B	20,000	High	Pink	Top
15	C	15,000	Low	Pink	Top
16	D	25,000	Medium	Pink	Front

4. Physical design of stimuli

After selecting the product concepts required for the conjoint analysis study, they need to be exposed to the consumers as stimuli. This may be done in a variety of ways, mainly depending on the demands of the situation and the convenience of the researcher. Of course, it would be most desirable to present real-life prototypes of the products according to the product concepts specified. These may be given to consumers for their usage or as trials. But such extreme ways of presenting the products may not always be possible or even necessary. In such cases, product models, diagrams, or even verbal descriptions may be adopted. In our example of washing machines, it may not be possible to produce the 16 prototypes and take them to consumers. It is possible that just their models or pictures will suffice.

5. Data Collection

Ease of data collection is a key feature of conjoint analysis. The consumers are asked only to assign ratings scores to each of the product

stimuli or even rank the “different concepts” presented to them. This is quite a realistic task and is similar to shopping experiences where the customer merely makes choices. He does not have to respond to each of the attributes separately. This feature of conjoint analysis is possible due to the use of a fractional factorial design of experiment before the collection of data and the use of conjoint analysis after the collection of the data. In other words, the use of the technique eases the burden on the respondents.

6. Determination of part worth utilities

The rating or ranking data obtained from the consumers is analysed next. Two methods are more popular for this purpose. In one method, the part worth utilities for each of the levels of each attribute is arbitrarily assigned. Based on these assumed values, the consumer’s overall rating or ranking (as the case may be) is estimated. These estimated responses may, understandably, be quite different from the actual data. After a few iterations, convergence is achieved, so that the part-worth utilities found approximate the estimate responses to the actual data the best.

In the alternative method, the part worth utilities are derived in one step. Here, an error function describing the difference between the estimated and actual data is defined. This function is then minimised.

After using any of the available methods, the output is obtained for each respondent separately. This is quite significant, as the disaggregated data can be combined in any way desired. But if the output was only at the aggregate level, then disaggregation might not have been possible.

In our example of a washing machine, the part worth utilities may be found for each of the attributes and their levels as shown in Table 12.3

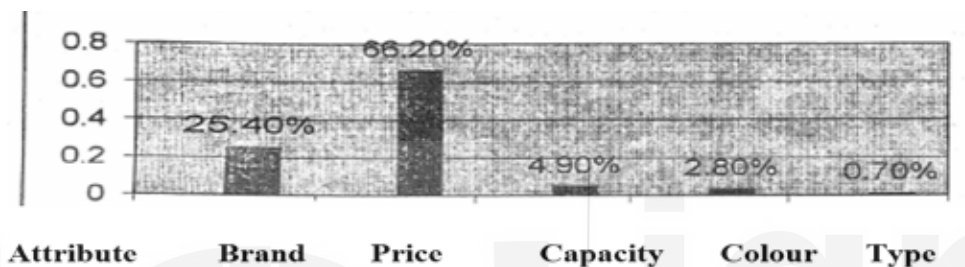
Table 12.3: Part worth utilities for the attributes and their levels

	Attribute	Levels	Part Worth Utility
1	Brand (25%)	A	2.2
		B	-2.3
		C	-0.8
		D	0.9
2	Price (66.2%)	Rs 15000	5.9
		Rs. 20000	-0.1
		Rs. 25000	-0.1
3	Washing Capacity (4.9%)	High	0
		Medium	0.1
		Low	-0.1
4	Colour	Steel Gray	-0.3
		White	-0.1
		Light Blue	0.1
		Pink	0.2

5	Loading Type	Front	-0.1
		Top	0.1

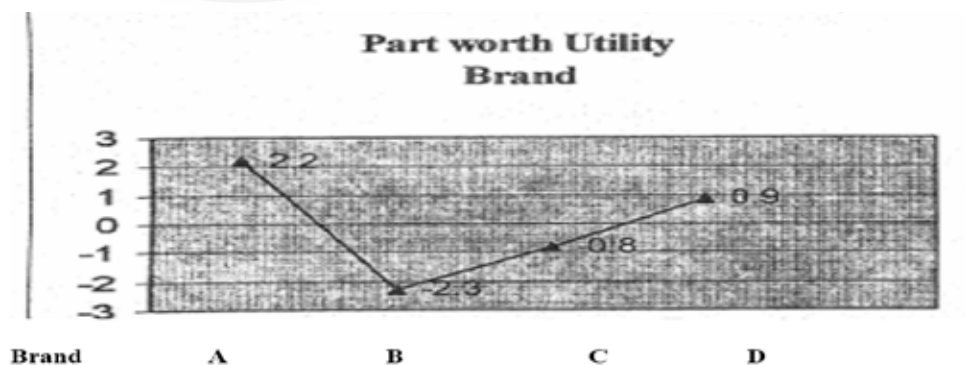
From the Table 12.3, we find that price plays the most important role (66.2%) in the minds of customers. This is followed by brand (25.4%), washing capacity (4.9%), colour (2.8%), and shape (0.7%). The relative importance values for the attributes were calculated by subtracting the maximum and minimum values of the respective attributes' part-worth utilities. The relative importance of different attributes has been shown in Figure 12.1.

Figure 12.1: Relative importance assigned by the customers to attributes of washing machine



Customers respond quite predictably to different price levels. They prefer the less expensive ones over the more expensive ones. Among brands, we find that “A” is the most liked brand and “B” is the least liked one. However, they do not prefer the “high” capacity the most. They like the “Medium” size best. In terms of colour, they prefer pink the most. Between the two types, front-loading washing machines are preferred more. However, the scores for wash capacity, colour, and shape are quite low. Figure 12.22 depicts a graphical representation of the conjoint analysis part worth utilities.

Figure 12.2: Part worth utilities assignment by the customers to different attributes levels of Washing Machine



Figures 12.3–12.6 depict the part-worth utilities for various attributes and their respective levels along a common Y-axis. As a result, the part-worth utilities become comparable both within and across attributes. This way, we can understand the trade-offs that consumers make.

Figure 12.3

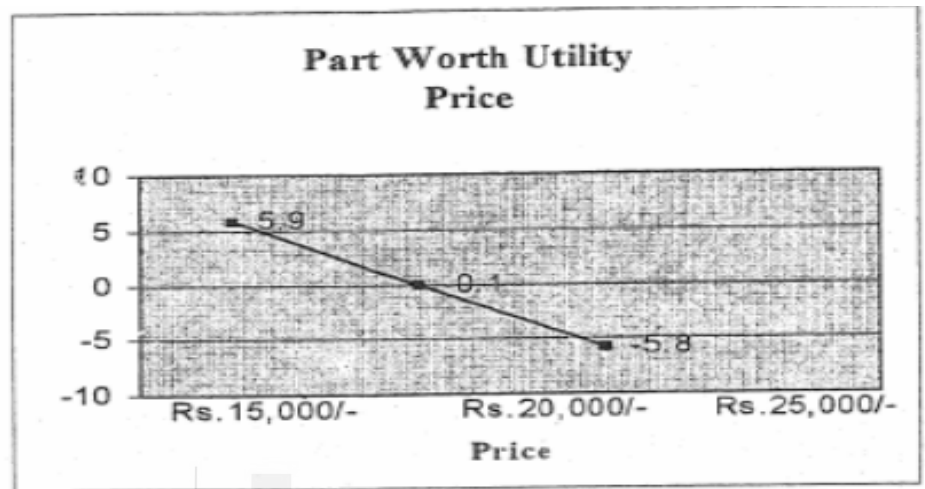


Figure 12.4

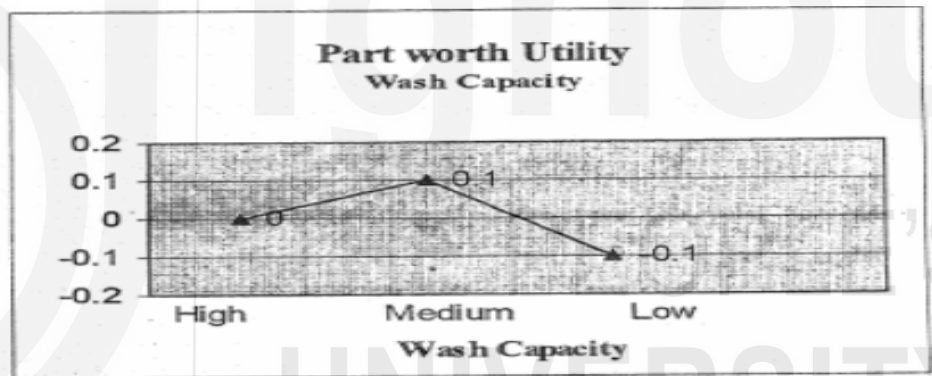


Figure 12.5

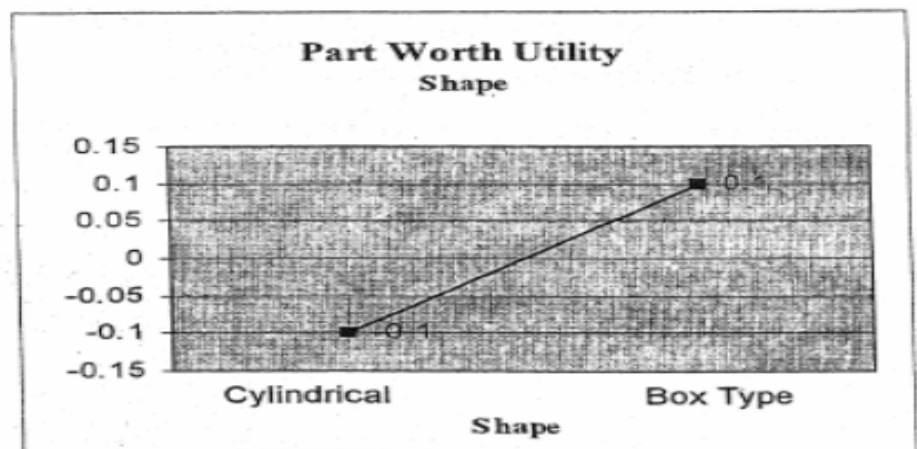
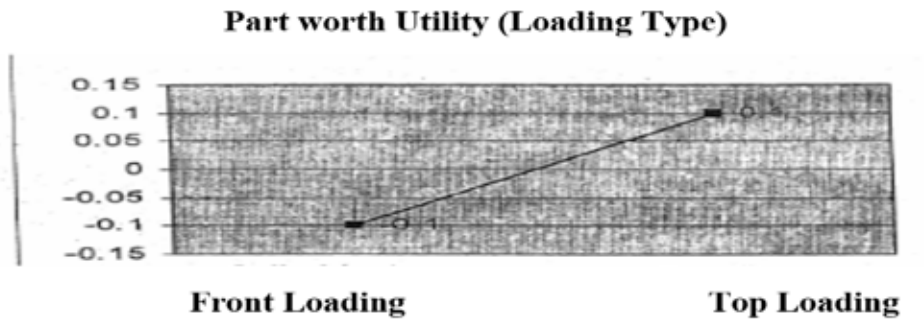


Figure 12.6



The part-worth utilities can now be added to determine the total utility for each of the possible product concepts. This allows us to scan the consumer preference pattern for all 288 product concepts, although he has been exposed to only 16 of them. We can now also rank all of these 288 product concepts.

7. Conjoint Analysis applications

Calculating the part-worth utilities is only the beginning of many interesting applications of conjoint analysis. In marketing, conjoint analysis has wide application. Most often, a marketing manager needs to understand how a customer makes a purchase decision and how they evaluate various tangible and intangible attributes associated with a particular product. For example, while making a purchase decision for a smart phone, the customer makes judgments about the product based on the combination of the brand name, price of different models, internet connectivity, screen size, embedded memory, ability to sync multiple email accounts, etc. The main objective of conjoint analysis is to find the most preferred attribute of the product.

Let us explain some important marketing applications of conjoint analysis.

i. Optimum Product Design

Since all possible product concepts can be compared after adding their respective attribute levels and part-worth utilities, it is possible to determine the demand for different products out of any given set of available products in the marketplace. The demand levels can be converted into profit figures as the cost of producing and marketing can also be calculated. These cost calculations are possible as the volume of operations and the features of the products are now known. As a result, the best product can be chosen based on profits (or any other given management goal). Customers' differential rates of purchase of products are also duly considered at this stage.

Quite often, a manager will want to know the effects of a minor change in any of his or her company's or a competitor's attributes. Conjoint Analysis facilitates this type of "What if" analysis with its data base of part-worth utilities. In fact, different kinds of scenarios can be simulated, and the manager can optimise not only the product but other aspects of his marketing strategy. Similarly, whenever there

is any change in the competitor's actions or in the environment, a fresh scenario can be drawn for simulation. Of course, the simulation will be limited to the attributes considered in the analysis. This feature also helps in increasing the shelf life of the conjoint analysis output.

ii. Market segmentation

Since the conjoint analysis is done at the individual customer level, the individual customer's identity can be retained throughout the analysis. Thus, consumers can be segmented according to their sensitivities to different product attributes.

It is also possible to identify the customer segments that would be most attracted to the proposed product. This helps in having a focused match between the chosen product position and the target customer segment. It can also help in identifying that part of a competitor's market that needs to be acquired in order to gain market share from them. Similarly, the same type of analysis can be done to identify the most vulnerable section of one's own market segment.

Sometimes, an additional product offer appears to be quite attractive. But this may come at the cost of cannibalization. Conjoint analysis can also help in estimating the effects of cannibalization. Thus, it helps in maximising the net profits of the organisation.

iii. SWOT Analysis

First and foremost, the brand's part worth utility can reveal information about its relative brand strength. Similarly, by looking at the other features of one's own and competitor's offers, conjoint analysis enables the marketer to conduct a detailed SWOT analysis.

iv. Estimating Customer Level Brand Equity

Conjoint Analysis bridges the gap between consumer perceptions and the financial worth of offers. This can be used for estimating the important parameter of brand equity at the consumer level. There is scope for differentiating the "loyal," "acceptors," and "switchers" for more accurate calculations of brand equity.

Activity 1

Speak with the marketing manager of your own company or any other organisation you are familiar with and discuss the important attributes of the products or services they offer to customers. Also, classify each attribute into the appropriate levels.

.....

.....

.....

.....

12.4 USE OF SPSS FOR CONJOINT ANALYSIS

Let us assume that a market research team wants to measure the relationship between the attributes of soft drinks and consumer preferences. For this, they

considered four-factor levels that characterize the “colour of the product”, six-factor levels for “brand”, three-factor levels that characterizes “price”, and two-factor levels for “label text,” as shown in the Table 12.4.

Table 12.4: Factors with levels for soft drinks

Factor	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6
Colour	Dark brown	Orange	Colourless	Yellow		
Brand	Coke	Mirinda	Fanta	Sprite	Pepsi	Maaza
Price	Rs 40/ litre	Rs 25/litre	Rs 30/litre			
Label	complex	simple				

Step 1: Objectives of conjoint analysis

As mentioned above, the objective of the study is to understand the consumers’ preferences towards the attributes of the mentioned soft drinks by determining the contributions of predictor variables and their levels using SPSS.

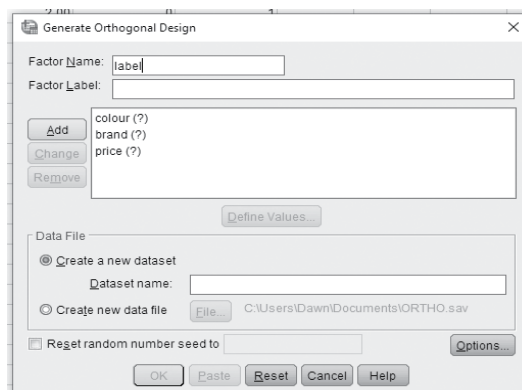
Step 2: Create an orthogonal design

In Table 12.4, there are 144 possible profiles of the product (4 x 6 x 3 x 2). The data analyst predicted that there would be decision fatigue among consumers when forced to choose such large profiles. Therefore, to minimize the customers’ burden, “orthogonal design” was adopted using SPSS to lower the number of combination profiles and thereby reduce the implementation difficulty and improve the response accuracy of the survey.

To create the combination of factor levels, we must first generate a representative subset profile known as an orthogonal array using SPSS. The below steps detail the procedure to create the combinations, or in other words, choice cards.

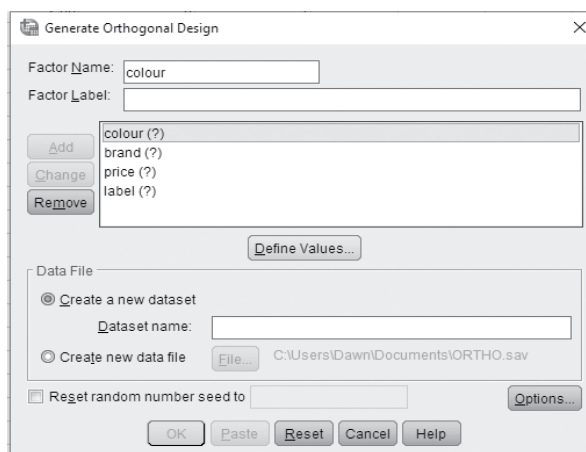
From the SPSS menu choose: Data → Orthogonal Design → Generate, and follow the below mentioned steps:

1. Selecting ‘Generate’ will take you to the ‘Generate Orthogonal Design’ dialog box. Feed the ‘Factor name’, ‘Factor label’ mentioned in the dialog box one by one., here the colour, brand, price, and Label text are the factors.

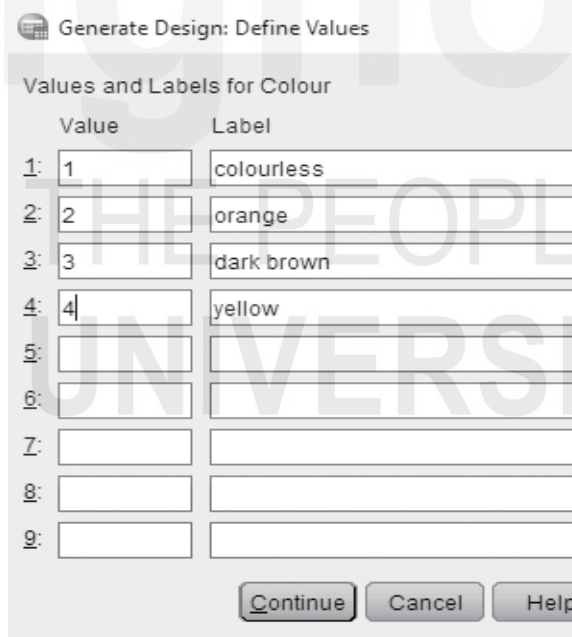


2. Click ‘Add’ to feed the second factor name (brand) and repeat the procedure until all factors are entered in the dialog box.

- Now, click on the 'factor name' that is already been fed in the dialog box and select 'Define Values'.



- A new dialog box named 'Generate Design: Define Values' appears where the levels are to be mentioned, here, the levels for factor "colour" is dark brown, orange, colourless and yellow. Repeat the procedure to feed all the levels of other factors considered.



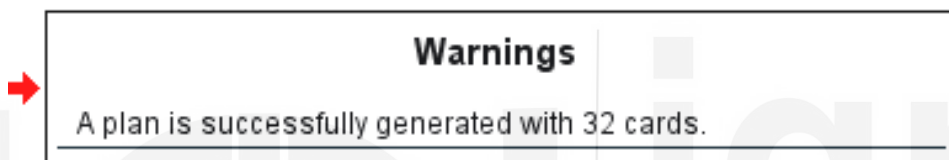
- Once the levels are mentioned, go to the data file section in the same dialogue box, "Generate Design: Define Values," and create a new data set name.
- After the data set name is defined, check that the reset random number seed is set to 200000.
- Now go to options and select the minimum number of cases to be generated. As per the thumb rule, the minimum number of cases (profiles) required to be considered is equal to the total number of levels across all factors - number of factors + 1 (In this example, as mentioned, we have four factors and 15 levels; therefore, the minimum

number of profiles required is 12)

8. After deciding on the minimum number of cases to be generated, the number of holdout cases needs to be determined. Most researchers suggest a ratio of 5:1. For example, if the minimum number of cases to be generated is 60, 10 cases are chosen as holdouts, and the remaining 50 cases are generated as choice cards.
9. Select “continue” followed by “OK.”

Step 3: Choice Card generation

To display the generated choice card or cases go back to the tool bar and from the menu, choose, Data → Orthogonal Design → Display. SPSS will now take you to the output file with the choice cards generated. In this example, 32 choice cards were generated, which is greater than the minimum number of required profiles (12). Hence, we can proceed.



Step 4: Data collection: Gathering preferences from respondents:

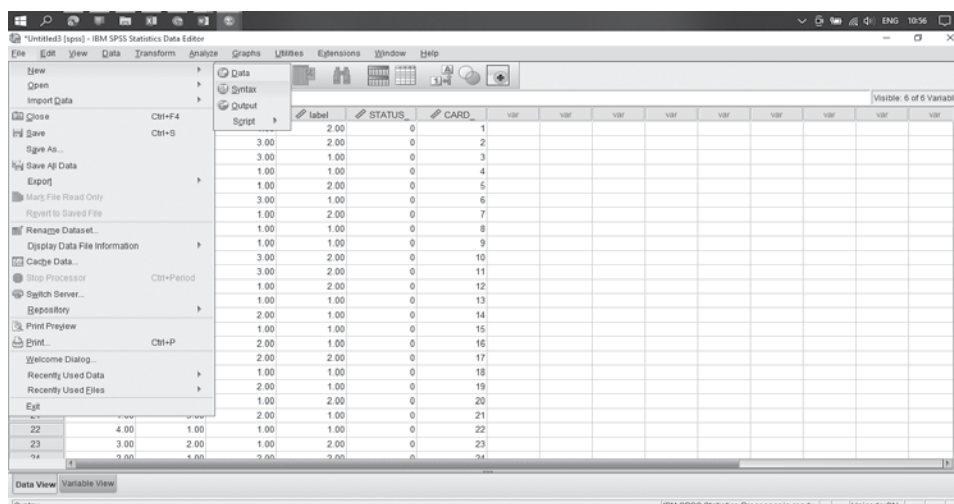
With the profiles or choice cards being generated in the SPSS, the next step is to gather preferences for each profile from customers through survey, either by ranking or rating preference.

Step 5: Perform conjoint analysis with CONJOINT syntax

To obtain the utility of the attribute levels the following steps are to be carried out.

1. Firstly, code the customers preference for each profile in SPSS obtained from the survey
2. After coding or importing the coded data into SPSS, perform the conjoint analysis through the following steps

☐ From the menus choose File → New → Syntax



- This opens a syntax window. Enter the command syntax for the CONJOINT command.

The following syntax is used to perform conjoint analysis when the collected preference is in terms of ranks.

```

1  CONJOINT PLAN = 'D:\softdrinks_choicecards.sav'
2      /DATA = 'D:\preference_softdrinks_choicecards.sav'
3      /SEQUENCE = Pref1 TO Pref32
4      /SUBJECT = ID
5      /FACTORS = colour (DISCRETE) brand (DISCRETE) price (LINEAR LESS) label (LINEAR MORE)
6
7

```

The actual Conjoint Analysis is performed with help of the procedure CONJOINT. There are various subcommands within this procedure: -

- The 'PLAN' subcommand tells CONJOINT, which file contains the data for the reduced design
 - The subcommand 'DATA' tells CONJOINT, which file contains the rankings of the respondents
 - 'SUBJECT' is used to define the serial number or ID of the customer.
 - 'FACTORS' is used to define the relationship between the factors and the preference values (ranks) of the interviewed persons. The relationships can be DISCRETE, LINEAR, IDEAL or ANTIIDEAL. In this example the factor 'colour' and 'brand' are discrete values, hence their relationship is mentioned as 'DISCRETE'. The relationship of factor 'price' is mentioned as 'LINEAR LESS' because lesser the price more the demand and for the 'label' it is 'LINEAR MORE' as from the conceptual knowledge customers trust products which have a detailed product label.
3. Now, run the command for the syntax generated using the 'run selection' (green left arrow icon) to obtain the output file with utilities for each level.

Step 6: Output file with utility for attribute levels

As mentioned in step 5., once the command is run from the Syntax Window, the utility is obtained through the following steps:

1. Check the significance of the estimated utility: When the choice tasks involve nonmetric rank-order data, correlations based on the actual and the predicted ranks (e.g., Spearman's rho or Kendall's tau) are used. When the choice tasks involve a rating (e.g., preference on a 0–100 scale), then a simple Pearson correlation. In this hypothetical case, since the preference of consumers were collected in rank-wise, Kendall's tau is used to check the significance of the model. Since the p-value is less 0.05, the resultant model is significant.

Correlations^a

	Value	Sig.
Pearson's R	.629	.000
Kendall's tau	.484	.000

a. Correlations between
observed and estimated
preferences

2. Estimate the overall utility: The Overall Utility is obtained from the summation of Part-Worth of level. For this example, the overall utility is estimated from the part-worth or utility for each level obtained in the analysis shown in the below table.

Utilities			
		Utility Estimate	Std. Error
colour	colourless	-.542	1.162
	orange	-.500	1.162
	dark brown	1.694	1.162
	yellow	-.653	1.162
brand	Pepsi	-.993	1.304
	Fanta	-3.062	1.304
	Coke	3.660	1.703
	Sprite	.549	1.703
	Mirinda	3.576	1.703
	maaza	-3.729	1.703
price	40	-.702	.809
	25	-1.404	1.618
	30	-2.106	2.427
label	complex	-1.264	1.341
	simple	-2.528	2.683
(Constant)		20.131	2.560

From the estimated utility/ part-worth obtained from the conjoint analysis, it is observed that in the factor colour, customers prefer 'dark brown' products, since the estimated utility (1.694) is highest for this level. Likewise, customers highly prefer Coke over to other brands. Rupees 40 and complex labels have the highest utility in the factor 'Price' and 'label'.

Overall highest utility = Part-Worth of dark-brown + Part-Worth of brand_coke + Part-Worth of the price_Rs 40+ Part-Worth of complex label.

i.e., Overall highest utility = 1.694+ 3.66 – 0.702 - 1.264 = 3.388

Overall lowest utility = Part-Worth of yellow + Part-Worth of maaza + Part-Worth of the price_Rs 30+ Part-Worth of simple label.

i.e., Overall lowest utility = -0.653 - 3.729 -2.106 – 2.528 = - 9.016

Stage 7: Interpretation of results by factor importance

From the analysis, the brand of the product is the most dominating factor with a factor importance of 52.24% along with the colour of the product with 27.47% of influence. Label is next important factor with 11.553%. The results also provide insights that price has the least importance about 8.744% while purchasing the cold drink.



Importance Values	
colour	27.467
brand	52.237
price	8.744
label	11.553
Averaged Importance Score	

From the results, it is seen that Maaza, despite being the favourite among mango lovers, still scored the lowest utility value of -3.729. This may be due to the characteristics of the soft drinks, where Maaza is the only non-aerated drink considered in the levels. This provides insight into why it is not significant to compare aerated and non-aerated drinks. Maaza is hence not included in the analysis.

Considering only the aerated drinks in the analysis, it is noted that Fanta has the lowest utility of -3.062. One of the interpretations that can be made is that sales of Fanta can be boosted by incorporating attributes such as a dark brown drink (which is practically not possible for an orange-flavoured drink), but with complex labels and a price of Rs. 40 per litre, sales can be boosted, which were found to have the highest utility. It is also noted that Mirinda, though an orange drink, is doing comparatively better than Fanta. Therefore, Mirinda is a strong competitor to Fanta. Through this hypothetical case, one can understand the attributes that contribute to the success or failure of a particular brand in a product category.

12.5 MULTIDIMENSIONAL SCALING (MDS)

Multidimensional scaling (MDS) or perceptual mapping (or positioning mapping) is a multivariate analysis technique that is particularly suitable for measuring human perceptions and preferences for products and brands. MDS is concerned with the spatial representation of relationships among behavioural data, i.e., buyer perceptions and preferences.

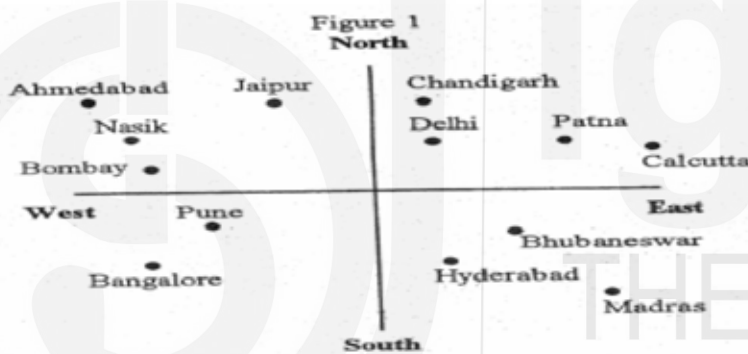
Multidimensional scaling is often used in marketing to identify key dimensions underlying customer evaluations of products, services, or companies. Other common applications include the comparison of physical

qualities (e.g., food tastes or various smells), perceptions of political candidates or issues, and even the assessment of cultural differences between distinct groups. The main purpose of MDS is to map the objects in a multidimensional space such that their relative positions in the space show the degree of perceived proximity or similarity amongst them. This map is termed a “perceptual map” or “spatial map.” Thus, the MDS involves the following aspects:

- It helps in the identification of attributes that influence how consumers perceive or assess brands or products.
- It enables the positioning of different products or brands on the basis of these attributes.
- It helps generate a perceptual map, indicating the location of the brands on the basis of attributes.

Let us explain the concept of MDS through spatial map. Consider the following geographic locations of various cities in our country:

Figure 12.7



The distance between any two points on the map is given by the formula

$$d_{12} = \sqrt{(X_1 - X_2)^2 + (Y_1 - Y_2)^2}$$

The distance between various cities could be calculated and put in a tabular form as shown in Table 12.5

Table 12.5

Cities	Bombay	Hyderabad	Delhi	Bhubaneswar	Patna
Bombay					
Hyderabad					
Delhi					
Bhubaneswar					
Patna					

In short, it is a relatively simple matter to go from the map in Figure 12.7 to the set of numerical distances in Table 12.5. However, the converse, or going from Table 12.5 to Figure 12.7, is very difficult.

Given different objects and their attributes, every consumer perceives the objects in a different manner. So, one must first understand the consumer's perceptions (distance) of products and brands and then plot a graphical representation of the same. The task is to work backwards. That is, we would have to determine the:

- Number of dimensions (which can be more than two in perceptual mapping) at the same time.
- Configuration (or pattern) of points in that perceptual space, so that their computed inter-point distances most closely match the input data of say, Table 12.5. This is the problem of metric MDS. Whenever input data is interval-scaled or ratio-scaled, it is called metric MDS.

The same input data in the format of Table 12.5 could be transformed into a rank-order list. Suppose we take the smallest distance between two cities and give it the rank of 1. The remaining distances between cities could then be arranged in ascending order and ranked from 2 to 13 (for example, the largest distance on the map could be between Calcutta and Bombay). As a result of using this method, we discover:

- Number of dimensions
- Configuration (pattern) of points in that dimensionality, so that the ranks of their computed inter-point distances matched the ranks of the input data as closely as possible. This is the problem of non-metric MDS. Whenever input data is in the form of nominal or ordinal form, it is called a non-metric MDS.

Whichever method for MDS one selects (metric or non-metric MDS), it does not change the relative distances of the points.

12.6 ATTRIBUTE SPACE - OBJECTIVE SPACE AND PERCEIVED SPACE

Any product or service can be visualized as composed of both objective and perceived (subjective) attributes, or "dimensions." A company has an objective description of its product in physical or chemical terms. Such an objective attribute as "space," in which various brands are viewed as points positioned in space, will usually not agree with the buyer's perception of the products. The buyer's perception of a product, service, or company is a sum total of not only the company that is selling the product, but also other factors in the market like competition and the products of competitors. advertising, after-sales service, etc.

Also note that the dimensions of perceived space need not be the same as those of objective space. As an example, for many consumer goods that are highly individualistic, like tea, coffee, toothpastes, perfumes, and shampoos, where strong individual preferences play a key role in buying decisions, the objective space of the company and the perceived space of the consumer may be widely different. But in the case of an industrial product, the perceived configuration of brands may agree rather closely with an objectively constructed configuration in which measurements of such characteristics as speed, reliability, and efficiency can be made rather

straightforwardly.

Ultimately, the view of the consumer is the most relevant one, as this is what would make the consumer go out and buy a particular product or brand.

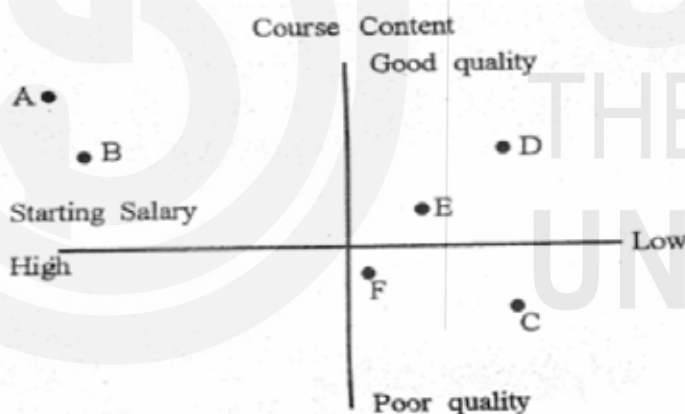
Example of Multidimensional Scaling

A study was conducted in a leading business school (A) among the first-year MBA students to find out how other business schools were “similar” or “different” from the “A” business school. Various attributes are used to evaluate a business school: faculty, faculty/student ratio, starting salaries of the MBA students, physical facilities like a library and computers, and so on. But which of the attributes do students “actually” use in making a competitive comparison between the business schools?

Let us consider six leading business schools as A, B, C, D, E, and F. Around one hundred students were given a questionnaire in which they were asked to respond to the questions that indicated various attributes of a management school, and students had to mark schools that were similar or dissimilar on the different counts.

From the information that was gathered from the questionnaire, what emerged was that course content and starting salary are the most important criteria by which students judge a business school (Figure 12.8).

Figure 12.8



The Multidimensional scaling technique is therefore very useful in plotting the perceived perception of the consumer about different brands, products, and services. This analysis could be used by a company’s marketing manager to develop additional marketing strategies.

12.7 APPLICATIONS OF MULTIDIMENSIONAL SCALING

Some of the typical marketing applications that emerge from the MDS technique are:

a) Market Segmentation

Market segmentation is the technique of trying to identify groups of consumers who exhibit commonality of perception of products

and preferences. One can use MDS techniques to identify present perceptions of products by consumers and use it to modify the company's product, package, advertising, and additional features so that the product offering of the company moves more and more closer to the "ideal" requirement of the consumer.

b) Advertisement Evaluation

The MDS technique could be used at the stage of advertisement pre-testing. Once an advertisement has been developed, it can be tested for similarity or dissimilarity with other advertisements in the same product category. As the ultimate objective of an advertisement is to effectively communicate with the target consumer, this is only possible if the advertisement is distinct in its message from the other competing advertisements.

c) Product Re-positioning Studies

If a company is interested in re-positioning its product or service (in the mind of the consumer), the first and foremost activity to be done is to assess the current perception of the product in the mind of the consumer. The classic repositioning case is that of Cadbury Chocolates, which kept on assessing its positioning platform and successfully moved chocolate from a product perceived as one for children to one that could be consumed by a person of any age at any time of the day and for varied occasions.

d) New Product Development

The MDS technique shows us the various perceived perceptions of the different brands. Spaces or gaps in product perceptions could be used to develop new offerings for the target consumers.

e) Test Marketing

The MDS technique can be used to identify cities that have similar demographic characteristics, and one could then identify a city that could represent a national character and use that city for test marketing. One can thus "observe" that MDS is a very useful technique to help understand the market and develop strategies for the future.

12.8 USE OF SPSS FOR MULTI DIMENSIONAL SCALING

Assume the Pepsico product manager employs perceptual mapping techniques to determine the relative position of the product and the major competitors. For this analysis, the market researcher considered six soft-drink brands, namely Coke, Miranda, Sprite, Fanta, Maaza, and their signature brand Pepsi.

Step 1: Objective(s) of the research: As mentioned above, the objective of the study is to identify the position of the Pepsi in the market and identify its major competitors using perceptual maps developed in SPSS.

Step 2: Identification of all relevant objects and methods to evaluate:

Since perceptual mapping is a relative positioning technique, the researcher must make sure that only relevant products, goods, services, or other items are included. In this case, the researcher considered six soft-drink brands namely Coke, Miranda, Sprite, Fanta, Maaza and Pepsi. After having selected the brands, three key decision needs to be made with respect to the product.

- **Base of analysis- similarity or preference data:** At this stage decision on the type of data intended to be collected is taken i.e., to compare the drinks based on similarity or preferences. For this analysis, the market researcher emphasized collecting data based on the similarity /dissimilarity of soft-drink for the use of MDS technique.
- **Using a disaggregate or aggregate analysis:** The next decision involves whether to use aggregate or disaggregate analyses. As mentioned earlier, aggregate analyses provide an overall perspective on the entire sample in a single analysis. Whereas disaggregate analyses allow for an individualized analysis, where all respondents can be analyzed separately and even portrayed with their own personal perceptual map. In this case study, analyses are presented at the aggregate level.
- **Selecting decompositional or compositional methods:** The next step is to choose between decompositional (attribute-free) or compositional (attribute-based) methods that revolve around the level of specificity the researcher desires. In this analysis both the methods are employed. Firstly, the decompositional method is employed by simply collecting the similarities among the soft-drink brands. Later, certain attributes were considered to understand the most determinantal attribute to soft drinks using conjoint analysis.

Step 3: Data collection: Collection of Preference Data: The starting point for data collection for the MDS analysis is considering the perceptions of the respondents concerning the similarity or dissimilarity of Pepsi and five other competing soft-drink brands in the market.

Similarity judgments were made with the comparison-of-paired-objects approach. The 15 pairs of soft-drink brands $[(6 * 5)/2]$ were presented to the respondents, which indicated how similar each was on a 5-point scale, with 1 being “not at all similar” and 5 being “very similar.” The results are tabulated for each respondent in a lower triangular matrix.

Step 4: Create the perceptual map using SPSS

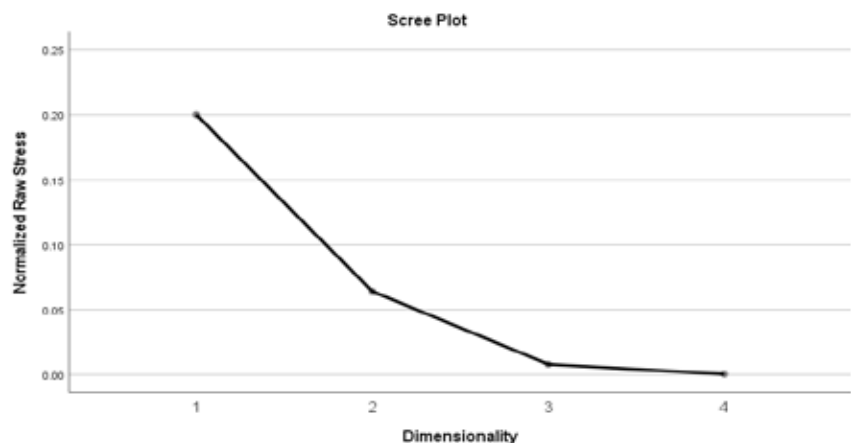
For the creation of perceptual maps for the collected data, the following steps need to be input first in SPSS.

1. Open the SPSS data file, enter or import the coded data to SPSS by going to File → Import data → Excel.
2. Once the data is coded in SPSS, and the variables are defined in the variable view window, the data is ready to be analysed.
3. For MDS analysis, go to menu and choose Analyse → Scale → Multidimensional Scaling (PROXSCAL). This will take you to ‘Multidimensional Scaling: Data format’ window with default

settings. Select 'Define' to open the dialog box to mention the proximities/products considered.

4. Now, in the new dialog box named 'Multidimensional Scaling (Proximities in Matrices Across Columns)' select all the variables, for example in this analysis Coke, Mirinda, Sprite, Maaza, Fanta, and Pepsi are the brands/variables.
5. After entering the variables in 'Proximities', now select 'Model'. A new dialog box opens named 'Multidimensional Scaling: Model' where three important categories need to be chosen as mentioned below:
 - Firstly, based on the type of data collected, the 'Proximity Transformations' needs to be selected. In this case study, the data taken are in intervals, therefore select the 'Interval' option.
 - Next, in 'Shape' based on how the data is entered, one category must be selected. In this case study, the results are tabulated for each respondent in a lower triangular matrix. Therefore 'Lower-triangular matrix' is chosen.
 - In 'Proximities' the measurement of similarities or dissimilarities must be chosen based on the framed objectives.
 - Finally, in 'Dimensions', the number of dimensions/ attributes to compare the product needs to be addressed. The minimum dimension is 1 and maximum is (n-1) where n being the number of products/ brands considered. In this example, first the analysis is carried out with 1 (minimum) and 5 (maximum) dimension.
6. The next step is to select 'Plots' and check 'Stress' in Plots to determine the number of dimensions and press 'Continue' and then 'OK'. For example, in the analysis, in the scree plot in Figure 12.9, the elbow indicates substantial improvement in the goodness-of-fit when the number of dimensions is increased from 1 to 2. Therefore, the best fit is obtained with a relatively low number of dimensions (two).

Figure 12.9: Scree Plot



7. Once the number of dimensions is determined, go back to 'Dimension' and mention the minimum number of determined dimensions. In

this case, the minimum number of dimensions determined is 2, and thereby maximum is 4 (a totally six products considered).

8. Now the final step, select 'Output' mentioned in the dialog box. Select the display option in the appeared dialog box and check the 'Common space coordinates' to get the perceptual map. Press 'Continue' and then 'OK'.

Step 5: Evaluation of Goodness of Fit: Firstly, in the output data, the Goodness of fit needs to be analysed with Stress and Tucker's coefficient. The stress measures the misfit of the solution, lower values of stress indicate a better fit. In an ideal case stress value will be 0, indicating excellent goodness of fit. Values for Tucker's coefficient can fall between 0 and 1. Models with strong fit typically have values that are close to 1. As shown in the below output table for Goodness of fit, the S-stress value is quite less i.e., 0.166 and closer to zero, and Tucker's coefficient value 0.9674, closer to 1. Thus, indicating a better fit.

Stress and Fit Measures	
Dimensionality: 2	
Normalized Raw Stress	.06414
Stress-I	.25327 ^e
Stress-II	.81028 ^e
S-Stress	.16585 ^f
Dispersion Accounted For (D.A.F.)	.93586
Tucker's Coefficient of Congruence	.96740
PROXSCAL minimizes Normalized Raw Stress.	
e. Optimal scaling factor = 1.069.	
f. Optimal scaling factor = .959.	

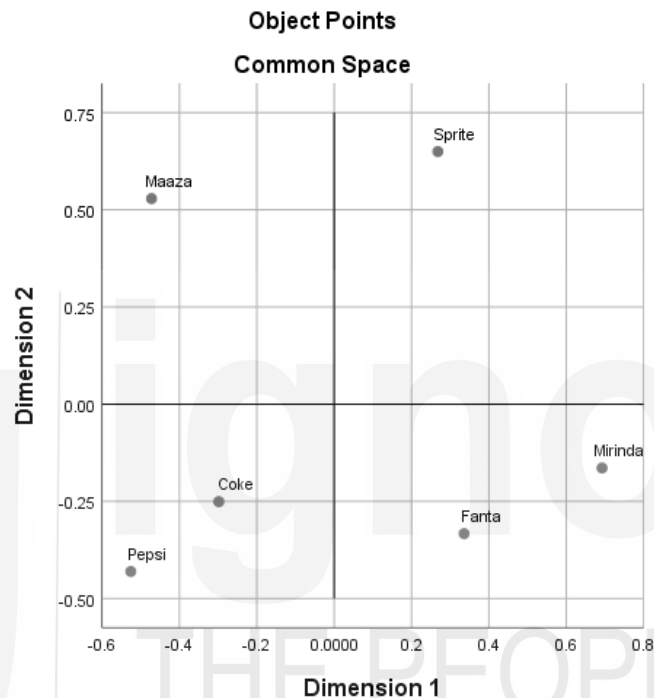
Step 6: Interpreting the MDS results: Once the perceptual map is established, we can begin the process of interpretation. In this example, Pepsico also gathered ratings of each soft-drink using three attributes such as colour, price and label. These attributes along with the brands were analysed using conjoint analysis in order to determine the most influential factors/attributes. Analysing the factor importance obtained in the conjoint analysis, colour and label text of soft-drink products were found to be the most influential attributes.

Therefore, here in MDS, the interpretation of perception of similarity between the soft-drink products, the two influencing determinants are assumed to be colour (dimension1) and label (dimension 2) by the data analyst.

From the below figure 12.10, it is observed that Pepsi is most similar to Coke, Mirinda to Fanta, while Sprite and Maaza have no similar soft-drink product. Even from the analysis based on the conceptual knowledge, it is

evident from the perceptual map that the respondents have perceived the similarity based on colour, as Pepsi and Coke have similar characteristics of dark-brown colour, Mirinda and Fanta to be orange, while Maaza, and Sprite are yellow and colourless respectively, thus, they lie in different quadrants. From conjoint analysis, it is identified that Coke is the major competitor along with Mirinda and Sprite for Pepsi, and Fanta to be the least in aerated drinks.

Figure 12.10: Perceptual Map



Step 7: Validation of the MDS results: The validation of results are done by applying both decompositional and compositional methods. In the decompositional method (attribute free), as mentioned above, from the conceptual knowledge it is assumed that colour is the most deterministic factor to which the respondents have perceived the similarity of a pair of soft-drink brands. This is validated by using a compositional method, where conjoint analysis was performed, and again colour and label of the product are found to be most influencing attributes.

Thus, the case demonstrates the application of multidimensional scaling in identifying the relative position of brand based on identified dimensions.

Activity 2

Speak with the marketing manager of your own company or any other organisation you are familiar with and discuss the positions that their products or services occupy in the market in comparison to the brands that are their competitors.

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12.9 DIFFERENCE BETWEEN MULTIDIMENSIONAL SCALING AND CONJOINT ANALYSIS

Differences between Multidimensional scaling and conjoint analysis are shown in Table 12.6

Table 12.6: Difference between Multidimensional scaling and conjoint analysis

Basis	Multidimensional scaling	Conjoint analysis
Purpose	The primary purpose of MDS is to identify the dimensions/attributes by which objects are perceived by the consumers.	Conjoint analysis is used to measure the joint effect of the identified or observed attributes that are found to be important from consumers point of view.
Grouping	In MDS, the products are mapped together based on the revealed similarities and preferences that the consumers perceive.	In conjoint analysis, the most influential attributes that have the highest part-worth/utility for the considered products are combined together to enhance the product.
Statistics	The objects/ products are mapped in a multidimensional space such that the relative position in the space show the degree of perceived similarity or preferences.	Here, the part-worth of each attribute are estimated and overall utility for the highest profile is determined.

12.10 SUMMARY

In this unit, we have discussed two of the techniques of multivariate analysis, viz., conjoint analysis and Multidimensional Scaling (MDS).

Conjoint Analysis is essentially a data decomposition technique that attempts to plot the output data on the joint space of each attribute's importance. The important point to note is that the consumer is not asked to assign scores to different attributes separately. The main steps in using conjoint analysis are determining the salient attributes for the given product from the consumer's perspective, assigning a set of discrete levels or a range of continuous values to each of the attributes, using a fraction factorial design of experiment to design the stimuli for the experiment, physically designing the stimuli, data collection, conjoint analysis, and determining part-worth utilities. The possible applications of conjoint analysis include product design, market segmentation, SWOT analysis, etc.

We also discussed the multidimensional scaling technique, also known as the perceptual map or positioning map, which is a multi-variate technique that is particularly suitable for measuring human perceptions and preferences for products and brands. Its applications include market segmentation, positioning studies, test marketing, etc.

12.11 KEY WORDS

Choice set: Set of profiles constructed through experimental design principles and used in the choice-based conjoint approach.

Conjoint analysis: is essentially a data decomposition technique that attempts to plot the output data in the joint space of each attribute's importance.

Level: Specific nonmetric value describing a factor. Each factor must be represented by two or more levels, but the number of levels typically never exceeds four or five.

Multidimensional Scaling (MDS): also known as the perceptual map or positioning map, which is a multi-variate technique that is particularly suitable for measuring human perceptions and preferences for products and brands

Part-Worth: Estimate from conjoint analysis of the overall preference or utility associated with each level of each factor used to define the product or service.

Trade-off analysis Method of presenting profiles to respondents in which factors (attributes) are depicted two at a time and respondents rank all combinations of the levels in terms of preference.

Utility: An individual's subjective preference judgment representing the holistic value or worth of a specific object.

12.12 SELF-ASSESSMENT QUESTIONS

1. Define conjoint analysis and state its potential applications.
2. What is part-worth or utility in conjoint analysis?
3. Describe some marketing research problems that you feel may be amenable to conjoint analysis and explain how you would use conjoint analysis in these situations.
4. What are the steps involved in conjoint analysis? Explain with the help of examples.
5. Design a conjoint analysis experiment for a product of your choice with at least four variables and two levels of each variable that are appropriate to a marketing decision.
6. Define multidimensional scaling and discuss, with the help of examples, the areas where multidimensional scaling can be applied to marketing.
7. Explain how a perceptual map can be created for supermarkets. Also, explain how this map helps the store manager of a particular supermarket keep his store ahead of the competition.
8. Differentiate between conjoint analysis and Multidimensional scaling.

12.13 PROJECT QUESTION

Consider the following products and try to develop a perceptual map of the various brands in that particular product category. Before attempting such an exercise, the student should understand the product category in its totality, i.e., target consumer, pricing strategy, advertisement strategy, distribution, usage, and attitude toward the product, and then try to develop a positioning map for the same. Once a product has been studied, the study should try to identify two dimensions along which “brands” could be clubbed as similar or dissimilar.

- 1) Toothpaste (Colgate Gel, Pepsodent, Babool, Close-up, Dant Kanti, Sensodyne, Meswak etc.)
- 2) Shampoo (Kesh Kanti Herbal, Dabur Vatika, Sunsilk, Clinic Plus etc.).

12.14 FURTHER READINGS

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